



De La Salle University-Dasmariñas

**POCKETPT: AN AI-DRIVEN REHABILITATION ASSISTANT  
WITH HUMAN POSE ESTIMATION AND PAIN RECOGNITION  
USING CNN AND LSTM**

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## APPROVAL SHEET

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AND LSTM*

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## ABSTRACT

In the Philippines, access to professional physical therapy is severely limited which leaves a vast population without guided rehabilitation for musculoskeletal injuries, often leading to prolonged recovery or re-injury. This study designed, developed, and evaluated PocketPT, an AI-driven mobile application that guides users through personalized rehabilitation programs. The system provides a comprehensive solution by integrating assessment, exercise guidance, and progress monitoring into a single, accessible platform. A Convolutional Neural Network (CNN) model was deployed for human pose estimation, tracking body joint movements to evaluate an affected muscle's range of motion (AROM), attaining an 85.4% accuracy and 87.8% mean Average Precision (mAP@0.5). This architecture significantly outperformed the initially proposed CNN-LSTM hybrid model's 69.5% accuracy. A standalone CNN was implemented for real-time Facial Expression Recognition (FER), analyzing micro-expressions to objectively detect user pain levels during exercises, achieving a 99.46% weighted accuracy with perfect 100% recall for severe pain detection. These are processed by a rule-based decision tree algorithm, which generates and personalizes rehabilitation plans based on user-input data (e.g., affected muscle, pain duration, rehabilitation goal) and the real-time analysis from the AI models.

*Keywords:* Artificial Intelligence, Machine Learning, Convolutional Neural Network (CNN), Human Pose Estimation, Facial Expression Recognition, Pain Detection, Rehabilitation System, Decision Tree Algorithm, Computer Vision



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## CHAPTER I

### INTRODUCTION

Injuries, muscle strain, and body aches are common experiences that affect a broad spectrum of individuals, from casual walkers and active students to professional athletes and fitness trainers. While some may dismiss these as minor inconveniences, their cumulative impact on health, mobility, and productivity can be significant. In the context of sports and physical activity, the risk of injury rises substantially, particularly in high-contact sports. This is especially relevant in countries like the Philippines, where physical activities such as basketball, football, volleyball, and martial arts are embedded in the culture and daily life.

According to the World Health Organization (2023), over 1.3 million people die annually due to road traffic and injury-related causes, while 20 to 50 million individuals suffer non-fatal injuries, many of which are musculoskeletal in nature. In the United States, the National Safety Council (NSC, 2022) recorded more than 3.2 million sports and recreation-related injuries that required emergency medical attention in a single year. In the Philippines, a report from the Department of Health (DOH, 2022) noted that 1 in every 4 adolescent and young adult athletes experiences sports-related injuries, with lower limb injuries such as sprains, ACL tears, and shin splints being the most common, often resulting in pain, limited mobility, temporary withdrawal from sports participation, and increased risk of long-term musculoskeletal complications if not properly managed.



The problem intensifies when the researchers consider the lack of access to appropriate treatment. According to the Philippine Physical Therapy Association (2021), there are only around 6,000 licensed physical therapists serving a population of over 110 million, leading to a staggering therapist-to-patient ratio of 1:18,000. This is far below global standards, and the shortage is even more severe in rural areas where access to rehabilitative care is virtually nonexistent. Many individuals resort to self-medication or home remedies such as applying ice packs or resting, which are often insufficient and may delay proper recovery, increase the risk of complications, or even worsen the injury due to the lack of professional assessment and appropriate treatment.

To address this gap, the integration of digital health tools and artificial intelligence presents an innovative and scalable solution. Our proposed application seeks to revolutionize injury management by guiding users through effective rehabilitation practices, even without direct access to a healthcare provider. Two core technologies support this vision: human pose detection and pain recognition through facial expressions.

Human pose detection is a computer vision technique that analyzes a person's body posture by identifying and tracking key skeletal joints in real time. With this technology, the application can monitor a user's movements during the initial assessment to determine the pain level of a certain user. This reduces the risk of further injury by providing the user with the appropriate rehabilitation exercises caused by too much intense workouts. Pose detection has been widely adopted in sports science, with studies showing its potential to increase exercise adherence and improve recovery outcomes (Chen et al.,



2022).

Complementing this is a feature of pain recognition based on facial expression, which utilizes machine learning to interpret subtle cues such as furrowing eyebrows, squinting eyes, or tight lips, common involuntary responses to pain, enabling the system to provide a more objective assessment of discomfort levels and support more accurate and timely intervention. Through this, the application can detect real-time discomfort and adapt the intensity, pace, or type of exercise being recommended. Research published in the Journal of Medical Systems (Zhou et al., 2021) demonstrates that AI-driven facial expression analysis can achieve over 85% accuracy in detecting pain levels, making it a reliable tool for real-time feedback in remote healthcare applications.

These innovations converge into a powerful, user-centric tool designed to assist students, athletes, trainers, and even non-athletes in managing injuries effectively. The application not only educates users about injury treatment protocols but also empowers them to engage in safe self-rehabilitation, especially in environments where professional help is out of reach. In doing so, it addresses an urgent public health need by bridging the gap between injury occurrence and recovery through the support of smart technology, promoting timely intervention, improved recovery outcomes, and greater accessibility to rehabilitative guidance.



## 1.1 PROJECT CONTEXT

Physical rehabilitation plays a pivotal role in restoring mobility, reducing pain, and improving the quality of life for individuals recovering from injuries or living with musculoskeletal conditions. However, access to consistent and supervised therapy remains a significant challenge for many. Financial limitations, geographic distance from health facilities, and conflicting schedules often prevent individuals from receiving timely and adequate care. While applications such as LUNA, Exer AI, and PT Pal have made strides in providing virtual rehabilitation services, these platforms still rely heavily on human therapists to monitor progress and assess form, thereby limiting their scalability, automation, and accessibility, especially for users in underserved regions (LUNA, 2023; Exer AI, 2023; PT Pal, 2022), and highlighting the need for more autonomous, intelligent systems capable of delivering consistent, real-time guidance without constant professional supervision.

To bridge this critical gap, modern research has turned to human pose detection and pain recognition through facial analysis, two artificial intelligence technologies that show immense promise in revolutionizing digital physical therapy. A study by Chen et al. (2022) highlights how pose estimation can be applied in physical therapy to ensure that patients perform exercises correctly, significantly reducing the risk of re-injury. Similarly, work by Cao et al. (2019) with OpenPose demonstrated how deep learning models can accurately detect body movements even in unconstrained environments, supporting its integration into health monitoring systems.



Recent studies have begun to merge both techniques into physical therapy contexts. For instance, Xu et al. (2023) introduced a system that combines pose tracking with facial expression analysis to adapt exercise recommendations based on a user's movement accuracy and discomfort levels. Their findings revealed improvements in both patient adherence and recovery time compared to traditional methods. This holistic, AI-driven approach enables applications to offer personalized, responsive care without the need for continuous human intervention.

By embedding these technologies, future physical therapy applications can become more than just digital instruction manuals, as they can evolve into empathetic and intelligent companions in the recovery journey. This paradigm shift aims to democratize rehabilitation by making it more accessible, cost-effective, and personalized, offering hope to millions who are currently underserved by traditional healthcare models and fostering a more inclusive and technology-driven approach to recovery.

## **1.2 PURPOSE AND DESCRIPTION**

PocketPT is a mobile application designed to support comprehensive self-guided rehabilitation for individuals experiencing muscle strains and related musculoskeletal injuries. Rather than enforcing a fixed three-stage sequence, the application offers users three distinct therapeutic approaches that can be selected according to their clinical needs and rehabilitation goals: alleviating pain, improving mobility, and strengthening muscle. The pain-alleviation option focuses on reducing or eliminating pain and discomfort through appropriate exercises and strategies; the mobility-improvement option targets



increase in joint range of motion and flexibility; and the muscle-strengthening option emphasizes building muscle strength and endurance to restore functional capacity and reduce the risk of re-injury. Before initiating any of these approaches, users complete a brief, tailored assessment that consists of four targeted questions: defining a rehabilitation goal, identifying the affected muscle group and specifying the target muscle, reporting the current pain characteristics, and indicating any prior injuries or conditions that might influence mobility. Based on this initial assessment and the selected approach, the application personalizes exercise recommendations to address the user's current functional status while supporting safe and progressive rehabilitation.

Before providing treatment, rehabilitation, and strengthening exercises, users must first complete an initial assessment consisting of four key questions designed by the developers. The first question prompts users to select the muscle group they wish to focus on, choosing from the upper extremity, lower extremity, or core muscles. Once a muscle group is selected, users must then specify the specific muscle within that group. The upper extremity includes muscles in the neck, deltoids, chest, triceps, and biceps. The lower extremity consists of the quadriceps, hamstrings, calves, ankles, and gluteal muscles, while the core muscles include the abdominals, obliques, lower back, diaphragm, and multifidus.

Following muscle selection, users must identify their rehabilitation goal by choosing one of three options, each validated by a certified physical therapist: alleviate pain and muscle tension, improve mobility and range of motion, or strengthen and



stabilize muscle function. The pain and muscle tension alleviation option focuses on reducing aches and discomfort. The mobility improvement option targets an increasing range of motion and flexibility, while the strengthening option is for users aiming to enhance muscle strength and endurance. Lastly, users must evaluate their pain level, selecting from low, moderate, or severe pain. Low pain refers to mild discomfort that does not interfere with daily activities, moderate pain signifies noticeable discomfort that affects movement, and severe pain involves intense pain requiring immediate attention. Based on these responses, the system generates a personalized plan that includes treatment recommendations, rehabilitation exercises, and strengthening routines tailored to the user's needs.

Building upon the foundation of PocketPT, the researchers aimed to enhance the application by integrating computer vision technology to improve pain assessment and rehabilitation tracking. Currently, pain levels are determined solely through self-reported user input. However, by incorporating human pose detection and facial expression analysis, the system would be able to objectively assess the severity of pain. Through facial expression analysis, PocketPT can detect micro-expressions and involuntary reactions associated with pain, such as grimacing or tensing of facial muscles. These insights allow the system to gauge discomfort levels with greater accuracy, rather than relying solely on user-reported pain scores, which can often be subjective and inconsistent. Additionally, human pose estimation enables the system to analyze the user's movements, identifying irregular motion patterns or signs of physical strain that may indicate underlying pain. For instance, if a user selects the triceps as their target muscle, the



application will analyze how fully and comfortably the user can raise their arms vertically while simultaneously evaluating facial expressions for signs of discomfort. This AI-driven system will monitor motion patterns to ensure that rehabilitation exercises are performed accurately and effectively, while facial expression analysis will provide real-time insights into the user's pain levels.

By leveraging artificial intelligence and machine learning, PocketPT will become a more intelligent and adaptive rehabilitation tool, dynamically adjusting treatment plans based on user feedback and physiological responses. This advancement has the potential to significantly benefit athletes, physical therapists, and medical professionals, providing objective pain assessments and ensuring that rehabilitation exercises are conducted safely and effectively. Furthermore, the integration of pain recognition technology into healthcare and sports rehabilitation represents a transformative innovation, bridging the gap between subjective pain reporting and objective assessment methods. By making AI-driven rehabilitation more accessible and personalized, this technology could revolutionize home-based therapy, injury prevention, and post-recovery treatment, ultimately improving patient care and recovery outcomes.

Despite progress in digital rehabilitation, current systems often fall short. Many lack seamless mobile integration, limiting accessibility outside of clinical or desktop settings. Others fail to provide real-time feedback on exercise accuracy or pain recognition, relying instead on delayed assessments or subjective user reports. These limitations reduce the effectiveness of at-home rehabilitation and place greater



dependence on professional supervision. This study addressed this gap by introducing PocketPT, a mobile application that integrates human pose detection and facial expression-based pain recognition to deliver adaptive, real-time, and accessible rehabilitation support.

While existing platforms like LUNA focus on real-time video therapy sessions with professionals, and Exer AI uses basic motion analysis for posture correction, PocketPT introduces a groundbreaking layer of intelligence through the integration of computer vision and artificial intelligence. Most notably, it incorporates human pose detection and facial expression-based pain recognition, two advanced technologies that bridge a critical gap in digital rehabilitation by enabling real-time movement analysis and objective pain assessment, thereby enhancing the accuracy, responsiveness, and overall effectiveness of virtual physical therapy interventions. Human pose detection uses computer vision to track skeletal movement and ensure accurate execution of rehabilitation exercises. Studies by Chen et al. (2022) and Cao et al. (2019) emphasize how real-time pose estimation can revolutionize at-home therapy by reducing errors and enhancing user safety. Meanwhile, pain recognition through facial analysis, backed by research from Zhou et al. (2021), provides objective insight into discomfort levels by identifying micro-expressions such as grimacing or facial tensing, thereby reducing reliance on subjective self-reported pain scores and improving the consistency and accuracy of pain assessment in digital rehabilitation systems.



## 1.3 STATEMENT OF THE PROBLEM

Injuries, particularly those requiring physical rehabilitation, demand personalized care plans and consistent monitoring to ensure effective recovery. However, many individuals face barriers to accessing timely, high-quality rehabilitation guidance due to limitations in clinical availability, cost, or logistical constraints. PocketPT, a mobile application powered by artificial intelligence and machine learning, aimed to address these challenges by offering an automated and adaptive approach to physical therapy. Despite the promise of such technologies, there remains a critical need to evaluate how accurately and effectively PocketPT can replicate the decision-making processes of licensed physical therapists, guide users through proper exercise routines, and adapt therapy plans based on real-time data such as pain expression, user performance, and feedback. This study sought to examine the extent to which PocketPT could provide a clinically relevant and user-centric rehabilitation experience by exploring key components such as personalized pain assessment, pose estimation for movement correction, facial pain recognition, and the application of decision-tree algorithms for dynamic treatment planning.

Specifically, this study aimed to answer the following questions:

1. How could the integration of user-specific pain characteristics, motion-based pain recognition, and datasets verified by certified physical therapists through machine learning algorithms enhance the personalization and accuracy of rehabilitation plan recommendations?



2. How could the movement tracking system using Pose Estimation (CNN-LSTM) be developed to accurately evaluate and correct user exercise performance?
3. How could Facial Expression Recognition (FER) using CNN be implemented to automatically recognize pain or discomfort during user assessment?
4. How would the rule-based decision tree algorithm offer accurate and effective method for generating personalized rehabilitation plans and dynamically adjust therapy based on user performance and real-time feedback?
5. How effective and accurate were AI-generated therapy plans when compared to the recommendations made by a Physical Therapist?
6. How could user feedback help in refining and improving the systems rehabilitation system to ensure continued relevance and performance?

## 1.4 HYPOTHESIS

This study established that PocketPT, a mobile and AI-powered rehabilitation application, provides a more effective and clinically practical solution for physical therapy compared to traditional methods by integrating real-time movement analysis, automated feedback, and pain recognition technologies that enhance accuracy, accessibility, and patient engagement throughout the recovery process. The findings reveal that the application significantly improves user satisfaction and engagement by offering personalized exercise routines, intuitive interfaces, and interactive feedback. Furthermore,



the system enhances the accuracy of physical therapy progress tracking through automated analysis of patient inputs, motion data, and pain indicators.

Clinically, PocketPT demonstrates strong feasibility by supplying healthcare professionals with reliable, real-time data on patient performance and adherence. This facilitates remote monitoring and decision-making, ultimately contributing to better patient outcomes and more efficient therapy management. As such, the integration of PocketPT into rehabilitation settings supports both patient autonomy and professional oversight.

To test these outcomes, the study utilized a combination of user surveys, usability assessments, comparative performance data, and expert evaluation in clinical environments. The evidence supports the conclusion that PocketPT is a viable alternative to conventional rehabilitation, offering benefits in both patient experience and therapeutic efficacy.

The null hypothesis posited that there would be no significant difference between PocketPT and traditional physical therapy in terms of user satisfaction, engagement, progress tracking accuracy, or clinical feasibility. It further assumed that the application would not provide sufficiently reliable data to support remote clinical monitoring or improve adherence to prescribed rehabilitation routines.



## **1.5 RESEARCH OBJECTIVES**

### **1.5.1 GENERAL OBJECTIVE**

To develop an AI-assisted physical therapy application that autonomously analyzes movement, detects pain, and generates personalized rehabilitation plans without therapist supervision.

### **1.5.2 SPECIFIC OBJECTIVES**

1. To gather patient-specific pain attributes, clinical datasets, and videos demonstrating Active Range of Motion (AROM) exercises;
2. To consult licensed physical therapists to validate relevant features (muscle groups, pain levels, functional goals);
3. To develop Facial Expression Recognition (FER) to automatically detect pain or discomfort during assessment;
4. To develop a CNN-LSTM based pose estimation model capable of accurately analyzing user movements in real time;
5. To design and implement a rule-based decision tree algorithm that evaluates exercise performance based on physical therapy guideline;
6. To integrate all functional modules into a unified PocketPT mobile system with seamless operation and user-friendly interaction;



7. To design an adaptive feedback loop that personalizes rehabilitation guidance corrective, real-time responses; and

8. To evaluate and validate the system through performance testing, expert assessment, and user feedback.

## 1.6 SIGNIFICANCE OF THE STUDY

This study aimed to develop an accessible and automated rehabilitation solution for individuals needing physical therapy. By eliminating the need for direct therapist supervision, PocketPT can expand rehabilitation access to those with limited resources while still providing structured and effective therapy.

For rehabilitation-driven individuals, the application dynamically generates and adjusts rehabilitation programs tailored to the user's pain levels, movement accuracy, and recovery goals. This ensures that users, ranging from amateur athletes to fitness enthusiasts, receive consistent, expert-backed guidance and personalized support, even in the absence of a human therapist, promoting safer and more effective recovery outcomes.

For educational institutions, students between the ages of 18 to 25, particularly those engaged in varsity sports, physical education programs, or maintaining physically active lifestyles, can integrate the application into sports departments and wellness programs to promote safe training practices and injury recovery support. The platform empowers students to independently monitor their recovery, adhere to tailored exercise plans, and receive feedback on their movement quality, all while learning to manage



physical well-being without the constant need for direct clinical supervision.

For healthcare providers, the application allows for remote observation of patient progress via computer vision analysis and intelligent pain detection, streamlining routine check-ins and follow-ups. This reduces the physical burden on therapists while still maintaining a high standard of care and providing actionable data for clinical decision-making.

For healthcare institutions such as hospitals, outpatient physical therapy clinics, and rehabilitation centers, PocketPT can be utilized as a complementary digital rehabilitation tool to enhance patient care. The application's AI-assisted movement analysis and pain detection capabilities can aid physical therapists in remotely monitoring multiple patients simultaneously, thereby reducing workload and improving the efficiency of care delivery. Moreover, it facilitates continuous patient engagement beyond the clinic by enabling structured home exercise programs, ultimately improving adherence rates and rehabilitation outcomes.

For young professionals, whether they recover from work-related injuries or sports strain, individuals in fast-paced environments can benefit from on-demand access to personalized rehabilitation support. The application's self-assessment and adaptive exercise features provide flexible rehabilitation without compromising on quality or safety.



## 1.7 SCOPE AND LIMITATIONS

This study focused on the development of a personalized mobile application designed to assist with physical therapy, specifically targeting rehabilitation for muscle and neurological systems. The application aimed to provide rehabilitation plans tailored to individual needs, utilizing the rule-based decision tree algorithm to offer dynamic, adaptive exercise routines based on user input. It focused on creating a practical, accessible, and intelligent tool for individuals recovering from soft tissue injuries, muscle strains, and neuromuscular impairments, addressing the need for efficient self-managed rehabilitation. Providing rehabilitation solutions specifically for muscle and neurological rehabilitation, with a focus on improving strength, mobility, and reducing pain through targeted exercise routines was another focal point of this study. The system was designed to guide users through various phases of recovery, from injury treatment to rehabilitation and strengthening, ensuring a comprehensive approach to muscular and neurological health. The pose estimation component utilized a standalone CNN architecture specifically optimized for detecting major body joints (shoulders, hips, knees, elbows). The facial pain recognition system employed a CNN model providing objective pain assessment during rehabilitation exercises. The system processed video input through standard RGB smartphone cameras without requiring specialized depth-sensing hardware, ensuring accessibility across a wide range of Android devices. The application was designed for adults aged 18 years and above residing in Cavite province, Philippines, who are either physically active individuals, persons experiencing muscular discomfort, or those recovering from soft tissue injuries. The system was intended for users who can



independently perform exercises with minimal assistance and who have access to Android smartphones. The rehabilitation plans generated were suitable for mild to moderate musculoskeletal conditions affecting major muscle groups and large joint complexes. The rehabilitation exercises and treatment protocols included in the application were validated by licensed physical therapists specifically for common musculoskeletal conditions affecting major muscle groups (biceps, triceps, deltoids, chest, quadriceps, hamstrings, calves, gluteal, core muscles).

While this research was designed to offer a comprehensive rehabilitation tool, certain limitations must be acknowledged. Diagnosing and treating bone injuries, such as fractures, dislocations, or osteoarthritis, requires specialized diagnostic methods like X-rays or CT scans, which the application does not support, limiting its use to musculoskeletal rehabilitation and non-invasive injury management rather than acute or severe bone conditions. As a result, the application could not provide rehabilitation plans for skeletal injuries or major conditions that require such imaging tools for diagnosis and treatment. Furthermore, the system could not address issues that extend beyond the muscular and neurological domains, such as cardiovascular rehabilitation or specific joint ailments. Another limitation was the absence of real-time interaction with professional physical therapists within the application, which may be a key consideration for users requiring more intensive or specialized care. Additionally, as the system relied on user-input data for the initial assessment (rehabilitation goal, target muscle, pain type, pain duration, injury history) in addition to AI-analyzed AROM video and facial expressions, variations in user input accuracy may have affected the appropriateness of treatment plans



generated. Users who inaccurately reported their pain levels, selected incorrect muscle groups, or provided incomplete injury history information may have received rehabilitation plans that are suboptimal for their actual condition. The system assumed users can accurately identify their own muscle groups and pain characteristics based on the provided descriptions and visual aids within the application interface. The application was optimized for Android devices and web browsers, with no iOS version available during the study period. Users must have smartphones with functional cameras capable of recording video at sufficient resolution and frame rates for pose estimation analysis. The application required periodic internet connectivity for cloud synchronization via Firebase, though core functionality including exercise recording and local data storage remains available offline through Hive local database. Lastly, the duration of this study lasted for two semesters – within the second semester of Academic Year 2024 – 2025, and the first semester of Academic Year 2025 – 2026. This means that the user testing and validation was limited to this timeframe, and broader external validation may be necessary for wider application deployment.



## CHAPTER II

### REVIEW OF RELATED LITERATURE

This chapter provides an understanding of a PocketPT system, including classification methods and deployment plans. Machine learning algorithms were utilized to examine user- provided data and make meaningful improvements to the ability to interpret injuries and provide therapy. In the real world, the system must provide correct navigation because physically active people's health is at risk when using the program. However, mobile devices and user input data can be used to implement the system issue in a cost-effective and generally accessible manner. This chapter defines PocketPT, along with the requirements for its deployment; aspects that were addressed include machine learning, data analysis, and design centered on users.

A variety of expanding sectors, medicine among them, has demonstrated that artificial intelligence (AI) and machine learning (ML) are effective. As the readers head into the subsequent part of the literature review, the researchers analyzed current research and the deployment of AI and ML within healthcare and its correlation with Pocket PT features.

#### 2.1 FOREIGN LITERATURE

The company Exer AI posted last 2023 about their mobile application called Exer Health, that it is largely dependent on the camera, which is a standard feature in most of



the current generation of smartphones. It is with the help of the built-in motion AI platform that the camera records the range of motion and the proper posture of the patient. The app generates the relevant reports once the patient finishes the recommended exercise. These reports are then reviewed by their PT on their own time. Besides, the Exer Health app does not restrict itself to at-home usage. Via Exer Health, physical therapists are facilitated to implement it during in-person sessions for various reasons such as to ensure correct posture and to get accurate calculations through range of motion as well as joint speed. It can trace up to 24 points of the body and may define possibly 100 types of movements and exercises. As the app is also mindful of busy schedules, it sends reminders to the patients to do their exercises every day. Exer Health intends to be reconciled with the PTs, instead of competing with them. This technologically advanced solution that uses an AI algorithm was created to provide comprehensive physical therapy benefits to patients, and it also gives novelties, very valuable metrics that doctors do not have. While Exer Health focuses on leveraging AI and motion tracking for both remote and in-person therapy, other digital platforms like Pt Pal Pro aim to strengthen the communication and consistency between patients and healthcare providers.

Pt Pal Pro is a patient engagement software designed to improve the rapport between healthcare providers and their patients. Patients can receive appointment reminders on their phones, therapy recommendations, and treatment plans stored on their platform. Instead of using handwritten notes or printouts with detailed exercise plans, patients can rely on Pt Pal Pro to provide them with exercise instructions. To enable smooth data transfer, the platform interfaces with the electronic health record (EHR)



system of practice. In-depth patient analytics is one of the app's key features. It keeps track of things like how consistently a patient completes their activities and inquiries about their mental and physical well-being, as posted in the article written by PT Pal last 2022. The increasing integration of digital tools like PT Pal Pro reflects a broader movement in healthcare, particularly in physiotherapy, toward remote care and patient empowerment, a trend that was significantly accelerated during the COVID-19 pandemic and continues to shape the adoption of technology-driven rehabilitation solutions.

The article, posted by the United Kingdom's National Health Service clinicians in 2020, highlights the growing role of physiotherapy apps in enhancing remote consultations and patient care. To reduce the risks of exposing patients to COVID-19, approximately 70,000 physiotherapists in the UK are being urged to implement remote consultations. Rising in popularity, physiotherapy apps have become an essential resource for physiotherapists aiming to enhance patient results. Applications can be utilized effectively to assist with assessment, adherence to treatment, symptom monitoring, or self-management of conditions, enhancing a patient's physical and mental wellbeing. Representing NHS organizations, ORCHA evaluates more applications than any other entity, focusing on clinical validation, safety, data protection, and user satisfaction. Here are two highly rated apps created to assist with remote physiotherapy. GPEP Physiotherapy Exercises was created with input from NHS GPs and physiotherapists to provide users with self-help guidance from experts, allowing individuals with back, muscle, and joint issues to begin their recovery sooner. AfterTrauma offers insights into the physical and emotional effects of significant injuries and has been collaboratively



developed with survivors of serious injuries, caregivers, medical professionals, and therapists. The application is designed to assist users in reconstructing their lives. While general physiotherapy apps have proven invaluable in supporting remote care and recovery, recent advancements have pushed the boundaries further by introducing AI-powered tools specifically tailored for sports injury rehabilitation.

The study, as presented by Zhan in 2024, explores the creation of a tailored evaluation system for sports injury rehabilitation powered by artificial intelligence (AI). Sports injury rehabilitation is a form of physical therapy aimed at treating musculoskeletal disorders, injuries, and pain in individuals of all ages. Sports rehabilitation enhances health and fitness, supports injury recovery, and reduces pain via movement, exercise, and physical therapy. In the event of a sports injury, rehabilitation has evolved into a distinct field that has progressively united sports doctors, sports physiotherapists, and orthopedic surgeons. Identifying the most effective methods to reduce recovery time, prevent injuries, and improve performance is essential for athletes in sports. The goal of this study was to create a tailored evaluation system for sports injury rehabilitation powered by artificial intelligence (AI). This research introduces a new advanced penguin search optimized efficient random forest (APSO-ERF) designed for rehabilitation in sports injury athletics training. This research utilized exercise movement imagery data to create customized rehabilitation for sports injuries. The data underwent preprocessing with a Wiener filter for reducing noise and restore the image. Convolutional neural networks (CNN) were employed to extract high-level features from images. The suggested approach was utilized to measure physical rehabilitation by analyzing patient performance while they performed



assigned exercises for sports injury rehabilitation. The suggested approach was evaluated against various conventional algorithms. Achieving 97.80% accuracy, 96.01% sensitivity, 97.90% specificity, 98.88% precision, 96.11% recall, and 97.50% F1-score, the APSO-ERF method outperformed traditional algorithms in customized sports injury recovery. The findings show that the suggested approach achieved excellent results in the accuracy of rehabilitation for sports injuries in athletic exercises. Building on the potential of AI-driven approaches in sports injury rehabilitation, recent innovations have also explored more accessible, smartphone-based solutions for targeted therapies such as shoulder rehabilitation.

Research conducted by Pereira, Cunha, Viana, Lopes, Melo, Ana, and Sousa in 2023 indicates that shoulder rehabilitation involves a series of physical therapy appointments to restore the movement of the injured limbs. Nevertheless, these sessions are frequently constrained by the accessibility and expense of specialized technicians, alongside the patient's travel to the session sites. This paper introduces an innovative smartphone-based method utilizing a pose estimation algorithm to assess movement quality and offer feedback, enabling patients to conduct self-directed recovery sessions. This article examines the latest advancements in wearable technologies and camera-based systems for detecting the human body and aiding rehabilitation. It also details the developed system that utilizes MediaPipe to obtain the coordinates of 33 specific points on the user's body, comparing these with reference videos created by professional physiotherapists through cosine similarity and dynamic time warping. This article additionally introduces a clinical study utilizing QTM, an optoelectronic motion capture



system, to verify the techniques employed by the smartphone application. The findings indicate that statistically significant variances exist among the three methods for various exercises, underscoring the necessity of choosing a suitable method for particular exercises. Extending beyond shoulder-specific recovery, recent advancements have also focused on general musculoskeletal rehabilitation by leveraging automated, vision-based assessment tools to enable accurate, at-home performance evaluation.

Priyadarshani stated in 2024 that rehabilitation exercises are essential for recovery and treatment following surgery for numerous musculoskeletal issues. An automated vision-based assessment for rehabilitation exercises is a mobile and affordable method for assessing patients' performance in a home environment, predicting a performance score by analyzing the sequences of correct and incorrect exercises executed by the patient. Recent studies have indicated that examining the spatial and temporal characteristics of skeleton data is crucial for this task. Typically, rehabilitation exercises require the movement of multiple body parts that are distantly situated. Hence, for a precise evaluation, it is essential for the model to comprehend the joint-specific functions and relationship between these non-local attributes. Nevertheless, most approaches treat every body joint uniformly and do not effectively capture the correlation data among all joints. The researchers therefore suggested an attention-enhanced spatial-temporal graph convolutional network that captures the temporal and spatial relationships among the body joints to overcome this restriction. Better assessment is made possible by integrating an attention mechanism to learn global information for the joint-specific roles. It accomplishes this by calculating the weighted total of the features at every point,



independent of their positional distance, to determine the response at that position. While vision-based technologies have advanced the evaluation of physical rehabilitation, similar innovations are now enhancing the interpretation of emotional states through facial expression analysis, especially when integrated with Internet-of-Things (IoT) systems.

Shanggua and Zixuan stated in 2024 that facial expressions communicate human feelings and can be classified into macro-expressions (MaEs) and micro-expressions (MiEs) according to their duration and intensity. Though MaEs are optional and readily identifiable, MiEs are automatic, swift, and capable of exposing hidden feelings. The combination of facial expression analysis and Internet-of-Things (IoT) systems holds considerable promise in various situations. IoT-augmented MaE analysis allows for instant observation of patient emotions, enhancing mental health treatment in intelligent healthcare. Likewise, MiE detection using IoT improves surveillance precision and threat identification in intelligent security systems. This study sought to deliver a thorough summary of advancements in facial expression analysis research and examine its incorporation with IoT systems. The researchers outline the differences between our research and current surveys, highlight improvements in MaE and MiE methods across different learning frameworks, and explore their possible uses in IoT. The researchers emphasize obstacles and future pathways for the integration of facial expression technologies and IoT systems, with the goal of encouraging innovation in this area. This study provides a structured insight into how recent advancements and real-world implementations of facial expression analysis can improve IoT systems in healthcare, security, and other fields. Beyond emotional recognition, intelligent systems have also



advanced toward identifying critical physical events, such as falls, which are vital for safeguarding vulnerable populations like the elderly.

Yang and Junkai (2024) state that prompt identification of falls and notifying medical assistance is essential for health surveillance in elderly people living alone. This article primarily addresses challenges like inadequate adaptability, privacy violations, and low recognition precision linked to conventional visual sensor-based fall detection. The researchers suggest an infrared video fall detection approach that employs spatial-temporal graph convolutional networks (ST-GCNs) to tackle these issues. The researchers employed a fine-tuned version of AlphaPose to obtain 2D human skeleton sequences from infrared video footage. Afterward, the skeleton data was depicted in Cartesian and polar coordinates and analyzed using a two-stream ST-GCN to swiftly identify fall behaviors. To improve the network's ability to recognize fall actions, the researchers enhanced the adjacent matrix of the graph convolutional units and incorporated multi-scale temporal graph convolution units. To enable practical implementation, the researchers refined the time window and network depth of the ST-GCN, achieving a balance between accuracy and speed. The experimental findings from a proprietary infrared dataset for human action recognition revealed that our suggested algorithm successfully detects fall behaviors with an impressive accuracy of 96%. Furthermore, our algorithm demonstrated strong performance, detecting falls in both near-infrared and thermal-infrared videos. Just as fall detection systems leverage spatial-temporal analysis to safeguard physical well-being, facial expression recognition technologies apply similar visual computing strategies to interpret emotional states, further broadening the scope of intelligent healthcare



applications.

Li, Jin, Zhou, Kubota, and Ju introduced a novel Convolutional Neural Network (CNN) architecture enriched with an attention mechanism in 2020 for facial expression recognition. A pivotal component of their approach includes the utilization of Local Binary Pattern (LBP) features that have been proven to outperform traditional features like HOG and Gabor due to their rotation and gray-scale insensitivity. These features make LBP particularly effective in extracting fine-grained texture details such as wrinkles and creases, which are important facial expression cues in varying illumination and orientation conditions. The attention module of their network architecture allows the model to selectively focus on informative areas of the input data, thus improving recognition performance by assigning greater weights to more informative facial features. To verify the efficacy of their method, they tested it on five widely used facial expression datasets: CK+, JAFFE, FER2013, Oulu-CASIA, and their own Nanchang University Facial Expression (NCUFE) dataset. The NCUFE dataset was specifically designed for this study and contains facial expression images of seven emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral. The dataset consists of images acquired from 35 master students (6 females and 29 males), captured by a Microsoft Kinect sensor, capturing both RGB and depth images. One image in the RGB has a size of  $1280 \times 1024$ , and one of the depth images has a size of  $512 \times 424$ . 245 sequences each were captured for all three image types, and each sequence was 110 images, for a total of 26,950 images. The students were approximately 100 cm from the Kinect and asked to replicate facial expressions on paper reference sheets during capture. This allowed the image capturing conditions to be



uniform and facilitated the development of a varied and exhaustive dataset for the evaluation of facial expression recognition performance. Building on human-centered technology, the study presents CNN with an attention mechanism for facial expression recognition. Likewise, in human-robot interaction, accurate sensory systems, particularly human pose estimation, are vital for safe and efficient collaboration. Comparing marker-less pose estimators like OpenPose and Detectron2 has become central in applications such as rehabilitation and surveillance, as these systems enable real-time tracking of body movements, enhance interaction precision, and reduce reliance on intrusive or cumbersome tracking equipment.

In 2021, Hernández, Morell, Ramon, and Jara investigated the performance of the affordable marker-less human pose estimators OpenPose and Detectron2 in assessing angles in the elbow and shoulder joints during upper-limb rehabilitation exercises. The study focused on four standard exercises: elbow side flexion, elbow flexion, shoulder extension, and shoulder abduction. Using two Kinect 2 RGBD cameras, they acquired accurate joint and skeleton estimations during the exercises. The research provided a numerical comparison (RMSE and MAE) of the angle measurements from OpenPose, Detectron2, and the actual ground truth. The findings revealed that OpenPose outperforms Detectron2 in these types of applications. Given OpenPose's superior performance in capturing upper-limb joint angles for rehabilitation tasks, subsequent studies have explored its utility in more complex scenarios, such as multi-tiered Human Action Recognition (HAR) systems.



Malik et al. (2023) discussed that Human Action Recognition (HAR) is a field within computer vision focused on recognizing human actions at different tiers, which include low level, action level, and interaction level. In the past, several HAR algorithms have been introduced that rely on manual techniques for recognizing actions. Nevertheless, the artisanal methods are ineffective when it comes to identifying interaction level actions due to their complexity. In contrast, conventional deep learning methods use the whole image as input and subsequently derive numerous features, which considerably raises the complexity of the systems; this leads to much greater computational time and resource consumption. This study emphasizes creating a highly accurate multi-view interaction level action recognition system that utilizes 2D skeleton data, aiming to lower computational complexity through deep learning architecture. The suggested system utilizes the OpenPose method to extract 2D skeleton information from the dataset. Subsequently, the obtained 2D skeleton features are fed directly as input to the Convolutional Neural Networks and Long Short-Term Memory (CNN-LSTM) framework for recognizing actions. To simplify the process, rather than sending the entire image, only the extracted features are provided to the CNN-LSTM architecture, which removes the necessity for feature extraction. The suggested approach was evaluated against other current methods, and the results validate the effectiveness of the proposed technique. The suggested OpenPose-CNNLSTM attained an accuracy of 94.4% on the MCAD (Multi-camera action dataset) and 91.67% on IXMAS (INRIA Xmas Motion Acquisition Sequences). The suggested approach also notably lowers computational complexity by limiting the input features to 50. As research continues to evolve in



recognizing complex human motions, attention has also turned to gait analysis, an area with profound implications for rehabilitation and medical diagnosis. Like HAR, gait recognition benefits greatly from deep learning and motion-capture technologies.

Ali et al. (2024) discussed that recent studies have concentrated on affordable gait analysis options for rehabilitation and home therapy. Alanazi et al. (2022) employed micro-Doppler spectrograms and Kinect V2 for gait recognition, attaining accuracy rates of 95.7%–98.8% with a multilayer CNN. Khokhlova and colleagues introduced an LSTM ensemble for unsupervised gait classification that utilizes joint orientation data, achieving 94% accuracy with their MMGS dataset. A different study employed Kinect V2 alongside a GRU classifier to identify pathological gaits, reaching an accuracy of 90.1% with complete body data and 93.67% by utilizing just leg joints. Zhu et al. (2025) presented a 2-stream CNN to forecast neurological disorders using joint data. Zheng et al. (2025) applied SVM to sensor data from 200 patients with Parkinson's and 100 healthy individuals, attaining 96.7% accuracy. Ramli et al. (2023) utilized temporospatial gait features to classify Duchenne Muscular Dystrophy (DMD) with an accuracy exceeding 91%. D'Angelo and Romano performed 3D analyses of gait in DMD patients, recognizing distinctive walking patterns such as anterior pelvic tilt and plantar flexion. The integration of CNNs and LSTMs has yielded impressive outcomes: Gao et al. (2024) attained 93.1% accuracy in identifying abnormal gait types with LCWSnet, whereas Albuquerque et al. (2024) employed VGG-16 alongside a bidirectional LSTM to achieve 91.4% accuracy on the GAIT-IT dataset.



## 2.2 LOCAL LITERATURE

In 2024, Dillague, Juico, Yu, and De Goma introduced a novel approach to Facial Emotion Detection by incorporating three features, Eyes, Mouth, and Eyebrows, into the dataset for image processing and emotion classification. Their research suggests that a more detailed segmentation of essential attributes improves overall detection accuracy compared to traditional two-feature segmentation. Using Artificial Neural Networks with a Backpropagation algorithm for training and testing, the study aimed to evaluate whether this additional segmentation could enhance precision and match the accuracy of earlier studies. The findings revealed that the three- feature segmentation significantly improved accuracy. The model's enhancements and the dataset used were restricted to open-source datasets available for research purposes.

These advancements in facial emotion detection, particularly through improved segmentation and neural network models, underscore the growing role of precise AI-driven analysis in understanding human expression, an approach that parallels the evolving use of mobile applications in physical therapy. Just as facial feature segmentation refines emotional assessment, mobile apps like PocketPT aimed to refine physical evaluations and treatments by offering reliable, accessible, and data-driven alternatives to traditional methods.

Thanks to technological progress, mobile apps have contributed to improving assessments, treatments, and exercise programs for key players in healthcare. Nevertheless, there is an absence of sufficient evidence regarding the reliability and



validity of these mobile applications compared to traditional tools/methods utilized in rehabilitation. Consequently, this systematic review seeks to identify and consolidate current research on evidence supporting the validity and reliability of mobile applications employed in physical therapy. Methods: This systematic review encompasses studies published in English that evaluated the PT mobile application in healthy subjects, compared it to gold standard devices, and assessed the app's validity and/or reliability. A literature review was performed across nine databases, utilizing two electronic tools - (1) Mendeley and (2) Rayyan. The eligibility of the articles will be evaluated for validity and reliability using the Brink and Louw (2012) Critical Appraisal Tool. A qualitative review along with meta-analyses were performed for the purpose of data synthesis. Anticipated Outcomes: This research enhanced existing understanding and medical practices by offering insights on credible and dependable PT applications, consolidating evidence on mobile applications that enhanced PT evaluations and treatments, along with which applications could be explored and enhanced further. In general, the findings of this study provide insights into how PT mobile apps could enhance traditional testing methods or procedures in physical therapy, including evaluations, treatments, and home exercise routines (Gonzaga, Blanco, Magdalena, Merquita, Santos, See, Tolentino, Tsai, Llanos, Versales, Fernandez, 2022). As the demand for evidence-based digital solutions continues to grow, emerging platforms begin to reflect this shift toward integrating technology into physical therapy practices.

Metro Pacific Health Tech Corporation (MPHTC), a subsidiary of Metro Pacific Investments Corporation, launched mWell in August 2022 as the Philippines' first fully



integrated digital health app. While the app offers core services like teleconsultations, e-pharmacy, emergency care, wellness programs, and a clinic management system, its most forward-looking feature lies in its support for potential telerehabilitation. With tools such as home care services, the upcoming mWell Wellness Wearable, which tracks steps, sleep, and exercise, and its seamless health monitoring ecosystem, mWell is positioned to evolve into a robust platform for remote rehabilitation. Though telerehabilitation is not explicitly mentioned, the app's direction strongly suggests future integration, aligning with its mission to make healthcare accessible, continuous, and personalized for every Filipino, as posted in the article written by Luna last 2022. To further understand the real-world potential and limitations of such digital health innovations, it is essential to examine user experiences and perceptions surrounding telerehabilitation.

Given the limited literature on telerehabilitation, this study explored participant perspectives on its facilitators and obstacles through a quantitative, descriptive survey using a researcher-made questionnaire. Results revealed that key facilitators included reduced travel time and costs (WM = 3.606) and lower exposure to health risks (WM = 3.545), while primary challenges involved technical difficulties (WM = 3.273) and limited internet access (WM = 3.242). Despite these barriers, most participants reported high satisfaction (WM = 3.338) and moderate compliance (WM = 2.798) with telerehabilitation. The findings suggest that telerehabilitation is a practical alternative to traditional physical therapy, enhancing accessibility and continuity of care, particularly during the COVID-19 pandemic, as posted in the article written by Ico, Bautista, Espirito, and Ramos last 2021. Building on this perspective, further research has examined broader



behavioral factors influencing the adoption of digital health solutions in the Philippines.

As posted in the article written by Ong, Kurata, Castro, De Leon, Dela Rosa, and Tomines (2022), telemedicine has the potential to improve access to healthcare and address system gaps, yet it remains underutilized in the Philippines despite global recognition. This study examined the behavioral factors affecting the adoption of telemedicine among 533 Filipino respondents using the UTAUT2 framework. Analysis through structural equation modeling and neural networks revealed that usage behavior (UB) was the strongest predictor of user intention and adoption. Key influences on UB included performance expectancy, effort expectancy, hedonic motivation, social influence, and resistance to use. As the first local study of its kind, the findings provide essential insights for advancing telemedicine in the Philippines and serve as a guide for healthcare technology implementation in developing countries. In line with the push for localized digital health solutions, another study explored the development of a mobile application tailored to assess physical fitness among Filipino students.

Martin (2022) conducted a descriptive developmental study. The research created and assessed a mobile app that could serve as an alternative method for measuring the Physical Fitness levels of senior high school students in Porac, Pampanga, Philippines. This descriptive developmental research involved intentionally chosen participants, including 72 senior high school students in Information and Communication Technology (ICT), 20 Physical Education instructors, and 5 ICT specialists from different institutions and areas. Participants responded to section 10.2 of the Department of Education's



Technical Evaluation Guidelines and Checklist, which is utilized for assessing computer programs, applications, and mobile apps. A descriptive statistical analysis was performed on the data. The study's findings led to the creation of a mobile app for physical fitness assessments for Android smartphones, based on the needs analysis results, the Department of Education manual, and the Philippine Physical Fitness Test standards. The mobile app for physical fitness was developed concerning Multimedia features, Control Behavior, System Information, Overall Interface, and Customization/Support for User Preferences. The mobile application created for assessing physical fitness received excellent evaluations from students, teachers, and ICT specialists. Thus, the suggested mobile app might serve as an effective method for monitoring and calculating the physical fitness levels of students. The findings of this research could be employed by future researchers focused on creating additional mobile applications aimed at fitness and physical activity. While mobile applications for physical education in schools have shown great promise, similar innovations are also emerging in the realm of wellness and personal fitness, such as yoga

Pre-recorded yoga lessons have gained popularity due to the advantages they provide. The disadvantage of learning yoga through these tutorials is that practitioners lack feedback on their performance, potentially resulting in serious injuries. To resolve this issue, software applications detect and rectify incorrect executions; nonetheless, they fail to offer a criteria-based assessment. To tackle this problem, this research introduces Motion, a mobile app that assesses the accuracy and consistency of a yoga practitioner's poses in real-time and offers a record of past performances. The app employs MediaPipe



Pose and ML Kit for Human Pose Estimation and Machine Learning to monitor and recognize yoga performance. Angular data of key body parts for each frame of a practitioner's performance are recorded and utilized to assess pose precision and uniformity. The assessment results are subsequently recorded in an SQLite database and shown in the performance history. The application's pose classifier attained an accuracy of 99.33%, showcasing its efficiency in the real-time categorization of yoga poses, and received an average modified PSSUQ score of 1.99, reflecting excellent usability. Given these implications and Motion's capability to deliver real-time assessment and progress monitoring, the application serves as an essential resource in yoga, health, and education. (Reyes, Gabutan, Villacarlos, Sta Romana, Aliac, 2023). The trend of health focused applications that can leverage real-time feedback and machine learning. A similar approach has been adopted in the field of telerehabilitation to address challenges in remote physical therapy.

The COVID-19 pandemic has hastened the uptake of telerehabilitation, uncovering both chances and difficulties in virtual physical therapy. This research presents TherapEase, an online platform aimed at improving the quality of virtual rehabilitation services by tackling concerns like exercise execution precision and communication effectiveness. TherapEase seeks to enhance, not substitute, traditional in-person physical therapy. A significant aspect of the system is its integration of pain evaluation during therapy sessions via emotion recognition, employing the You Only Look Once (YOLO) algorithm to analyze patients' facial expressions. Created with an agile approach, the application was assessed according to the ISO 25010:2011 software quality model,



emphasizing Functional Suitability, Performance Efficiency, Interaction Capability, and Reliability. Findings show significant satisfaction across all assessed areas, with average scores varying from 3.58 to 3.67 out of 4. These results indicate that TherapEase successfully combines multiple elements to offer a cohesive and effective telerehabilitation solution, which may enhance the provision of remote physical therapy services while also supporting conventional in-person care. Additional studies are required to enhance the precision of the pain evaluation component (Centeno, Mahusay, Cortez, Rana, Diaz, Sison, 2024). Expanding on the use of emotion recognition technologies in healthcare applications, recent studies have further explored innovative approaches to facial expression analysis for real-time and accurate emotion detection.

De Ocampo (2023) explored the application of Haar features as an additional feature extraction step for a deep neural network to reduce input dimensionality in facial detection before recognizing expressions, which is utilized in multi-scale facial expression recognition (FER). Identifying emotions can assist in understanding and conveying non-verbal cues. While deep neural networks excel at autonomously extracting and understanding high-level representations of facial features from extensive multidimensional data, a face detection model that operates quickly for real-time applications is still necessary. Hybrid methods are employed to detect and classify facial characteristics, where the first method involves Haar feature extraction paired with CNN-based models. The suggested integration of Haar Cascade with CNN showed a recognition rate surpassing 90% for all facial expressions, with the highest accuracy attained in detecting “Happy” and “Surprise” expressions. While advancements in facial expression



recognition have significantly improved real-time emotion detection in health-related technologies. The broader adoption of innovations in telerehabilitation are faced with challenges in the developing countries like the Philippines.

Although recognized internationally as a feasible substitute for in-person consultation and therapy, telerehabilitation has not yet fully developed in developing nations such as the Philippines. During the coronavirus disease 2019 pandemic, as social distancing hindered in-clinic rehabilitation services, Filipinos sought to investigate telerehabilitation. Nonetheless, various obstacles were noted, particularly in the pre-implementation phase of telerehabilitation, prompting a reassessment of current local evidence. Our goal was to identify the difficulties encountered by telerehabilitation in the Philippines. The researchers conducted a search until March 2020 across PubMed, Scopus, Embase, Cochrane Library, and HeRDIN for telerehabilitation-related publications involving Filipinos either as investigators or as part of the population. Due to the presumed limited number of scientific publications on telerehabilitation in the area, the researchers conducted handsearching in gray literature and incorporated pertinent papers from various rehabilitation-related professional organizations in the Philippines. The researchers examined the articles and identified the human, organizational, and technical obstacles to telerehabilitation or telehealth overall. The researchers examined 21 published papers and 4 unpublished ones, which primarily included reviews, feasibility studies, or case reports/series. Twelve of the 25 studies involved patients and doctors in remote teleconsultations, teletherapies, telementoring, or telemonitoring. Patients pursued telemedicine or telerehabilitation for various medical issues, chronic ailments, mental



health disorders, orthopedic concerns, neurological conditions, speech disorders, and cardiovascular diseases. Results in the previously mentioned studies primarily encompassed telehealth acceptance, enabling factors, obstacles, and satisfaction levels. Additional research focused on telehealth regulations, legal aspects, and ethical concerns. The researchers discovered 18 unique human challenges, 17 organizational challenges, and 18 technical challenges associated with telerehabilitation in the Philippines. The primary challenges included slow internet speeds, legal issues, and doubts. There is a lack of information on telerehabilitation in the Philippines. Local initiatives can concentrate on investigating or tackling the most urgent human, organizational, and technical obstacles to the development of telerehabilitation in the nation (Leochico, Espiritu, Ignacio, Mojica, 2020).

## 2.3 SYNTHESIS

**Table 2.3.1**

*Synthesis Table*

| Related Study /<br>Technique(s) Used | Goal | Similarities | Differences |
|--------------------------------------|------|--------------|-------------|
|--------------------------------------|------|--------------|-------------|



|                                                                                                        |                                                                                        |                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Malik et al. (2023) - OpenPose and Convolutional Neural Networks and Long Short-Term Memory (CNN-LSTM) | Human Action Recognition (HAR) at multi-view interaction level using 2D skeleton data. | Uses LSTM (in a CNN-LSTM framework) with 2D skeleton features (from OpenPose) for analyzing human actions, which is related to understanding poses over time. | This focuses on general Human Action Recognition, not specifically injury detection. Combines CNN and LSTM for the action recognition task itself, whereas the proposed application plan LSTM for pose/injury and a separate CNN for facial recognition. Their CNN component is part of HAR, while the application is for facial. Utilizes OpenPose for initial skeleton data. |
| Ali et al. (2024) - LSTM ensemble                                                                      | Unsupervised gait classification using joint orientation data.                         | Employs LSTM for analyzing human movement (gait) based on joint data, which is a form of pose/movement analysis over time.                                    | Focus is specifically on gait classification and is unsupervised. The proposed application aims for broader pose estimation for injury detection and is likely supervised.                                                                                                                                                                                                     |



|                                                                                                          |                                          |                                                                                                                                                        |                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gao et al. (2024) - CNNs and LSTMs (LCWSnet)                                                             | Identifying abnormal gait types.         | Integrates LSTMs (with CNNs) for analyzing human movement patterns (gait) to detect abnormalities, conceptually similar to detecting issues from pose. | The focus is specifically on gait. Combines CNN and LSTM for the gait analysis, while PocketPT separates their main functions (LSTM for pose, CNN for facial).                 |
| Albuquerque et al. (2024) - VGG-16 (CNN) and Bidirectional LSTM                                          | Gait recognition on the GAIT-IT dataset. | Uses Bidirectional LSTM (with a CNN like VGG-16) for analyzing human movement (gait), relevant to temporal pose sequence analysis.                     | The focus is specifically on gait recognition. Combines CNN and LSTM for gait analysis.                                                                                        |
| Li, Jin, Zhou, Kubota, and Ju (2020) - CNN with attention mechanism, Local Binary Pattern (LBP) features | Facial expression recognition.           | Uses CNN architecture specifically for facial expression recognition.                                                                                  | Aims for general facial expression recognition (seven emotions). The proposed application targets pain scaling via facial recognition.<br><br>Uses LBP features and a specific |



|                                                                                                      |                                                                                                                                                   |                                                                                                                                 |                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                      |                                                                                                                                                   |                                                                                                                                 | attention module, which differ from PocketPT's planned CNN approach.                                                                                                                            |
| Centeno, Mahusay, Cortez, Rana, Diaz, Sison (2024) – You Only Look Once (YOLO) algorithm (CNN-based) | Pain evaluation during therapy sessions via emotion recognition (analyzing facial expressions) in a virtual rehabilitation platform (TherapEase). | Directly uses a CNN- based algorithm (YOLO) for analyzing facial expressions to evaluate pain, aligning closely with your goal. | This is a component within a larger online telerehabilitation platform. PocketPT is a standalone mobile application and has a different overall architecture.                                   |
| De Ocampo (2023) - Haar features and CNN-based models                                                | Multi-scale facial expression recognition (FER).                                                                                                  | Employs CNN- based models for facial expression recognition.                                                                    | Uses Haar features as a pre-processing step for dimensionality reduction before the CNN. Focus is on general FER with high accuracy for "Happy" and "Surprise," while PocketPT's focus is pain. |

The integration of technology in physical therapy (PT) has grown substantially in recent years, with a focus on mobile applications and machine learning to enhance



rehabilitation processes. A significant challenge, however, lies in the implementation of human pose estimation algorithms in clinical settings. Local literature highlights the need for advanced systems that not only manage complex data and algorithms but also provide easy application for clinical professionals. These tools, which facilitate the monitoring of patients through gait analysis, pose estimation, and movement tracking, show promising potential for improving rehabilitation and movement science. Despite this progress, technical barriers such as algorithm complexity, rapid domain changes, and privacy concerns still pose challenges to broad clinical adoption. Moreover, the reliability and validity of mobile applications for physical therapy remain under-researched, particularly in comparison to traditional methods.

Several studies emphasize the importance of mobile applications in modern rehabilitation practices. For instance, a systematic review consolidates findings on the validity and reliability of PT apps, indicating that while mobile platforms have shown potential, their efficiency in comparison to traditional methods remains inconclusive. Furthermore, advancements such as the mWell app by Metro Pacific Health Tech Corporation (MPHTC) in the Philippines have illustrated the role of mobile applications in holistic healthcare, integrating telemedicine, e-pharmacy, and wellness services into a single platform. Such innovations demonstrate the shift towards comprehensive digital healthcare ecosystems, which also extend to the rehabilitation sector, offering enhanced access to physical therapy services remotely.

In the context of telerehabilitation, the recent surge in interest highlights both its



facilitators and obstacles. Key advantages include reduced travel and external risk exposure, making it an attractive alternative during periods like the COVID-19 pandemic. However, challenges like technical issues and internet access have been identified as significant barriers. While telerehabilitation holds promise, its adoption is still limited, particularly in the Philippines, where telemedicine faces lower acceptance rates despite its proven benefits globally. Research on Filipino perspectives using the Unified Theory of Acceptance and Use of Technology (UTAUT2) framework further underscores the influence of performance expectancy, ease of use, and social factors on telemedicine adoption.

In light of these studies, the PocketPT app presents an opportunity to address existing gaps in the field. While there have been mobile applications aimed at improving physical therapy assessments, a consistent theme in the literature is the lack of robust evidence supporting their effectiveness. PocketPT could differentiate itself by focusing on evidence-based functionality and user accessibility, ensuring reliable tracking of physical therapy progress, integration with existing healthcare systems, and leveraging machine learning to tailor rehabilitation exercises. Its development could fill the void left by current apps by offering a comprehensive, data-driven, and clinically validated solution that supports both patients and healthcare providers in managing physical therapy routines effectively.



## CHAPTER III

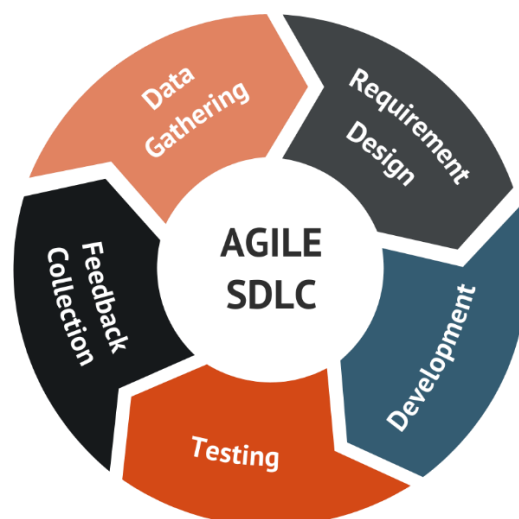
### RESEARCH METHODOLOGY

This chapter describes the methods and frameworks used in the development and evaluation of the project. This chapter explains the methodology adopted in the software development process, conceptual framework, and algorithms guiding system design and functionality. The operational framework, use cases, and activity diagrams represent workflows and user interaction; however, respondents of the study offer an essential element in validation. All these elements collectively delineate the structured approach adopted in realizing the objectives of the study.

#### 3.1 SOFTWARE DEVELOPMENT METHODOLOGY

**Figure 3.1.1**

*Agile SDLC Model of PocketPT*





The PocketPT mobile application was developed utilizing the Agile Software Development methodology, a framework that prioritizes adaptability, user-centric design, and incremental progress. Agile promotes the division of development work into smaller, manageable iterations, each culminating in a functional deliverable that can be tested and evaluated. Throughout the project, continuous feedback loops were established with users and stakeholders, ensuring that the application's features and functionalities were directly aligned with real-world needs and expectations.

By embracing Agile principles, the development team maintained a high degree of flexibility, allowing them to efficiently adjust to evolving requirements, address unforeseen challenges, and incorporate improvements based on feedback collected at every stage of the development cycle. This iterative approach not only enhanced the quality and relevance of the final product but also significantly reduced the risks associated with rigid, linear development models. Ultimately, Agile enabled PocketPT to be crafted as a dynamic, responsive application that could genuinely meet the rehabilitation needs of its users.

### **3.1.1 DATA GATHERING**

Licensed physical therapists were recognized as the cornerstone in the formulation of rehabilitation plans for PocketPT. Their expertise was sought to identify the essential components of an effective rehabilitation program. These components included prescribing specific exercises for targeted muscle groups, designing treatment pathways for varying levels of pain, and structuring step-by-step recovery plans. All insights



gathered from these consultations were carefully documented to ensure that the application's recommendations remained clinically valid, relevant, and beneficial to users.

Upon collecting the necessary information, data was systematically organized into a structured dataset in CSV format. Data collection involved constant communication with the partnered physical therapists to consolidate CSV files that would be used by the application for generating the user's rehabilitation plan. The exercises.csv file contains a comprehensive list of exercises with the following structured attributes: Exercise\_ID (a unique identifier assigned to each exercise), Exercise (the name of the exercise), Exercise\_Description (a brief description of the exercise and its purpose), Muscle\_Involved (the primary muscle group targeted by the exercise), Pain\_Level (the level of pain at which a certain exercise is allowed and assigned to), Functional\_Goal (the rehabilitation goal to which a certain exercise is assigned), Repetition (the number of repetitions that the user must execute per set), Set (the number of sets that the exercise must be performed), and Other\_Muscles (additional muscle groups besides the primary one that are majorly involved in the exercise). The treatments.csv file contains a list of treatments with the following structured attributes: Treatment\_ID (a unique identifier assigned to each treatment), Treatment (the name of the treatment), Treatment\_Description (a brief description of the treatment and its application), Muscle\_Involved (muscle group or groups where a certain treatment is assigned), Pain\_Level (the level of pain at which a certain treatment is assigned to), and Pain\_Duration (the period of time that has passed since the pain was first experienced by the user).



The COCO-Pose Dataset was employed during the data collection and development phase to facilitate the training and validation of the human pose estimation component integrated into the application. The dataset's rich diversity, encompassing various lighting conditions, camera angles, and degrees of body occlusion, ensures that the model learns to generalize effectively to real-world scenarios. This dataset, derived from the Common Objects in Context (COCO) dataset, is a large-scale benchmark dataset widely utilized in computer vision research, particularly for human pose estimation tasks. It provides a comprehensive collection of images depicting individuals in diverse real-world settings and activities, each annotated with precise key points corresponding to human body joints. These key points include major anatomical landmarks such as the head, shoulders, elbows, wrists, hips, knees, and ankles, enabling detailed analysis and modeling of human poses and movements. The pose estimation component was developed utilizing a hybrid CNN-LSTM architecture designed to capture both spatial and temporal features of human movement. The primary spatial feature extraction was performed using a YOLO11s-pose backbone initialized with ImageNet pretrained weights, leveraging transfer learning to accelerate convergence and improve generalization. This model was selected for its efficiency on mobile platforms while maintaining competitive accuracy metrics, achieving a mean Average Precision at 0.5 IoU (mAP@50) of 72.3 percent and a key point accuracy of 85.4 percent on the standalone spatial branch.

The architecture comprises 17 key points representing anatomically significant joints: head, neck, shoulders (left and right), elbows (left and right), wrists (left and right), hips (left and right), knees (left and right), and ankles (left and right). These key points



enable precise assessment of exercise form and movement quality. To enhance temporal consistency and reduce frame-to-frame jitter inherent in frame-level CNN predictions, a Long-Short Term Memory (LSTM) module was incorporated to process sequential key point data across a temporal window of 16 frames, approximately 0.5 seconds of video. The LSTM processes flattened key point coordinates in the shape of batch by temporal by spatial dimensions, learning temporal dependencies that smooth erratic predictions while preserving movement authenticity. This temporal smoothing achieved an angle error reduction of 69 percent while maintaining temporal consistency scores of 88.7 percent, demonstrating that the hybrid architecture effectively combines the strengths of spatial precision from CNNs with temporal coherence from sequential models.

For pain recognition, data collection involved assembling a comprehensive facial dataset specifically curated for pain detection. The dataset was carefully selected based on criteria including diverse facial expressions spanning the pain spectrum, representation of multiple demographics to ensure generalizability, varied lighting conditions reflecting clinical environments, and images captured from multiple camera angles to account for patient positioning variations during assessments. The dataset distribution reflected realistic proportions observed in clinical assessments, with approximately 46,000 samples labeled as low pain representing mild discomfort with minimal facial expression changes, approximately 1,400 samples labeled as moderate pain representing moderate discomfort with noticeable facial expressions such as brow lowering and eye tightening, and only approximately 330 samples labeled as severe pain representing extreme discomfort with pronounced grimacing and facial contraction. This clinically representative distribution,



while creating severe class imbalance, accurately reflected the reality that pain assessment scenarios in rehabilitation environments typically present predominantly non-painful or low-pain states, with severe pain instances being comparatively rare.

## 3.1.2 REQUIREMENT DESIGN

The features of PocketPT were meticulously designed to optimize user experience while ensuring effective functionality. The user input interface was developed in the form of an interactive quiz, capturing comprehensive user data including rehabilitation goal, affected muscle group, pain description, and injury history. These goals include pain management, mobility improvement, and muscle strengthening. Additionally, pain severity was classified into three levels: low, moderate, and severe. This is to establish a solid foundation for rehabilitation plans generated by the rule-based decision tree. This method ensures that the rehabilitation plans generated are highly personalized, addressing both the user's present condition and their specific recovery goals. A navigation system was also implemented, presenting these personalized rehabilitation plans within a user-friendly interface that allows for seamless browsing of recommended treatments and exercises.

Beyond these core functionalities, licensed physical therapists recommended additional feature enhancements to enrich the overall application. These suggested improvements included a progress tracking system to monitor user adherence and progress over time, exercise reminders to promote consistency, and analytics capabilities to identify recovery trends. Other proposed features involved the integration of a detailed medical



history section, a customizable pain scale ranging from one to ten, the ability to indicate the length and spread of pain across different body parts, and the classification of pain types. These enhancements aim to deepen user engagement and contribute to the long-term success of rehabilitation outcomes.

### **3.1.3 DEVELOPMENT**

The front-end development of PocketPT was accomplished using Flutter, an open-source UI toolkit widely recognized for building visually appealing, high-performance, and user-friendly applications. Prioritizing an intuitive and accessible user experience, the front-end implementation concentrated on key features such as dynamic forms for quiz-based data collection, seamless navigation menus to enhance user flow throughout the application, and interactive dashboards that allow users to effectively track and monitor their rehabilitation progress.

On the other hand, the backend architecture was developed using Python, capitalizing on its robust libraries and frameworks to implement advanced machine learning models. Specifically, Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) algorithms were incorporated to analyze and track user movements, identifying the affected body parts where pain manifests. Additionally, another standalone Convolutional Neural Network (CNN) model was employed to extract and interpret patterns from users' facial expressions, focusing on the detection of pain and discomfort to support the dynamic assessment of the user's pain scale.



For data storage and management, Firebase Firestore was integrated as the database solution, offering secure and efficient handling of user information. Beyond basic user profile management, the database also stores data collected through the front-end quizzes and rehabilitation plans generated by the rule-based decision tree. Its flexible structure ensures that the retrieval, updating, and synchronization of records are consistently reliable and seamless. By adhering to high security and consistency standards, the application guarantees that user data remains protected while maintaining quick and dependable access for real-time application needs.

### **3.1.4 MODEL TRAINING**

The preprocessing pipeline incorporated multiple image processing techniques systematically applied to enhance model robustness. Hard augmentation techniques were specifically applied to oversampled minority class instances to force the model to recognize the structural features of pain expressions independent of environmental variations. Random rotations within a range of plus or minus 10 degrees were applied to ensure the model developed invariance to head tilt variations common during patient assessments. Color jitter operations randomly altered brightness, contrast, and saturation within plus or minus 20 percent, and hue within plus or minus 10 percent, simulating the diverse lighting conditions inherent in clinical environments ranging from well-lit examination rooms to naturally-lit spaces. Gaussian blur effects were applied to simulate out-of-focus frames common in mobile video capture, particularly relevant given that the application functions in real-time mobile scenarios where perfect focus cannot always be



guaranteed. Downscale-upscale operations randomly shrunk and enlarged images to simulate distance variability, ensuring the model could recognize pain expressions whether patients were close to or distant from the camera. These augmentation strategies operated on the principle that genuine pain recognition requires learning the semantic structure of pain expressions, such as facial grimacing patterns and characteristic eye closure, rather than memorizing pixel-level details susceptible to environmental variation.

The datasets were partitioned into training, validation, and testing subsets using stratified splitting to ensure representative class distributions across all three sets. For the pose estimation model, the stratified approach maintained balanced representation of various pose types, distance ranges, and environmental conditions across training, validation, and test sets. For the pain recognition model, stratified splitting proved particularly critical given the severe class imbalance, ensuring that minority classes (moderate and severe pain) were proportionally represented in all three partitions rather than being concentrated in a single set. The training set was used to adjust model parameters through iterative gradient descent optimization, the validation set was used to monitor model performance during training and prevent overfitting through early stopping mechanisms, and the held-out test set of 7,260 samples remained completely isolated from all training processes to provide unbiased estimates of deployment-phase performance.

A conservative modeling approach was implemented as a safety-first mechanism, particularly critical given the clinical application context. The system employs confidence gating whereby any key point prediction with a confidence score below 0.5 (50 percent)



is discarded entirely rather than displayed to the user. This strict thresholding mechanism prioritizes precision (89.2 percent) over recall (76.8 percent), ensuring that only statistically reliable key point detections were presented to users. This approach eliminates erratic or "flying limb" predictions that would otherwise degrade the user experience and potentially provide inaccurate feedback for exercise form assessment. Furthermore, the model demonstrated robust performance across environmental variations, maintaining 90.63 percent accuracy in far-range scenarios where bounding box area fell below 3 percent of the image, validating its applicability to diverse user distances and clinical settings.

The pose estimation component was developed utilizing a hybrid CNN-LSTM architecture designed to capture both spatial and temporal features of human movement. The spatial feature extraction was performed using a YOLO11s-pose backbone initialized with ImageNet pretrained weights, leveraging transfer learning to accelerate convergence and improve generalization without training from scratch. Transfer learning provided the models with foundational knowledge of visual features such as edges, curves, and shadow patterns before exposure to specialized pose or pain recognition tasks, significantly reducing training time and improving sample efficiency. The YOLO11s-pose model was selected for its efficiency on mobile platforms while maintaining competitive accuracy metrics, achieving a mean Average Precision at 0.5 IoU (mAP@50) of 72.3 percent and a key point accuracy of 85.4 percent on the standalone spatial branch. To enhance temporal consistency and reduce frame-to-frame jitter inherent in frame-level CNN predictions, a Long-Short Term Memory module was incorporated to process sequential key point data



across a temporal window of 16 frames, approximately 0.5 seconds of video. The LSTM processed flattened key point coordinates in tensor dimensions of batch by temporal by spatial, learning temporal dependencies that smoothed erratic predictions while preserving movement authenticity. This temporal smoothing achieved a reduction in angle error of 69 percent while maintaining temporal consistency scores of 88.7 percent, demonstrating that the hybrid architecture effectively combined the strengths of spatial precision from CNNs with temporal coherence from sequential models.

The pain recognition architecture employed a teacher-student knowledge distillation framework, leveraging a hybrid training strategy fundamentally different from standard supervised learning. The teacher model, EfficientNet-B4 initialized with ImageNet pretrained weights, served as the knowledge source and was frozen during training to optimize computational efficiency. The student model, ResNet18 also initialized with ImageNet pretrained weights, represented the deployed model with approximately 26.4 million parameters, designed for real-time inference on mobile devices. The knowledge distillation training process minimized a combined loss function incorporating both standard classification loss and distillation loss. The Cross-Entropy loss with an alpha weight of 0.5 provided direct supervision from ground truth labels, while Knowledge Distillation loss based on Kullback-Leibler divergence with a temperature parameter of 4.0 constrained the student to match the teacher's probability distributions. Rather than learning hard classifications of simply low, moderate, or severe pain, the student model learned subtle relationships between pain classes by mimicking the teacher's confidence distributions. For instance, when the teacher output probability



estimates such as 85 percent low pain and 15 percent moderate pain, the student was penalized not merely for disagreeing with the ground truth label but for failing to match this nuanced distribution. This approach successfully captured what researchers term, dark knowledge, enabling the model to learn the ambiguous gray areas between pain classes rather than treating classification as a discrete decision problem.

### 3.1.5 TESTING

The testing methodology for the developed PocketPT application involves validation of plans generated by the application against assessments made by a licensed physical therapist to ensure the highest degree of accuracy and clinical reliability. The evaluation focuses on three critical areas: the application's ability to correctly identify the location of pain within the user's body, the accuracy of its assessment of the user's pain intensity based on a standardized pain scale, and the appropriateness of the exercise recommendations generated for rehabilitation. Successful alignment between the application's outputs and the professional assessments would serve as the primary indicator of the application's effectiveness and completion of its testing phase.

The CNN-based facial pain recognition model was evaluated through the use of accuracy, precision, recall, and F1-score, which collectively determined its reliability in classifying different levels of pain intensity from facial expressions. Inference latency was also measured to verify that the model could function efficiently in real-time scenarios on mobile platforms without causing delays in user interaction. The CNN/LSTM-based human pose estimation model underwent a similar evaluation process, where accuracy,



precision, recall, and F1-score was applied to assess its effectiveness in detecting joint positions and movement patterns that are relevant to pain localization and rehabilitation monitoring. Inference latency was likewise monitored to ensure that the model provides timely and responsive feedback during user exercise sessions. Additionally, any discrepancies identified during the testing process were documented and analyzed to refine the application further, ensuring that PocketPT could reliably support users in managing and improving their rehabilitation journey with minimal professional oversight.

### **3.1.6 DEPLOYMENT**

The PocketPT application was scheduled for deployment in the vicinity of Cavite to evaluate its versatility and effectiveness. Although the application was primarily designed for rehabilitation purposes, its integrated mobility and strengthening exercises also make it highly suitable for use in fitness and gym environments. To ensure the application's clinical relevance and practical usability, licensed physical therapists assessed the accuracy and appropriateness of the rehabilitation plans generated by the system. Simultaneously, individuals currently undergoing rehabilitation participated in user testing, providing valuable insights into the app's usability, accessibility, and overall functionality. Feedback from both professional practitioners and users is crucial, as it identified areas requiring improvement, guided further development, and ensured that PocketPT consistently meets its therapeutic and user experience objectives.



## 3.1.7 FEEDBACK ITERATION AND COLLECTION

User feedback was systematically gathered and thoroughly analyzed, recognizing its critical role in guiding the continuous improvement of the PocketPT application. To ensure structured and comprehensive feedback collection, researchers administered a carefully designed survey targeting key aspects such as user experience, functionality, accuracy of rehabilitation recommendations, and overall satisfaction with the application. The survey was distributed to users through QR Code, capturing a diverse range of perspectives from individuals undergoing rehabilitation as well as those engaged in mobility and strengthening exercises. This dual- environmental approach ensures that the application's performance is assessed across different contexts of use, providing richer, more actionable insights.

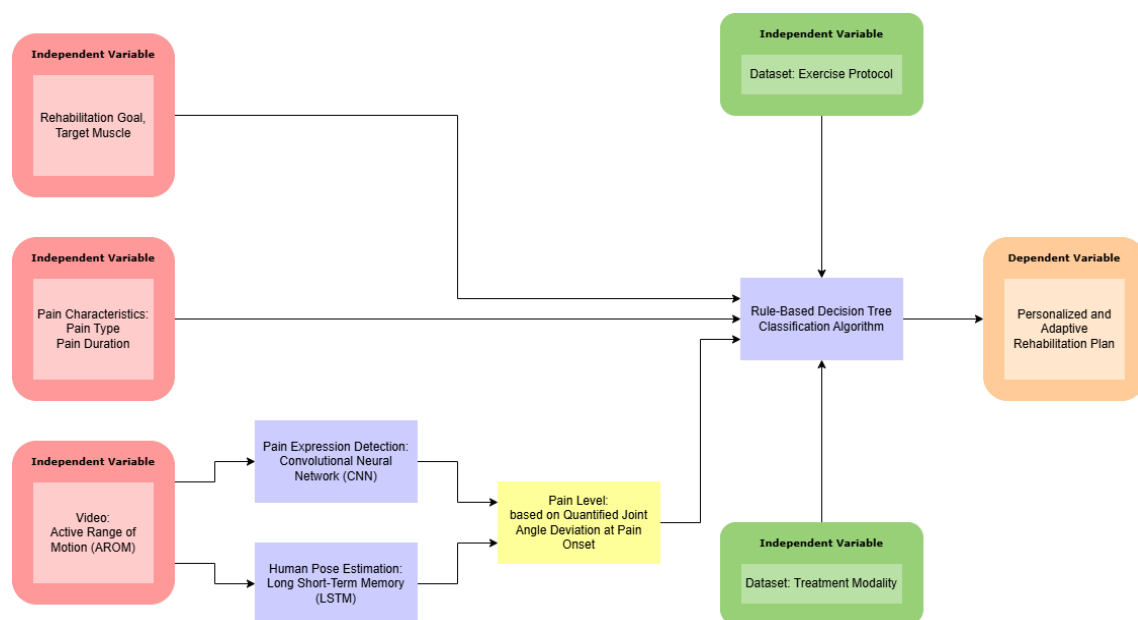
All feedback collected was carefully reviewed, categorized, and prioritized based on urgency and impact. The development team incorporated these insights into successive updates, addressing identified issues, enhancing features, and optimizing the user experience. This iterative refinement process is intended to ensure that PocketPT evolves in alignment with real-world user needs and maintains high standards of effectiveness, usability, and clinical relevance.



### 3.2 CONCEPTUAL FRAMEWORK

**Figure 3.2.1**

*Conceptual Framework of PocketPT*



This framework outlines the design and functionality of a sophisticated system that generates personalized and adaptive rehabilitation plans, aimed at enhancing the rehabilitation process through the integration of multiple data inputs and advanced technological methodologies. The system leverages several key components to process various pieces of information, primarily independent variables, and through a series of well-structured steps produces a customized rehabilitation plan, which serves as the ultimate dependent variable.



The system initiates its operation by gathering vital data from the user. This crucial information serves as the foundational input for the entire process. The system collects detailed user data, including the user's rehabilitation goals, the affected body part, and additional personal reports regarding the nature of the injury and the pain experienced. Specifically, the user provides information about the type of pain (e.g., sharp, dull, burning) and the duration of the pain since the injury first occurred. This initial set of inputs is vital in forming a comprehensive understanding of the user's condition and tailoring the rehabilitation plan accordingly. A particularly valuable input is a video recording of the patient's Active Range of Motion (AROM) exercises. These recordings offer dynamic insights into the patient's ability to move and perform physical exercises, which is an essential factor in determining the severity of the injury and identifying appropriate exercises and therapies.

Once the user data is collected, the framework refines this information by leveraging a predefined knowledge base. This knowledge base consists of independent variables, which are systematized datasets containing crucial information about suitable rehabilitation exercises and various treatment modalities. The dataset categorizes different exercises that are customized to the user's rehabilitation needs, while another dataset provides information about the various treatment modalities, including physical therapy techniques, manual therapies, and medical interventions that can be applied to the user's specific condition. The predefined knowledge base allows the system to organize and process these independent variables in a manner that ensures the information is both accurate and contextually relevant to the individual user's rehabilitation journey. The



datasets are curated based on established medical knowledge and clinical best practices, ensuring the system's recommendations are grounded in proven methodologies.

A critical step in the framework is the analysis of the AROM video data, which occurs through the application of advanced artificial intelligence techniques. The system utilizes a Convolutional Neural Network (CNN) to analyze the video frames for visual cues of pain. This AI model focuses on detecting facial expressions and changes in the user's movement patterns, which may indicate discomfort or pain during the execution of exercises. By identifying these subtle signs of pain, the system gains further insights into the user's current condition and level of discomfort during rehabilitation exercises.

Additionally, the system integrates a Long Short-Term Memory (LSTM) network, a type of deep learning algorithm that specializes in processing sequential data. The LSTM network tracks the user's joint movements over time, enabling the system to perform human pose estimation. This involves analyzing joint locations and the trajectories of movement throughout the video, which provides valuable information on the user's range of motion and movement patterns. The LSTM network is particularly effective in understanding the temporal sequence of movements, allowing it to detect changes in motion that might indicate improvements or setbacks in the user's recovery.

The combination of pain expression detection data (via the CNN) and pose estimation data (via the LSTM network) is crucial in calculating the user's pain level during exercises. This data is analyzed in real-time to assess changes in joint angles and to track any deviations that may signify pain or discomfort. The system uses the output



from the CNN and kinematic analysis (such as the detection of restricted or altered movement patterns) to determine the level of pain the user is experiencing.

Pain level determination is based on a multi-step process that includes evaluating the user's movement behavior, such as the extent of joint flexion or extension, and comparing it to predefined thresholds indicating pain. The system integrates the analysis of facial expressions and body posture to create a comprehensive assessment of the user's physical state. This allows the system to dynamically adjust its recommendations based on real-time data and to monitor the user's progress throughout the rehabilitation process.

After the pain and pose estimation data is processed, the system integrates all the collected information, such as the user's rehabilitation goals, pain characteristics, pain level, and data from the exercises and treatment modality datasets, into a central processing unit. This information is then passed to the core decision-making engine, which utilizes a Rule-Based Decision Tree Classification algorithm. This algorithm applies a structured set of rules to evaluate the combined inputs and determine the most appropriate rehabilitation plan for the user.

The Rule-Based Decision Tree classification algorithm is designed to ensure that the system makes decisions based on a clear, logical framework. It evaluates the user's input data against predefined rules that have been established based on clinical guidelines, medical research, and expert recommendations. The algorithm processes the combined inputs to identify the best course of action for the user, which may include suggesting specific exercises, recommending treatment modalities, and adjusting the rehabilitation



plan based on ongoing assessments.

The output of the Rule-Based Decision Tree classification algorithm is a personalized rehabilitation plan tailored to the user's specific condition and rehabilitation goals. This output represents the dependent variable in the system, as it is the final result produced by processing all the inputs and applying the decision-making algorithm.

The personalized rehabilitation plan generated by the system is dynamic and adaptive, meaning it can be adjusted as the user progresses through their rehabilitation journey. As new data is collected, whether from follow-up AROM videos, changes in pain levels, or shifts in the user's rehabilitation goals, the system can adapt the rehabilitation plan accordingly. This ensures that the plan remains relevant and responsive to the user's needs, providing a continuously optimized path to recovery.

Ultimately, the system's ability to integrate a wide range of inputs and utilize advanced AI technologies ensures that the rehabilitation process is not only personalized but also adaptable, offering patients a tailored approach that evolves as their condition improves or changes over time.

### **3.3 ALGORITHM**

The development of the application incorporates three primary algorithms: Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN), and Rule-Based Decision Tree. These algorithms serve as the core components of the application, providing the essential functionality required for its operation.



The LSTM algorithm, a type of Recurrent Neural Network (RNN) used in deep learning, is designed to analyze sequential data, such as video frames, time series, or motion patterns over time. LSTM is capable of retaining patterns from previous histories, making it ideal for understanding complex movements. Instead of analyzing individual frames, LSTM observes the entire sequence of a person's movement from start to finish. It then determines whether the user's movement is correct or if any body part is displaying abnormal behavior. Within the application, LSTM tracks the user's physical movement, identifies potential injury locations by monitoring joint motion over time, compares the motion against healthy standards, and predicts which specific body part may be restricted or injured.

The CNN algorithm, a specialized neural network for pattern recognition in images, employs filters (convolutions) to detect features such as edges, shapes, and textures, which enable the network to understand the content of an image. In this application, CNN is utilized for facial recognition, specifically to analyze facial expressions and detect signs of pain or discomfort. This allows the system to assess the user's emotional state in response to physical activity and potential injury.

The Rule-Based Decision Tree algorithm is a machine learning approach that utilizes if- then rules derived from data to make decisions or classifications. The algorithm splits the collected data into branches based on various features, such as pain type, target muscle, and pain duration. Each branch evaluates conditions such as whether the pain duration exceeds or falls below seven days. As new data is collected, whether from follow-



up AROM videos, changes in pain levels, or shifts in the user's rehabilitation goals, the system updates and adapts the rehabilitation plan accordingly. These decision rules are derived from training data and can be manually adjusted to refine the decision-making process.

### **3.3.1 FORMULAS AND MATHEMATICAL REPRESENTATIONS**

#### **3.3.1.1 LSTM: LONG SHORT-TERM MEMORY**

Long-Short Term Memory (LSTM) networks represent a specialized variant of Recurrent Neural Networks (RNNs) developed specifically to address critical limitations of traditional RNNs in processing sequential data. Traditional RNNs suffer from the vanishing gradient problem, where gradients propagated backward through many time steps become exponentially smaller, causing the network to lose the ability to learn long-term dependencies. As sequential data grows longer, standard RNNs progressively forget earlier information in the sequence, failing to capture temporal relationships necessary for tasks requiring understanding of long-range contexts. LSTMs overcome this fundamental limitation through an architectural innovation: the introduction of a memory cell that maintains a persistent state across time steps, enabling selective information flow through multiple time steps without degradation.



**Table 3.3.1.1.1**

*Long Short-Term Memory: Notation*

| Symbol    | Description                                             |
|-----------|---------------------------------------------------------|
| $x_t$     | To input time step $t$                                  |
| $h_{t-1}$ | Hidden state from the previous step                     |
| $C_{t-1}$ | Cell state from the previous step                       |
| $W, b$    | Weights and biases                                      |
| $\sigma$  | Sigmoid function that has an output between 0 and 1.    |
| $\tanh$   | Hyperbolic tangent that has an output between -1 and 1. |
| $\odot$   | An element-wise multiplication                          |

The LSTM architecture processes sequential input through a series of computationally distinct but functionally interdependent gates that regulate information flow. At each time step  $t$ , the LSTM receives three primary inputs: the current input  $x_t$  from the sequence, the hidden state  $h_{t-1}$  representing learned information from all previous time steps, and the cell state  $C_{t-1}$  representing persistent memory maintained across time steps. These three inputs are processed through gate mechanisms that determine which information should be retained, updated, or discarded.



**Table 3.3.1.1.2**

*Long Short-Term Memory: Gates and States*

| Gate                 | Mathematical Formula                           |
|----------------------|------------------------------------------------|
| Forget Gate          | $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$ |
| Input Gate           | $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$ |
| Candidate Cell State | $C_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$  |
| Update Cell State    | $C_t = f_t \odot C_{t-1} + i_t \odot C_t$      |
| Output Gate          | $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$ |
| New Hidden State     | $h_t = o_t \odot \tanh(C_t)$                   |

The forget gate, the first gating mechanism, operates as a sigmoid-activated neural network layer that outputs values between 0 and 1 for each element of the cell state. This gate evaluates both the current input and previous hidden state to determine which elements of the previous cell state should be retained and which should be forgotten. The forget gate output is element-wise multiplied with the previous cell state, effectively zeroing out dimensions corresponding to gate outputs near 0 and preserving dimensions where gate outputs approach 1.

The input gate and candidate value computation work together to determine what new information should be added to the cell state. The input gate decides which dimensions should be updated, while a candidate cell state is computed independently using the hyperbolic tangent activation function with output range  $[-1, 1]$ . The candidate cell state represents potential new information derived from the current input and previous



hidden state. These are combined through element-wise operations to update the cell state. This formulation allows the network to independently control forgetting old information and adding new information.

The output gate determines which elements of the updated cell state should be exposed to the hidden state output. The final hidden state combines the output gate decision with a nonlinear transformation of the cell state. This hidden state becomes the output of the time step and serves as input to subsequent time steps or to downstream processing layers.

The elegance of the LSTM architecture lies in how these gating mechanisms solve the vanishing gradient problem. The addition operations in cell state updates create shortcut paths for gradient flow during backpropagation, preventing gradients from vanishing even across many time steps. Each gate can learn to preserve or discard information independently, allowing LSTMs to flexibly capture both short-term and long-term dependencies without requiring manual specification of relevant temporal scales.

### **3.3.1.2 CNN: CONVOLUTIONAL NEURAL NETWORK**

Convolutional Neural Networks (CNNs) have become a cornerstone of modern image processing due to their unique architecture, which enables them to excel in tasks involving visual data. Unlike traditional fully connected neural networks that treat all inputs uniformly, CNNs are specifically designed to process data with grid-like structure, such as images. This specialized architecture enables CNNs to automatically detect



patterns and features such as edges, textures, and shapes at multiple levels of abstraction through a hierarchical learning process.

The core strength of CNNs lies in their ability to detect spatial hierarchies within images through a principle known as local connectivity. This means that CNNs can recognize low-level features like edges or corners in the initial layers and gradually build up to more complex structures such as objects, scenes, or in the context of this application, facial expressions and human body configurations in deeper layers. This hierarchical feature extraction process allows CNNs to capture the spatial relationships between pixels and the spatial layout of objects within an image without requiring manual feature engineering.

CNNs consist of several key components working in concert to extract and classify visual information. Convolutional layers apply learnable filters, also referred to as kernels, across the input image. During training, these filters are optimized to detect features most relevant to the classification task. As these filters slide across the input image in a process called convolution, they generate feature maps that highlight the presence and location of important features at different spatial positions. Pooling layers, typically employing maximum pooling or average pooling operations, reduce the dimensionality of the feature maps while preserving the most important information. This dimensionality reduction decreases computational complexity while maintaining feature saliency. Fully connected layers, positioned after convolutional and pooling layers, interpret the high-level abstract features extracted by earlier layers and make final predictions or classifications. The



combination of these components creates a powerful architecture capable of learning hierarchical representations directly from raw pixel data.

The convolutional operation itself can be expressed mathematically as a discrete convolution where a learnable filter  $w$  is applied across the input feature map  $x$  with stride and padding parameters controlling the filter movement and boundary handling. The resulting feature map highlights regions of the input where the filter pattern matches strongly. By stacking multiple convolutional layers and varying filter sizes, CNNs can learn multi-scale representations capturing both fine-grained details and coarser structural information. This multi-scale processing is particularly valuable in applications requiring detection of features at various sizes or levels of detail.

The efficiency of CNNs in detecting spatial hierarchies and learning discriminative features makes them invaluable for a wide variety of image-related tasks including image classification, object detection, facial recognition, and medical imaging. In the context of this application, CNNs excel at two distinct yet related problems. For pain recognition, CNNs learn to detect subtle facial features associated with different pain levels, including the spatial patterns of brow lowering, eye closure, and facial contraction that correlate with increasing pain intensity. For pose estimation, CNNs learn to identify and localize anatomical key points such as joints by detecting learned features specific to each body part's visual appearance and spatial context. The ability to automatically discover relevant features from raw image data without manual specification makes CNNs superior to hand-crafted feature approaches, particularly in complex domains like facial analysis and



human pose understanding where feature importance and discrimination vary across individuals and environmental conditions.

**Table 3.3.1.2.1**

*Convolutional Neural Network: Layers and Formula*

| Description                                 | Formula                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Basic Convolution                           | $Y(i, j) = \sum_{m=0}^{kh-1} \sum_{n=0}^{kw-1} K(m, n) \cdot X(i + m, j + n)$                                           |
| General Convolution with Stride and Padding | $Y(i, j) = \sum_{m=0}^{kh-1} \sum_{n=0}^{kw-1} K(m, n) \cdot X(i \cdot s + m - p, j \cdot s + n - p)$                   |
| Multi-Channel Convolution for RGB images    | $Y(i, j) = \sum_{c=0}^{C-1} \sum_{m=0}^{kh-1} \sum_{n=0}^{kw-1} K(m, n) \cdot X_c(i + m, j + n) + b$                    |
| Convolution with Activation and Bias        | $Y(i, j) = f \left( \sum_{c=0}^{C-1} \sum_{m=0}^{kh-1} \sum_{n=0}^{kw-1} K_c(m, n) \cdot X_c(i + m, j + n) + b \right)$ |
| Pooling Layer                               | $Y(i, j) = \max_{(m, n) \in R(i, j)} X(m, n)$                                                                           |
| Flattening Formula                          | $x[n] = X(i, j, c), n = c \cdot (H \cdot W) + i \cdot W + j$                                                            |
| CNN as a Composition of Layers              | $CNN(X) = F_n(\cdots F_2(F_1(X)) \cdots)$                                                                               |

The basic convolution operation forms the foundational computation in all convolutional neural networks. Expressed mathematically as the double summation  $Y(i, j) = \sum \sum K(m, n) \cdot X(i + m, j + n)$  where the summation indices  $m$  and  $n$  range from 0 to



$kh-1$  and  $kw-1$  respectively, this formula represents the sliding of a small filter kernel  $K$  across an input image  $X$ . At each position within the image, the kernel is element-wise multiplied with the corresponding region of the input, and the results are summed to produce a single output pixel. This operation detects features such as edges, textures, and shapes in the input by computing weighted sums that respond to local patterns. Each output pixel represents a response to the local visual pattern in the input region covered by the kernel at that position, enabling the network to learn spatial feature detectors.

In practical implementations, the basic convolution operation is enhanced through the introduction of stride and padding parameters that provide control over the output feature map dimensions and computational efficiency. The general convolution formula with these parameters is expressed as  $Y(i, j) = \sum \sum K(m, n) \cdot X(i \cdot s + m - p, j \cdot s + n - p)$  where  $s$  represents the stride parameter controlling the step size as the filter slides across the input, and  $p$  represents the padding parameter controlling how the input boundaries are handled. The stride parameter allows down sampling of the input, reducing output spatial dimensions and computational cost while capturing larger receptive fields. Larger stride values produce smaller output feature maps with correspondingly reduced memory and computational requirements. The padding parameter allows zero-padding to be added around the input boundaries, controlling whether the output size matches the input size or becomes progressively smaller with each convolution. These parameters provide essential control over the spatial dimensions of feature maps flowing through the network, allowing network architects to balance model capacity, computational efficiency, and feature map resolution according to application requirements.



For applications processing colored images or multi-channel feature representations, the convolution operation extends naturally to process multiple input channels simultaneously. The multi-channel convolution formula is expressed as  $Y(i, j) = \sum_c \sum_m \sum_n K(m, n) \cdot X_c(i + m, j + n) + b$  where the summation over channel index  $c$  ranges from 0 to  $C-1$ , enabling processing of all  $C$  input channels. The kernel processes each input channel independently through element-wise multiplication, and the results across all channels are summed together with an added bias term  $b$ . This formulation enables learning from color information in RGB images where each channel represents red, green, and blue color intensities, as well as learning from deeper feature representations in higher network layers where each channel represents different learned feature maps from previous convolutions. Summation across channels allows the network to learn feature detectors that integrate information from multiple input channels, enabling the detection of complex visual patterns that depend on combinations of colors or combinations of features from different conceptual categories.

In modern neural networks, convolution operations are followed by nonlinear activation functions and bias terms that enable learning of complex nonlinear patterns beyond simple linear transformations. The complete convolution with activation and bias formula is expressed as  $Y(i, j) = f(\sum_c \sum_m \sum_n K_c(m, n) \cdot X_c(i + m, j + n) + b)$  where  $f$  represents a nonlinear activation function such as the Rectified Linear Unit (ReLU) function. The bias term  $b$  shifts the linear combination output independently of the input, allowing the network to learn thresholds for feature activation. The activation function  $f$  introduces nonlinearity by applying a nonlinear transformation to the summed result, enabling the



network to learn complex patterns and nonlinear relationships that extend far beyond what would be possible with purely linear transformations. Without nonlinearity, stacking multiple convolutional layers would produce only linear transformations that could be collapsed into a single linear layer, eliminating the representational power that deep networks provide. The ReLU activation function, commonly used in modern CNNs, applies the simple yet effective transformation  $f(x) = \max(0, x)$ , passing positive values through unchanged while zeroing negative values, providing computational efficiency while maintaining sufficient nonlinearity for learning complex patterns.

Following convolutional feature extraction, pooling layers downsample feature maps while preserving the most important information, reducing both spatial dimensions and computational requirements. The pooling operation is expressed mathematically as  $Y(i, j) = \max_{(m, n) \in R(i, j)} X(m, n)$ , where the maximum value is selected from a local region  $R(i, j)$  of the input feature map. This maximum pooling operation replaces each local region with its maximum value, effectively compressing information while retaining the most important activations from each region. Pooling layers serve multiple critical functions in CNN architecture. They reduce spatial dimensions, decreasing memory requirements and computational cost while maintaining feature information content. They provide translation invariance to the network by making output robust to small shifts or translations in the input, ensuring that slight changes in feature position do not substantially alter network predictions. This invariance property proves particularly valuable in rehabilitation assessment applications where slight variations in camera positioning or patient movement between frames should not substantially alter pose or



pain assessments. Pooling also helps prevent overfitting by forcing the network to extract only the most salient features from each region, regularizing the learned representations.

After extraction and pooling of spatial features through convolutional layers, the multi-dimensional feature maps must be converted into vector form suitable for processing by fully connected layers that perform final classification or regression. The flattening operation is expressed mathematically as  $x[n] = X(i, j, c)$  where  $n = c \cdot (H \cdot W) + i \cdot W + j$ , mapping a three-dimensional feature map with height  $H$ , width  $W$ , and  $C$  channels into a one-dimensional vector. The flattening operation preserves all information from the feature map by arranging it into a sequential vector where the linearization index  $n$  is computed based on the original spatial position  $(i, j)$  and channel index  $c$ . This linear arrangement ensures consistent ordering of feature information across different network instances. Flattening serves the critical function of bridging feature extraction layers from convolutional stages with decision-making layers implemented as fully connected neural networks. The fully connected layers interpret the flattened feature vectors and produce output predictions such as classification probabilities or regression values. This architectural design separates the spatial feature extraction phase, where convolutional layers automatically discover relevant features from pixel data, from the pattern recognition phase where fully connected layers learn how to interpret extracted features for the target prediction task.

The entire convolutional neural network can be conceptualized as a composition of successive layers, each performing a specific transformation on its input. This layered



composition is expressed mathematically as  $CNN(X) = F_n(\dots F_2(F_1(X))\dots)$  where each  $F_i$  represents a transformation function performed by layer  $i$ , and the nested function composition indicates that the output of each layer serves as input to the subsequent layer. This compositional structure means that the first layer  $F_1$  processes raw pixel input  $X$  through convolution and activation to produce intermediate feature representations. The second layer  $F_2$  processes these intermediate representations through convolution and pooling to extract higher-level features at coarser spatial scales. Subsequent layers continue this process, progressively extracting increasingly abstract features while reducing spatial dimensionality. The final layer  $F_n$  produces the output prediction by processing the most abstract features extracted by earlier layers. This hierarchical composition enables CNNs to learn increasingly complex patterns by progressively building abstract representations from raw pixels, implementing a form of hierarchical feature learning where early layers learn low-level features such as edges, middle layers learn intermediate features such as shapes and textures, and deeper layers learn high-level semantic features such as objects or facial expressions relevant to the classification task.

### **3.3.1.3 RULE-BASED DECISION TREE**

The rule-based decision tree algorithm is a widely used machine learning technique for both classification and regression tasks. In classification, it assigns categories or labels to input data, while in regression, it predicts continuous values. The algorithm works by recursively splitting the dataset into subsets based on specific criteria derived from input features. Each split is governed by a condition, typically an "if-then"



rule, that evaluates a particular attribute or feature of the data.

**Table 3.3.1.3.1**

*Rule-Based Decision Tree: Formula*

| Description   | Mathematical Formula                         |
|---------------|----------------------------------------------|
| Gini Impurity | $Gini(D) = 1 - \sum_{k=1}^n p_k^2$           |
| Entropy       | $Entropy(D) = - \sum_{k=1}^n p_k \log_2 p_k$ |

At the core of the decision tree's structure are nodes, where each node represents a decision point based on a feature, and branches, which represent possible outcomes or decisions based on that feature. The decision-making process starts at the root node, and as the algorithm processes the data, it traverses through the branches, applying the if-then rules to decide the best split. This process continues until the data is divided into smaller subsets, ultimately resulting in a set of leaves that provide the final classification or regression result.

The effectiveness of the rule-based decision tree lies in its simplicity and interpretability. Each decision rule is easy to understand, making the model transparent and providing insight into how decisions are made. The splitting criteria used at each node can be based on various metrics, such as information gain, Gini impurity, or mean squared error, depending on whether the task is classification or regression.

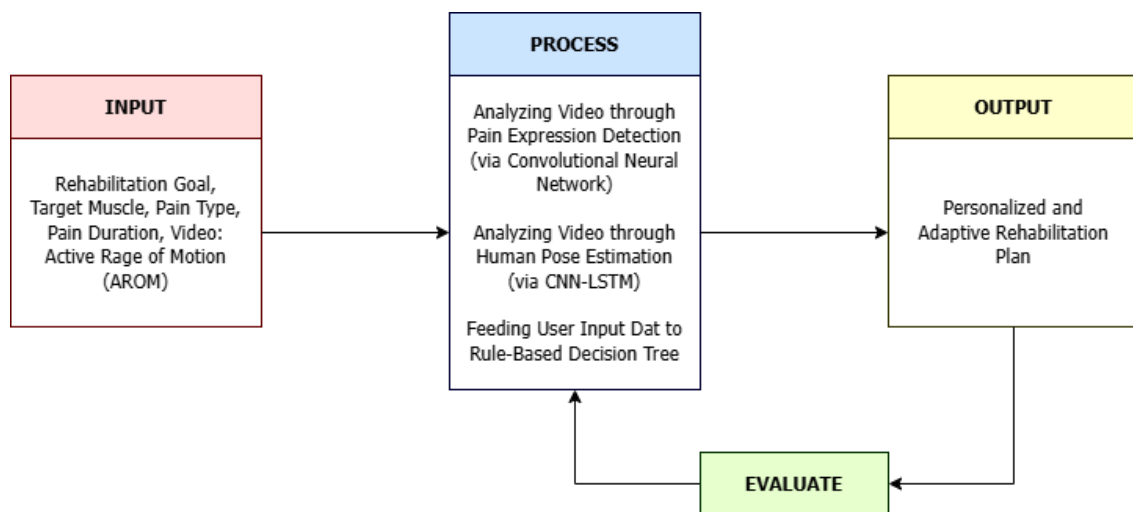


One of the main advantages of the decision tree algorithm is its ability to handle both categorical and continuous data without requiring extensive data preprocessing. Additionally, decision trees can capture non-linear relationships in the data, making them flexible and powerful for a wide range of applications, from predictive modeling to customer segmentation and risk assessment.

### 3.4 OPERATIONAL FRAMEWORK

**Figure 3.4.1**

*Research Paradigm for the PocketPT: An AI-Driven Rehabilitation Assistant with Human Pose Estimation and Pain Recognition using CNN And LSTM*



This diagram illustrates the core workflow of the PocketPT application, demonstrating how the system processes user data to generate a personalized and adaptive



rehabilitation plan. The process begins with the user's input, which includes the rehabilitation goal, target muscle, pain type, pain duration, and an Active Range of Motion (AROM) video. These inputs are then systematically processed through a series of AI-driven mechanisms.

The first step in the process involves the Long Short-Term Memory (LSTM) algorithm, which analyzes the user's movement through human pose estimation. This allows the system to identify the affected body part where pain is likely located. Following this, the Convolutional Neural Network (CNN) analyzes the user's facial expressions, detecting signs of pain or discomfort based on visual cues, with augmentation techniques (brightness, contrast, rotation) applied to account for varied lighting and angles. Once both algorithms have completed their analysis, the data provided by the user, along with the results from the two algorithms, are input into the Rule-Based Decision Tree Algorithm. This algorithm evaluates all the available information to determine the most appropriate rehabilitation plan.

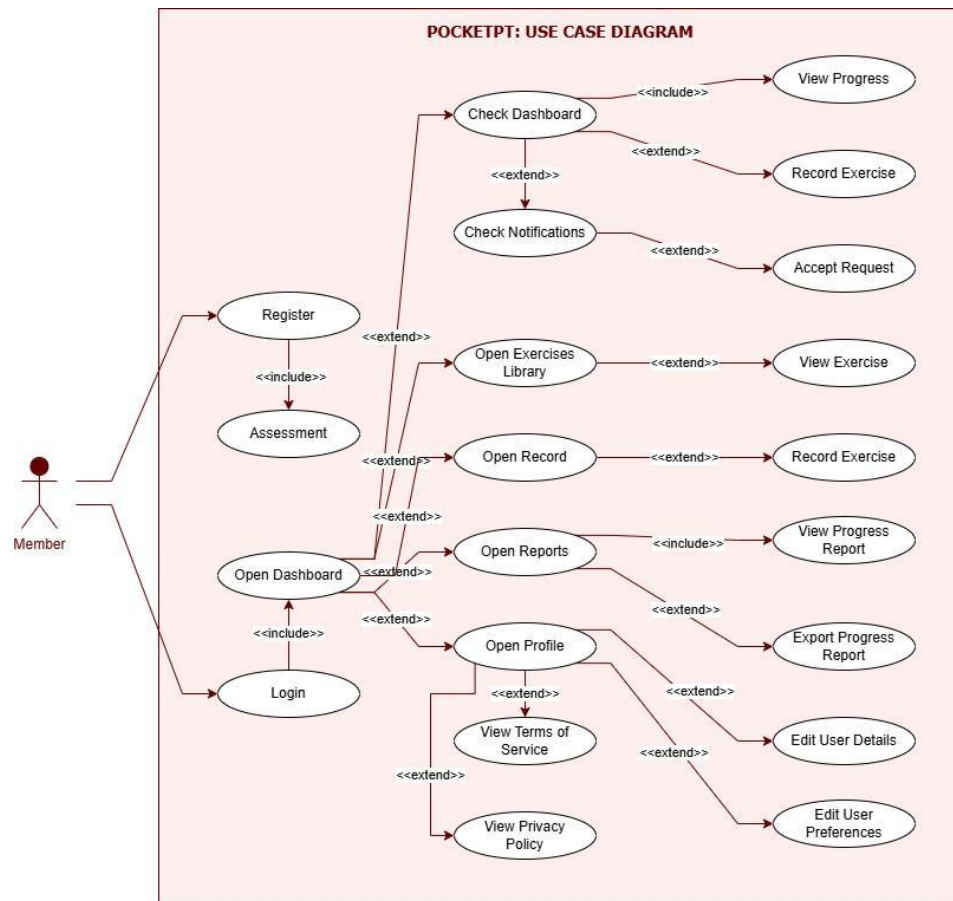
The final output of the system is a customized rehabilitation plan that is specifically tailored to meet the user's physical needs and can adjust over time based on ongoing data analysis. This dynamic approach ensures that the rehabilitation process is both effective and responsive to the user's progress and changing physical condition.



### 3.5 USE CASE/ FLOWCHART / STATE DIAGRAM

Figure 3.5.1

*Use Case Diagram for the PocketPT: An AI-Driven Rehabilitation Assistant with Human Pose Estimation and Pain Recognition using CNN And LSTM*



The use case diagram for PocketPT defines a single principal actor: the Users. Users are individuals who engage with the application. Their first interaction involves creating an account by providing their first name, last name, email address, and password,



and consenting to the Terms of Service and Privacy Policy. The information they submit during registration serves important functions: their name will be used in the generation of future progress reports, while their email address and password serve as their authentication credentials for subsequent logins. However, before completing registration, users are required to undergo a comprehensive guided assessment. This assessment collects essential information regarding the user's rehabilitation goals, the specific muscle region they wish to target, descriptions of any pain they are experiencing, including its level, type, and duration, and any history of recent or past injuries.

For the evaluation of pain levels, PocketPT employs an Active Range of Motion (AROM) method integrated with pain recognition and human pose estimation technologies, which precisely determine the maximum angle a user can stretch without discomfort. In the interest of user privacy, there is also an option to manually input a pain scale rating from 1 to 10. Based on the collected assessment data, PocketPT formulates a personalized rehabilitation plan utilizing a curated dataset verified by licensed physical therapists and processed through a rule-based decision tree algorithm designed to ensure the safe and effective selection of exercises and treatments. Once the assessment is completed, the user's registration is finalized.

As registered users, these people gain access to several core features of the application, including the ability to log in, and navigate the Dashboard, Exercises Library, Record, Reports, and Profile pages. Logging in requires the same email and password used during initial registration. Upon successful authentication, users are directed to the



Dashboard, which prominently displays their ongoing rehabilitation progress. The Dashboard highlights the user's current exercise, recent notifications, exercise streak (reflecting the number of consecutive days they have adhered to their rehabilitation routine), the total number of completed exercises, and the cumulative amount of time spent exercising. Users can also approve access requests from therapists seeking to monitor their progress and have the option to incorporate additional exercises outside of their prescribed rehabilitation plan. Through the Dashboard, they may also initiate or resume their rehabilitation journey using a dedicated Start/Resume button.

The Exercises Library provides users with access to an extensive list of exercises stored within the application's database. Each exercise card showcases an accompanying image, the targeted muscle group, and the acceptable pain threshold for that specific exercise. Expanding a card reveals detailed information such as step-by-step instructions, the rehabilitation goals addressed by the exercise, and the recommended number of sets and repetitions. This feature empowers users to explore alternative exercises beyond those included in their personalized rehabilitation plans.

Recording progress is facilitated through the Record feature. Initially, users are directed to a Pre-Record page, which offers a live camera view, displays the upcoming exercise, its sequence number, required sets and repetitions, and the targeted muscle group. Upon initiating the recording session, users proceed to the Record page, where they can monitor their exercise duration, pause the timer as needed, and navigate between exercises. Upon completing the final exercise in the sequence, users are prompted to either



save or discard their recorded progress via the Confirm Progress page.

The Reports page serves as a comprehensive summary of a user's rehabilitation activity. It compiles data from the initial assessment and records details such as exercise dates, sets, repetitions, and times. Users have the capability to export this report as a professionally formatted PDF, utilizing terminology consistent with medical standards, making it suitable for presentation during consultations with healthcare professionals.

Within the Profile page, users can manage their personal information and application settings. Options are available to update their name, email address, or password, and to configure notifications for daily exercise reminders and streak preservation alerts. The Terms of Service and Privacy Policy are accessible here, reaffirming that PocketPT is intended for informational purposes only and emphasizing the development team's commitment to data privacy, security, and ethical research practices. The application ensures that all information entered by users, and therapists' remains confidential and secure.

## **3.6 RESPONDENTS OF THE STUDY**

This study involved individuals with varying levels of physical activity, as their experiences and conditions are directly relevant to the research objectives. Participants include those who are physically active, such as individuals engaging in sports or maintaining regular physical activity like jogging or walking, as well as those requiring physical rehabilitation due to muscle strains, injuries, or general muscular discomfort.



Their input provided valuable insights into both the challenges and outcomes associated with physical activity and rehabilitation.

The participants were recruited within the Cavite area, targeting a minimum of 50 respondents. The sample was balanced by sex, comprising both male and female participants, and included at least 35 patients and 15 experts aged 18 years and above. In addition, certified physical therapists practicing in both gym and clinical environments were included to offer professional perspectives, enriching the study's findings with expert knowledge.

To collect relevant data, participants completed a structured questionnaire based on the ISO 25010 Software Product Quality Standards. The instrument was designed to gather both demographic profiles and multi-dimensional feedback on the PocketPT application. The questionnaire captured essential demographic information, including age, sex, residency in Cavite, and engagement in physical activities, to contextualize the user base. The core of the instrument consisted of quantitative and qualitative sections assessing six key quality dimensions: Functional Suitability (e.g., effectiveness of physical therapy support and dynamic plan adjustment), Performance Efficiency (e.g., accuracy and smoothness of pose detection and facial recognition), Usability (e.g., navigation clarity and ease of use), Reliability (e.g., system stability and data persistence), Security (e.g., authentication and data protection measures), and Portability (e.g., installation process and data export capabilities). Finally, open-ended questions solicited direct feedback on the application's strengths, weaknesses, and areas for improvement.



The researchers upheld the highest standards of ethical conduct to ensure the protection of participants' rights, maintain public trust, and safeguard the privacy of all individuals whose data may be involved in this undergraduate thesis. A strong commitment was made to preserving confidentiality, transparency, and academic integrity throughout the entire research process. All data obtained throughout the phases of data collection, model development, and system evaluation were used strictly and solely for academic purposes, specifically within the defined scope of this research project. At no point was any portion of the acquired data disclosed, disseminated, or utilized for purposes beyond the objectives of the study.

**Informed Consent and Privacy.** Prior to any data collection, the study underwent review and secure approval from the appropriate Ethics Review Committee. All participants were provided with a clear and comprehensive informed consent form. This document outlined the purpose and objectives of the study, described the data collection procedures, clarified the duration and scope of the research, and detailed the rights and responsibilities of the participants. Participation only commenced upon obtaining explicit consent from each respondent.

**Anonymity and Confidentiality of the Respondents.** The researchers acknowledge that sensitive and potentially identifiable information may be gathered during the course of this study. In response, strict measures were implemented to ensure the confidentiality and security of all collected data. These measures are designed to prevent unauthorized access, protect against data breaches, and ensure that personal



information remains anonymous and safeguarded at all stages of the research.

**Voluntary Participation.** Participation in this study was entirely voluntary. Respondents had the freedom to skip any question they were uncomfortable answering and may withdraw from the study at any point without penalty. This approach guarantees that participants are treated with respect and retain full autonomy throughout their involvement in the study.

**Responsible Use of Artificial Intelligence.** As the application developed in this study utilizes advanced AI technologies, specifically Convolutional Neural Networks (CNN) for facial expression analysis and Long Short-Term Memory (LSTM) models for pose estimation, the researchers affirm that all datasets used for model training were legally acquired and ethically sourced. Data collected from human participants was used exclusively to analyze physical pain indicators and body movements for rehabilitation purposes. No data was shared with third-party services or used for unrelated technological applications.



## CHAPTER IV

### PROJECT DEVELOPMENT

This chapter presents an overview of the methodologies and procedures employed throughout the development of the project. It begins with a detailed introduction to the web application, emphasizing its objectives, functionalities, and intended impact. The subsequent section on project development elaborates on the technical tools, frameworks, and implementation strategies utilized to realize the application's design and features. Finally, the development planning segment outlines the systematic strategies, workflow organization, and timeline management adopted to ensure the project's timely and successful completion.

#### 4.1 DESIGN AND METHODOLOGY

This study utilized a mixed-method research design, integrating both quantitative and qualitative approaches to comprehensively evaluate and improve the PocketPT mobile application. The adoption of a mixed-method approach allowed the researchers to capture both objective performance metrics and subjective user experiences, providing a holistic understanding of the application's effectiveness and usability.

The quantitative component involved the documentation of the performance metrics of the trained human pose detection and pain recognition models, including confusion matrices and accuracy scores, to evaluate the functionality and reliability of the



application objectively. Systematic collection and analysis of measurable data were used to assess both user satisfaction and system performance. Structured feedback from respondents regarding the application's strengths, weaknesses, and areas for improvement was categorized under the quantitative domain, as it allowed for comparative and numerical evaluation. Furthermore, survey questions pertaining to specific features of the application were measured using a Likert scale, providing quantifiable data that could be analyzed statistically. These quantitative data were compiled into tabular and graphical formats to facilitate clear interpretation and to support evidence-based conclusions regarding the system's performance. The qualitative component of the study focused on identifying specific changes and improvements required within the mobile application based on user feedback collected through surveys and direct observations. This approach enabled the researchers to explore user preferences, feature expectations, and additional recommendations in depth. Insights obtained from this qualitative analysis provided contextual understanding of user behavior and guided the development team in implementing modifications aimed at enhancing the overall user experience.

The research design of the PocketPT application followed a structured experimental framework which ensures systematic data collection, model training and evaluation, and providing reliable results. The study employed an experimental research design to explore the accuracy of utilizing Convolutional Neural Networks (CNN) and/or Long-Short Term Memory (LSTM) in estimating human poses and the detection of pain found in the face. Additionally, the study integrated these models into a mobile application for real-time human pose estimation and facial pain detection for usage in the assessment



process to understand the user's condition and situation - an attempt to mimic the process of assessing a patient done by medical professionals such as physical therapists.

The pain recognition system classifies facial expressions into three clinically relevant categories: low pain representing mild discomfort with minimal facial expression changes, moderate pain representing moderate discomfort with noticeable facial expressions such as brow lowering and eye tightening, and severe pain representing extreme discomfort with pronounced grimacing and facial contraction. The model achieved class-specific precision scores of 98.15 percent for low pain, 95.87 percent for moderate pain, and 92.43 percent for severe pain, demonstrating particular success at the challenging moderate pain category. This achievement is attributable to the knowledge distillation approach, which forced the model to learn the ambiguous gray areas between pain classes rather than treating classification as a discrete decision problem.

The convergence of validation accuracy (99.02 percent) with test accuracy (98.93 percent) provides empirical evidence of robust generalization and effective prevention of memorization. If the model had memorized training samples despite augmentation efforts, test accuracy would have significantly underperformed validation accuracy. Instead, the negligible divergence confirms that the learned features represent genuine pain recognition concepts rather than sample-specific memorization patterns. The close alignment validates that augmented samples provided sufficient synthetic variation to train a model capable of recognizing pain on novel faces encountered during deployment, satisfying the clinical requirement for real-world applicability.



The sampling technique for this study employed a non-probability approach, primarily utilizing purposive and convenience sampling techniques to ensure that respondents met the predefined criteria. Participants were targeted based on specific qualifications, including being residents of Cavite aged 18 years and above, physically active individuals, individuals experiencing muscular pain, or medical professionals such as physical therapists with an understanding of physical rehabilitation. The survey links were distributed through social media platforms, including Facebook group chats and private Messenger messages, as well as via direct referrals from partnered physical therapists. To further expand participation, respondents were encouraged to share the survey with acquaintances meeting the eligibility criteria, integrating elements of snowball sampling into the methodology.

This approach enabled the collection of a relevant dataset of responses aligned with the study objectives. The data were then processed and analyzed using both quantitative and qualitative methods. Quantitative data analysis involved descriptive statistics to summarize participant demographics, assess responses to Likert-scale items, and identify trends regarding the usability, strengths, and areas for improvement of the PocketPT application. Inferential statistics, including frequency distributions and cross-tabulations, were employed to examine relationships between respondent characteristics and their evaluation of the application. Concurrently, qualitative feedback from open-ended questions was analyzed to provide deeper insights into user experiences and specific recommendations for application enhancement. This mixed-methods approach ensured a thorough evaluation of PocketPT, integrating both structured metrics and contextual user



perspectives to inform evidence-based improvements.

The study ensured internal validity by employing a rigorous development process, including the use of established datasets such as the MPII Human Pose Dataset and the Pain Recognition from Facial Expression Dataset for model training, and by implementing advanced algorithms like Convolutional Neural Networks (CNN) for facial pain recognition and Long Short-Term Memory (LSTM) networks for human pose estimation. External validity was addressed by designing the PocketPT application for deployment in both clinical (e.g., ProBio Medcare Doctors Clinic) and gym-based settings (e.g., Standard Insurance Corp. Technical Training Center), and by targeting a diverse participant pool from Cavite, including individuals of varying activity levels, ages, and rehabilitation needs, thereby enhancing the generalizability of the findings. Reliability was maintained through systematic data collection methods, consistent use of preprocessing techniques such as data augmentation for facial expression analysis, and the application of standardized evaluation metrics, including accuracy, precision, recall, and F1-score, to ensure that model performance was both measurable and reproducible across different testing scenarios.

The research adhered to ethical standards to protect participant rights and welfare. Participation was entirely voluntary, with explicit guarantees for the right to withdraw at any time without penalty, thereby upholding the principle of autonomy. Confidentiality was prioritized through secure data storage, anonymized identifiers, and restricted access to collected information, though the use of a digital platform necessitated

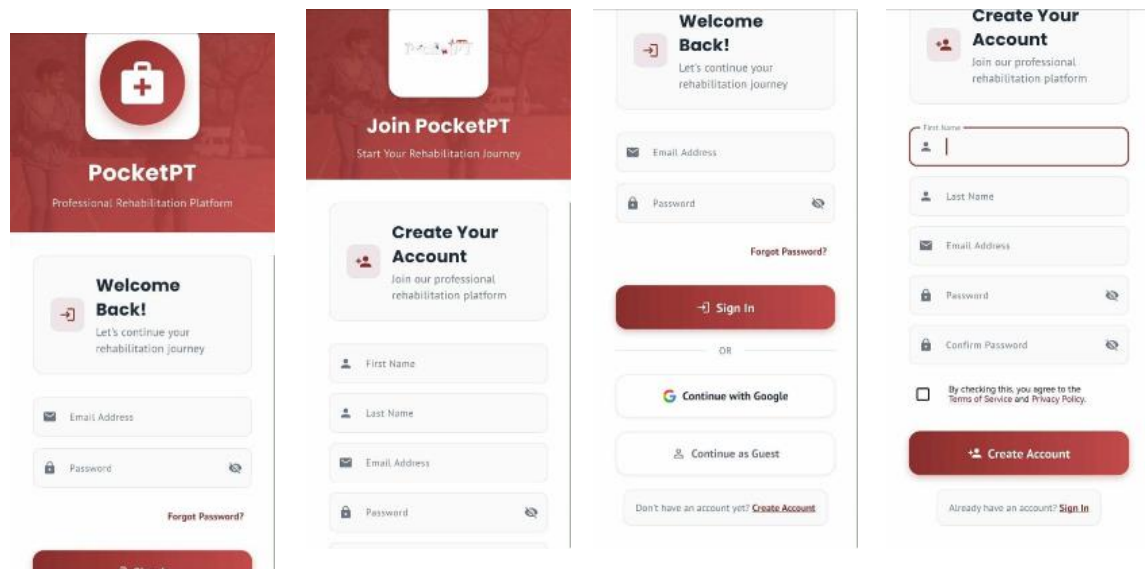


acknowledgment of minimal risks related to accidental data disclosure. Potential physical risks, such as muscle soreness, were mitigated by designing rehabilitation exercises under the guidance of licensed physical therapists. Furthermore, the study-maintained transparency by clearly communicating its purpose, procedures, and potential benefits, including personalized rehabilitation support, while committing to share aggregated results with participants after the study.

## 4.2 APPLICATION INTRODUCTION

**Figure 4.2.1**

*Application: Log-in and Sign-up Page*



If the user just installed the application, these are the pages that the user will first see. These pages consist of the Login page, Create Account page, Forgot Password page,



and E-mail Verification page. All these pages contain a header with the application's logo and name.

The login page consists of fields for the e-mail address and password for signing in, an option for forgot password if you want to reset your password, an option to register, and a button to continue as guest if you want to proceed with the application without an account but with limited features.

The Create Account page consists of five (5) input fields: First Name, Last Name, E-mail Address, Password, and Confirm Password. The first two fields for name is being requested for usage in the report containing the user's assessment information and history of exercises and tracked pain levels. The other three fields for the e-mail address and password are for account credential purposes. The password adheres to the security standards of requiring a symbol, a number, and the usage of both uppercase and lowercase letters.

Once the user has registered, the user will be redirected to the E-mail Verification page. The application requires the user to verify the e-mail address by sending a confirmation link that needs to be opened in order to proceed with the initial assessment process. If the user cannot see the confirmation e-mail in the Inbox, the user may check the Spam folder.

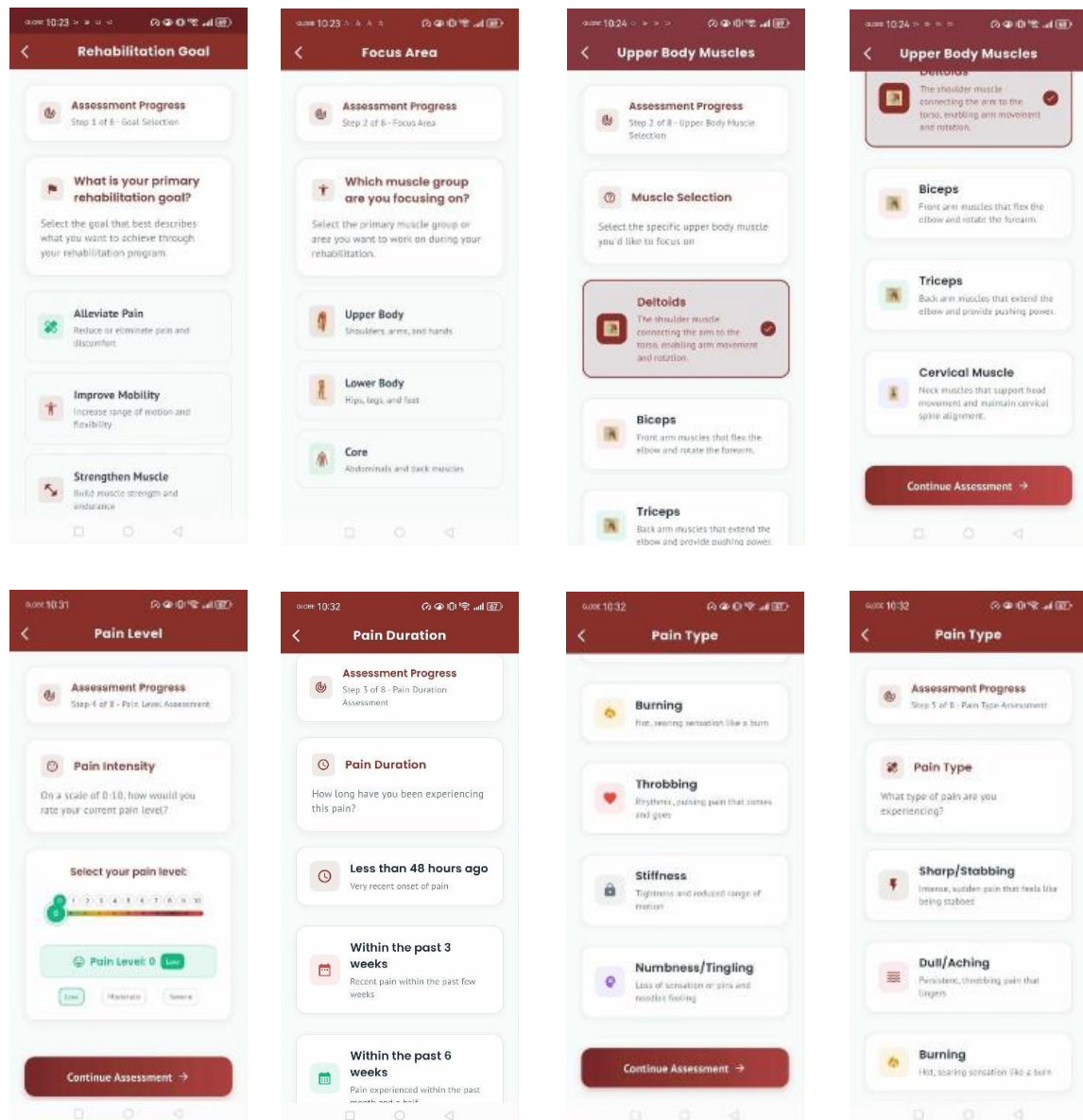
If the user forgot the password of the account, they may press the Forgot Password text from the Login page to redirect the screen to the Forgot Password page. The

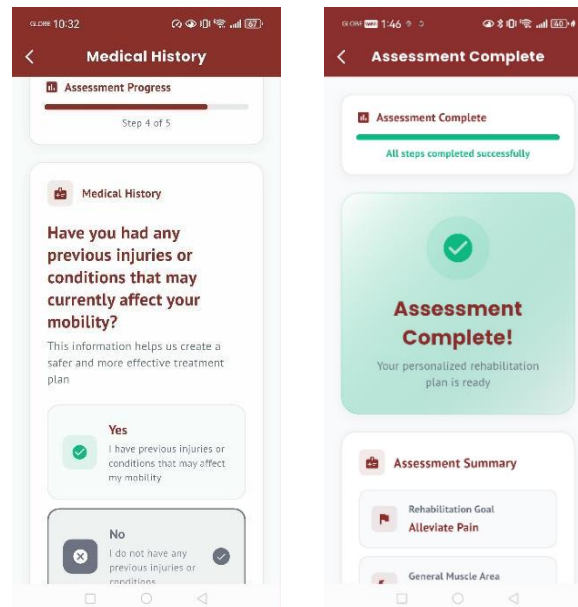


application will send an e-mail first to the user's e-mail address confirming if the user truly wants to reset the account's password. If the user inputs a non-registered e-mail address to the page, the application ignores the request.

**Figure 4.2.2**

*Application: Assessment Pages*





Once the user has registered and verified the account, the applications direct to the preliminary page of the assessment. The assessment pages are divided into four sections: rehabilitation goal, focus area, pain description, and injury history. These pages attempt to mimic the way medical professionals like physical therapists initially assess their patients.

The application asks the user first the primary goal of the rehabilitation. The user is given three (3) options which are validated by the partnered physical therapists. The “Alleviate Pain” option is selected if the user only plans to reduce or eliminate pain or discomfort in a specific muscle of choice. The “Improve Mobility” option is selected if the user plans to improve the range of motion and flexibility of a particular muscle of the body. Lastly, the “Strengthen Muscle” option builds strength and endurance on a particular muscle.



After stating the rehabilitation goal, you are asked to specify as to where the muscle in question is located in the body: the upper body, the lower body, or the core muscles. Once the user has chosen the general area of the muscle, the user is then asked to select the specific muscle that is to undergo rehabilitation. The upper body muscles include the ones from the shoulder and arm area such as deltoids, biceps, triceps, and cervical muscles. Abdominals, chest, lower back, and multifidus are considered as part of the core muscles. The lower body muscles include the ones from the hip and leg area such as the obliques, quadriceps, hamstrings, gluteals, and calf.

Since the user has specified the muscle in question, the user is now in the pain description section of the assessment process. The user is requested to undergo a process called range of motion (ROM) where a person is assessed which is used to measure the extent of movement around a joint. This evaluation helps identify joint mobility limitations caused by factors like injuries or conditions affecting muscles, tendons, and ligaments, and it's used to check the effectiveness of treatment interventions. The user is given the option to undergo this process (which was discussed in the next section) or skip to the manual feedback of the pain level from the user (Pain Level page).

After undergoing the ROM process or skipping it, the user is requested to adjust the pain level based on the user's opinion. The application provides a slider from 0 to 10 with help text containing description of each level if it aligns with the current pain the user is experiencing. It follows a color coding where low pain is assigned with the green color, moderate pain with yellow color, and severe pain with the red color.



After specifying and adjusting the pain level, the user is asked to describe the type of pain being experienced. This is a particular feature that the partnered therapists have requested to be added besides scaling the pain, together with the next description of pain. The application offers six (6) varied types of pain that a person may experience: sharp/stabbing, dull/aching, burning, throbbing, stiffness, and numbness/tingling. As common people may not have the medical knowledge to understand each type of pain, each option has brief descriptions explaining the pain in simpler terms.

Once the user has specified the type of pain, the user is requested of the period of time as to since when the user has started experiencing the pain. There are four (4) options provided on the page: less than 48 hours ago, within the past 3 weeks, within the past 6 weeks, and within the past 3 months. One of the significances of this feature is filter out users who have experienced the pain for less than 2 days. The partnered therapists have specified that these people should not be given any exercises yet and must focus on being given treatments first.

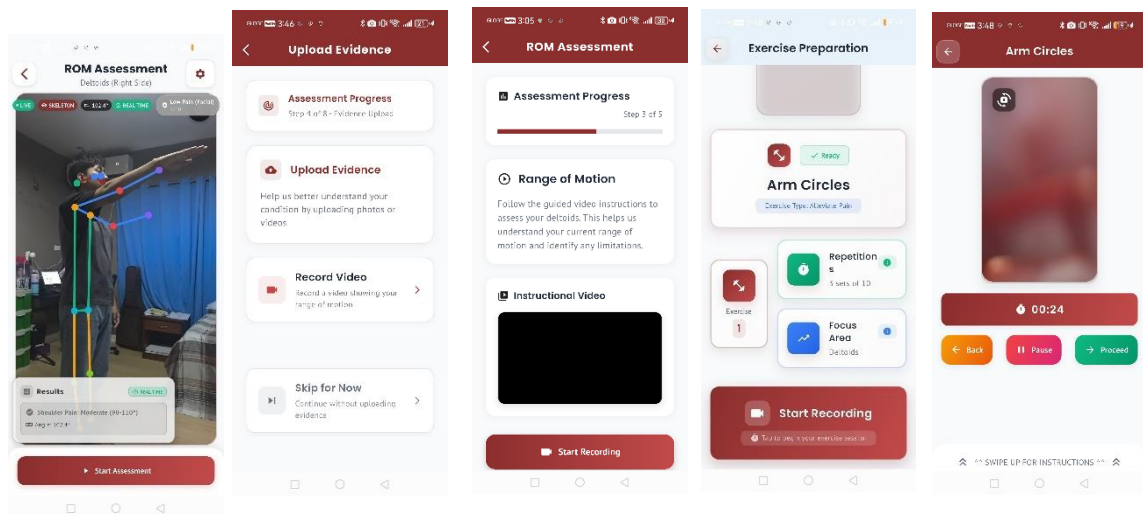
Once the user has described the pain experienced, the user is asked if any other muscles of the body have experienced previous injuries or has a condition that may hinder mobility on that particular muscle. If the user selected “Yes”, the user is displayed with the list of muscles and may opt to select one or more muscles with pain level indication to the selected muscles. If the user has indicated severe pain (8-10) in a particular muscle, all exercises where that muscle is majorly involved are filtered out from the list of exercises to be potentially recommended to the user as part of the rehabilitation plan.



Finally, after the assessment process, the user is provided with a summary of the selected choices throughout the overall process. Options selected by the user such as the rehabilitation goal, selected muscle, pain level, pain type, pain duration, and injury history are to be confirmed first by the user if all information is correct, before proceeding to the generation of the rehabilitation plan in the Generate Plan page.

**Figure 4.2.3**

*Application: Camera Interface*



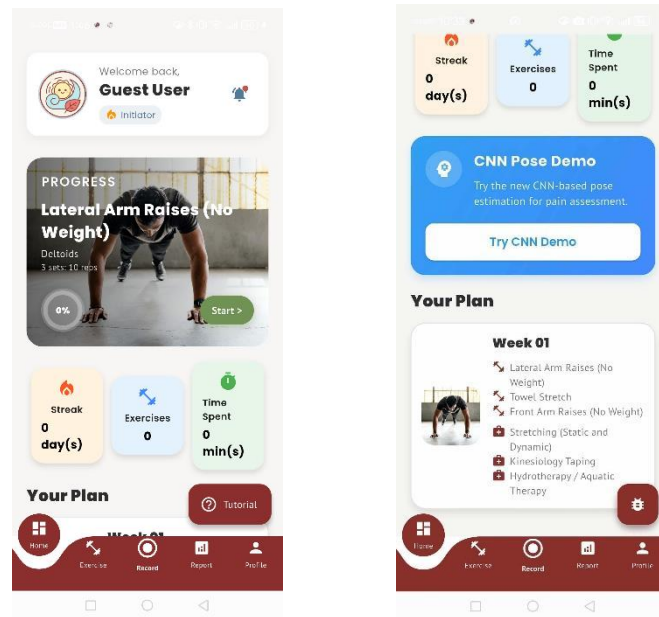
The Camera Interface serves as the primary feature of the mobile application, acting as the central tool for assessing and analyzing user data. It is designed to offer two key functionalities aimed at predicting relevant information when focused on a specific muscle or area of the body. The interface is divided into two main components. The first component is pain recognition, which utilizes the camera to detect and analyze the user's facial expressions. By interpreting subtle facial cues, the application can estimate the



user's pain level with high accuracy. The second component focuses on body movement detection and tracking, monitoring the user's physical motions to assess mobility, posture, and muscle engagement. By integrating these two functionalities, the Camera Interface can provide a comprehensive evaluation of the user's condition. This enables the application to deliver personalized recommendations for pain management, targeted exercises, and other therapeutic interventions, ensuring that the user receives the most appropriate and effective treatment.

**Figure 4.2.4**

*Application: Dashboard Page*



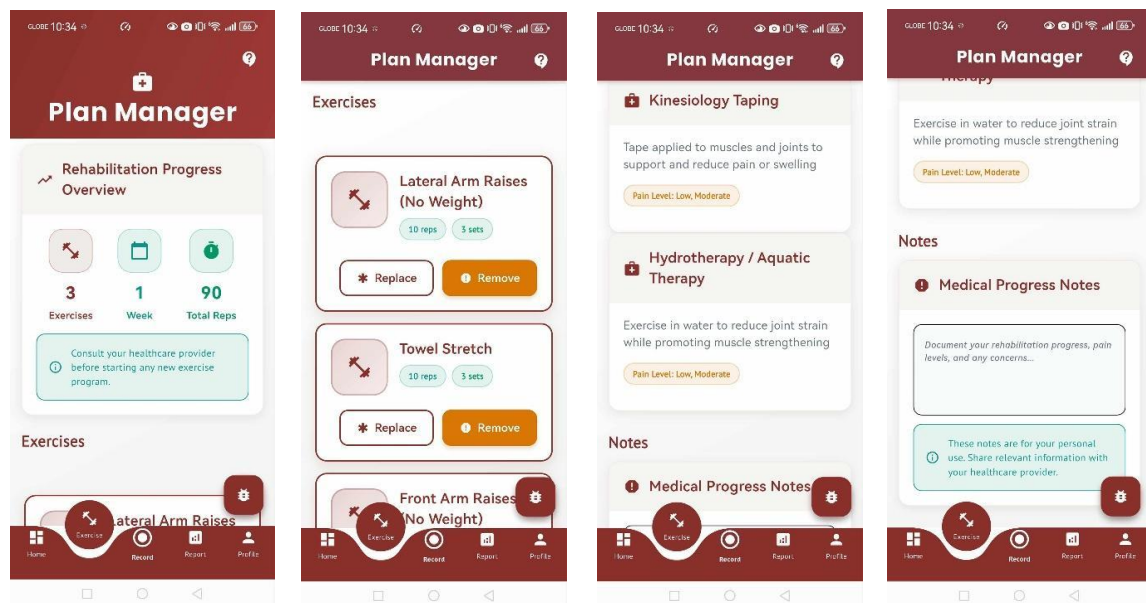
The Dashboard page serves as a central hub designed to help users monitor and track their progress over time. It provides a clear and intuitive overview of their activities through several interactive icons and indicators. Key tracking elements include Streaks,



Exercises, and Time Spent. The Streak icon displays the number of consecutive days the user has completed their exercises, encouraged consistency and fostered motivation. The Exercises icon shows the total number of exercises assigned and completed, helping users stay aware of their daily and overall goals. Meanwhile, the Time Spent icon records the duration of time the user has dedicated to performing their exercises, providing insight into their commitment and engagement. Additionally, the dashboard presents a personalized exercise and treatment plan generated by the application. This plan outlines the specific exercises and therapeutic activities the user should follow, ensuring that their program is tailored to their individual needs. By consolidating progress tracking and actionable guidance in a single interface, the dashboard empowers users to stay organized, motivated, and on track toward achieving their health and rehabilitation goals.

**Figure 4.2.5**

*Application: Plan Manager Page*





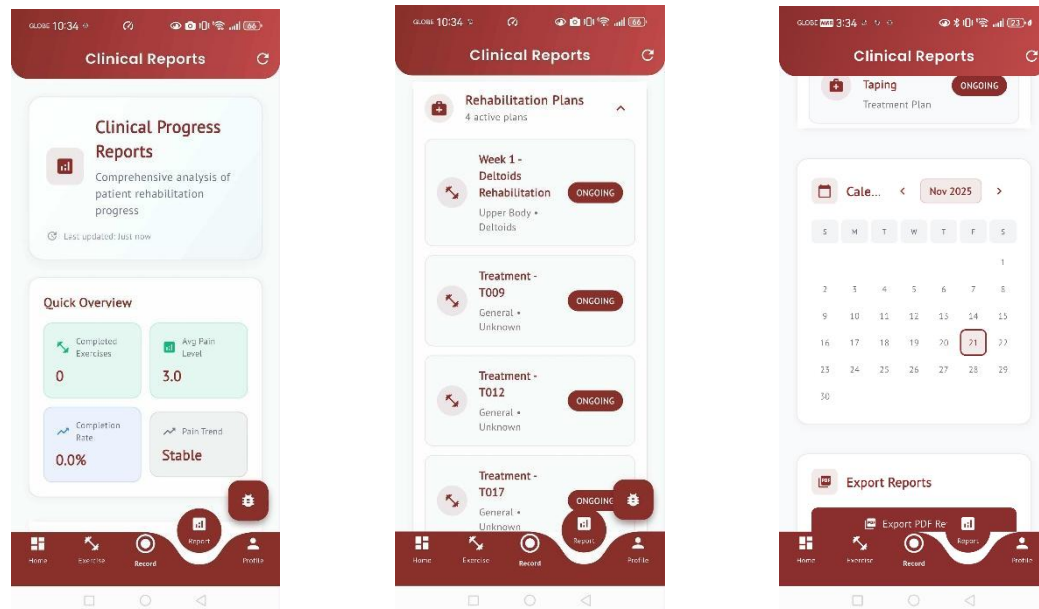
The Plan Manager page is designed to provide users with a comprehensive overview of their rehabilitation journey. Its primary purpose is to allow users to monitor their overall progress while clearly displaying the exercises and activities they are required to complete as part of their personalized rehabilitation plan. By consolidating this information in one place, the page helps users stay organized, adhere to their regimen, and maintain a clear understanding of their ongoing therapy.

In addition to tracking exercises, the Plan Manager includes a Notes feature, enabling users to record observations, comments, or any relevant health information. These notes can be shared with healthcare providers, facilitating better communication and ensuring that medical professionals have up-to-date insights into the user's progress and condition. By combining progress tracking with personalized documentation, the Plan Manager supports both the user's adherence to their rehabilitation program and effective collaboration with healthcare professionals.



**Figure 4.2.7**

*Application: Report Page*



The Report Interface offers users a detailed and comprehensive analysis of their rehabilitation progress, consolidating key data to provide a clear picture of their overall performance and improvement. This interface is designed to help users understand their progress over time and make informed decisions about their ongoing rehabilitation program.

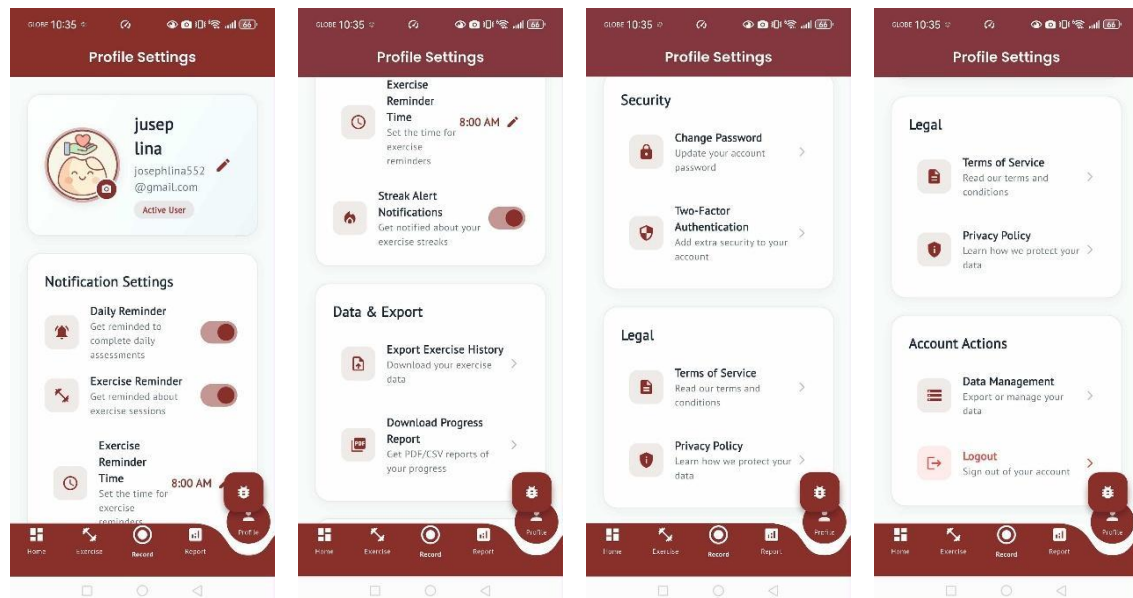
A central feature of the Report Interface is the “Export PDF” function, which allows users to generate and download a complete progress report. This report includes results from the initial assessment as well as a detailed history of completed exercises and other relevant rehabilitation data. By providing this functionality, the interface ensures that users have easy access to their personal health information, making it possible to



monitor progress, identify trends, and evaluate the effectiveness of their rehabilitation program. Moreover, the ability to export and share reports with healthcare providers enhances communication and collaboration, allowing medical professionals to make informed recommendations and adjustments to the user's treatment plan. Overall, the Report Interface serves as a powerful tool for tracking, analyzing, and sharing rehabilitation data, supporting both self-monitoring and professional oversight.

**Figure 4.2.8**

*Application: Profile Page*



The Profile Page serves as the central hub for users to manage and personalize their account settings. Within this interface, users can customize various aspects of their profile, including personal information, notification preferences, and security settings, ensuring that their experience is both tailored and secure.



Additionally, the Profile Page includes links to the application's Terms of Service and Privacy Policy, ensuring that users are informed about their rights, data usage, and the application's policies. By combining customization, data management, and transparency, the Profile Interface empowers users to take control of their experience, monitor their progress, and maintain secure and informed use of the application.

## 4.3 PROJECT DEVELOPMENT

The development of PocketPT followed the structured and iterative principles of the Agile Software Development Life Cycle (SDLC), allowing the proponents to systematically manage complexity and adapt to emerging challenges throughout the project's progression. The workflow was organized into multiple sprints, each dedicated to the development of core functional components such as the human pose estimation module, the facial pain recognition system, and the user interface. During the initial planning and analysis phase, the researchers identified the system requirements, refined the intended functionality, and established the key features necessary to support personalized rehabilitation. This phase also involved aligning technological capabilities with clinical expectations to ensure that the system would eventually produce relevant and credible outputs.

Despite effective planning, the development team encountered several technical constraints. One challenge was the insufficient processing capability of the initial development device, which hindered the training and testing of computationally intensive machine-learning models. This issue was addressed by acquiring a higher-performance



device from within the research team, ensuring that training pipelines for the CNN–LSTM pose estimation model and the CNN-based facial expression classifier could be executed without compromising accuracy or stability. Another obstacle arose from Flutter’s outdated support for TensorFlow Lite dependencies, which restricted model integration and compatibility. This limitation was resolved by transitioning to PyTorch, which provided more consistent support for mobile deployment and ensured smoother integration with the application’s backend. These adjustments highlight the adaptability and problem-solving rigor embedded in the Agile process.

Following the resolution of these challenges, the project advanced into the design and implementation stages, with a strong emphasis on creating a responsive and user-centered interface optimized for Android devices. The system architecture was developed in parallel with the integration of machine-learning pipelines, enabling iterative refinement through continuous testing at the end of each sprint. This approach allowed the team to validate model outputs, improve interface responsiveness, refine AROM measurement features, and ensure seamless interaction between frontend elements and backend processing. Once internal testing confirmed system stability, accuracy, and functional coherence, the proponents conducted structured demonstrations and evaluative sessions to verify the application’s performance in real-world scenarios. These evaluations ensured that PocketPT met both its technical expectations and its user-centered goals, ultimately resulting in a functional, reliable, and clinically aligned rehabilitation assistant.



## 4.4 DEVELOPMENT PLANNING

### 4.4.1 SOFTWARE SPECIFICATIONS

The software used for the building of this project was mentioned as follow:

1. **Visual Studio Code.** A versatile and lightweight source-code editor developed by Microsoft, was utilized as a complementary development environment throughout the creation of the PocketPT application. It served as an essential platform for writing, organizing, and managing the project's source code, particularly during the integration and testing of backend services, machine learning models, and Flutter components. Visual Studio Code (VS Code) provided extensive support for multiple programming languages and frameworks, such as Python, Dart, and JavaScript, through its vast library of extensions, making it highly adaptable to the project's multi-technology development stack.
2. **Android Studio.** It was employed as the primary integrated development environment (IDE) for designing and developing the Android-based PocketPT application. This platform offered a comprehensive suite of tools for writing, debugging, and testing code, as well as emulating the application across multiple Android devices and operating system versions. Android Studio was particularly essential in evaluating the performance, design consistency, and



overall functionality of PocketPT, especially after integrating advanced AI-driven features such as human pose estimation and facial pain recognition. Its robust simulator allowed the researchers to observe how the application responds to real-world usage scenarios, helping identify potential issues, optimize layouts, and ensure smooth navigation throughout the application.

3. **Flutter.** An open-source UI software development toolkit developed by Google, was used to build PocketPT's front-end interface. Its widget-based architecture allowed for the creation of a visually appealing, smooth, and highly responsive user interface that adapts seamlessly to various screen sizes and resolutions. Flutter enabled the implementation of interactive elements, dynamic navigation components, and subtle animations that enhance user engagement and usability. Moreover, Flutter's cross-platform capabilities offer future opportunities for expanding PocketPT to iOS devices without complete redevelopment, increasing accessibility and reach. By integrating Flutter with other technologies like Firebase and Hive, the researchers ensured that the interface could effectively communicate with the backend and local data storage, providing users with real-time feedback and progress tracking.

- a. **Hive.** A lightweight and high-performance key-value database



designed for Flutter applications was used as the local storage solution for PocketPT. Hive allowed the app to store critical data locally, enabling offline access to rehabilitation exercises, assessment histories, and user progress. The system could efficiently synchronize this local data with Firebase once an internet connection became available, ensuring that users' progress and records remain consistent across sessions. This dual storage approach, combining Hive for offline access and Firebase for online synchronization, enhanced the reliability, resilience, and usability of the application.

4. **Firebase.** A cloud-based platform developed by Google, was implemented as the online backend service for PocketPT. It provided essential functionalities such as real-time database storage, user authentication, and cloud synchronization. Firebase enabled secure and reliable management of user data, including assessment results, rehabilitation progress logs, and personalized plans. Its compatibility with Flutter facilitated seamless integration between the mobile front-end and cloud services, allowing data to be accessed and updated in real-time. By leveraging Firebase, the system could maintain continuity across devices, offer cloud backups, and provide scalability for potential future expansion of PocketPT to a broader user base.



## 4.4.2 LANGUAGED USED

This application was developed using a combination of programming languages catered to both the frontend and backend needs of the system. The languages used for building this project were mentioned as follows:

1. **Python.** A high-level and versatile programming language was employed for building and training the AI models responsible for human pose estimation and facial pain recognition. Its simplicity, readability, and extensive libraries, including PyTorch, made it an ideal choice for rapid development and experimental testing of AI architecture. Python enabled efficient training, validation, and optimization of models, ensuring that the AI components could deliver accurate, real-time performance within PocketPT.
  - a. **PyTorch.** A free and open-source machine learning framework developed by Meta AI, was utilized for building and training the deep learning models that form the core functionalities of PocketPT. PyTorch's dynamic computation graphs, flexible API, and support for GPU acceleration made it an ideal tool for developing AI modules, particularly for human pose estimation and facial pain recognition. The framework allowed the researchers to experiment with model architecture, optimize performance, and fine-tune parameters



efficiently. Once trained, the models were converted and embedded into the Android application, enabling real-time, AI-driven feedback during rehabilitation assessments. This integration ensured that users receive accurate monitoring and objective data analysis without requiring continuous manual oversight.

2. **Dart.** The primary programming language for the Flutter framework was used to develop PocketPT's user interface and application logic. Dart facilitated the creation of a dynamic, responsive, and interactive front-end that effectively displayed exercise guidance, user assessments, and progress tracking. Its seamless integration with Flutter ensured that the AI-driven features, such as pose estimation and pain recognition, could be incorporated smoothly, resulting in an intuitive and cohesive user experience. Dart's efficiency and scalability also allowed for easier maintenance and future expansion of the application.
3. **CSV.** These files were utilized to maintain structured datasets containing exercise instructions, treatment protocols, and rehabilitation routines. These datasets formed the basis for generating personalized exercise plans and AI-driven recommendations within PocketPT. Using CSV allowed the researchers to easily update and expand rehabilitation content without modifying the source code, offering a flexible and scalable approach to managing therapeutic datasets.



## 4.5 EVALUATION OF THE PROJECT

### 4.5.1 RESPONDENTS OF THE STUDY

This study engaged a diverse cohort of participants from the Cavite area to ensure the findings are both comprehensive and applicable to real-world rehabilitation scenarios, targeting a minimum of 50 respondents aged 18 years and above. The participant pool was strategically designed to include individuals with a spectrum of physical activity levels and needs; this encompassed physically active individuals, such as amateur athletes and fitness enthusiasts engaged in activities like sports, jogging, or gym workouts, as well as those currently experiencing musculoskeletal issues, including muscle strains, injuries, or chronic muscular discomfort requiring rehabilitation guidance. To critically validate the application's clinical relevance and safety, the study also incorporated the professional expertise of certified physical therapists practicing in diverse settings, including clinical environments and fitness centers. The sample was deliberately balanced by sex and comprised a minimum of 35 end-users (active individuals) and 15 experts, thereby ensuring the collected data captures a holistic view that integrates firsthand user experiences with authoritative clinical perspectives on the challenges and outcomes associated with physical activity and AI-assisted rehabilitation.



## 4.5.2 RESEARCH INSTRUMENTS AND TECHNIQUES

The researchers employed Google Forms as the primary research instrument for designing and distributing online survey questionnaires. This platform enabled the structured formulation of both quantitative and qualitative questions, ensuring a systematic approach to data gathering. Through its user-friendly interface and built-in analytics, Google Forms allowed for efficient monitoring, organization, and analysis of responses, making it suitable for academic research that required both numerical data and descriptive feedback.

In addition, the researchers collaborated with at least five licensed physical therapists who assisted in referring the survey to their patients and colleagues. This approach introduced a snowball sampling technique; wherein, initial participants were encouraged to share the survey with others who also fit the inclusion criteria. Combined with purposive sampling (targeting qualified respondents based on predefined characteristics) and convenience sampling (utilizing easily accessible online channels), this hybrid non-probability sampling method ensured a diverse yet relevant pool of participants.

## 4.5.3 DATA GATHERING

The process of data gathering served as a critical component of the study, ensuring that relevant and accurate information was collected to support the objectives and guide the development of the PocketPT application. The data



collection focused on obtaining both quantitative and qualitative feedback from respondents who met the study's inclusion criteria, individuals aged 18 years and above residing in Cavite, who were either physically active, experiencing muscular discomfort, or practicing physical therapy.

To reach the target population, the researchers disseminated the survey through Facebook and Messenger, leveraging these platforms' accessibility and widespread use among residents of Cavite. This approach facilitated the convenient and timely distribution of the survey, allowing participants to respond at their own pace using mobile devices or computers. The researchers also utilized private messages and group chats to directly share the survey link with potential respondents who met the study's inclusion criteria, individuals aged 18 years and above residing in Cavite, who were either physically active, experiencing muscular pain, or professionals such as physical therapists familiar with rehabilitation practices.

#### **4.5.4 QUESTIONNAIRES AND SURVEYS**

The researchers employed a questionnaire as the principal instrument for collecting data essential to the evaluation of the PocketPT application. The questionnaire was designed to obtain both quantitative and qualitative feedback from the target respondents, allowing for an assessment of the system's performance, usability, and overall user satisfaction. To ensure methodological rigor and alignment with international software evaluation standards, the



questionnaire was structured following the ISO/IEC 25010 quality model. This framework provided a standardized basis for measuring key software attributes such as functionality, reliability, usability, efficiency, maintainability, and portability, ensuring that the assessment captured both the technical and experiential dimensions of system performance.

To effectively quantify user perceptions and responses, the survey utilized a five-point Likert scale, with response options ranging from Strongly Disagree to Strongly Agree. This scale enabled the researchers to measure the degree of user agreement toward various statements concerning the system's design, effectiveness, and ease of use. The structured nature of the Likert-scale items allowed systematic data analysis, facilitating the identification of patterns and trends that reflect user experiences with the PocketPT application.

In addition to the structured questions, the survey also included open-ended items that invited respondents to provide detailed feedback and recommendations for system improvement. This combination of closed and open-ended questions enriched the dataset, enabling the researchers to capture both measurable indicators and nuanced perspectives from participants.

|                              |                 |                |              |                           |
|------------------------------|-----------------|----------------|--------------|---------------------------|
| <b>1</b>                     | <b>2</b>        | <b>3</b>       | <b>4</b>     | <b>5</b>                  |
| <b>STRONGLY<br/>DISAGREE</b> | <b>DISAGREE</b> | <b>NEUTRAL</b> | <b>AGREE</b> | <b>STRONGLY<br/>AGREE</b> |



## 4.4.5 INTERNET

The Internet played a vital role in the project's evaluation phase by serving as the primary medium for research, data collection, and participant engagement. Online academic databases and digital repositories provided extensive access to scholarly literature, enabling the researchers to ground the study in established methodologies and relevant empirical findings. This ensured that the evaluation of the application was theoretically supported and aligned with recognized standards such as ISO/IEC 25010. Furthermore, web-based platforms like Google Forms facilitated the creation, distribution, and management of structured questionnaires, streamlining the data collection process while minimizing human error and expediting data organization for analysis.

In addition, social media platforms such as Facebook and Messenger were used to recruit and engage participants efficiently, reaching physically active individuals, persons experiencing muscular pain, and licensed physical therapists within Cavite. This online approach allowed for wider accessibility and inclusivity, overcoming geographical constraints associated with traditional survey methods. Overall, the Internet's integration throughout the evaluation process significantly enhanced the study's efficiency, reliability, and data accuracy, ensuring a comprehensive and methodologically sound assessment of the PocketPT application.



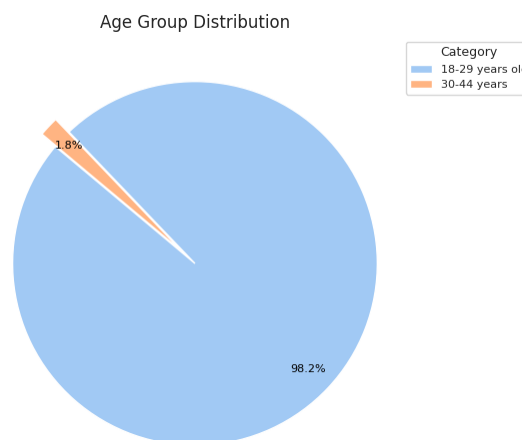
## CHAPTER V

### RESULTS AND DISCUSSION

This chapter presents the comprehensive findings from the survey questionnaire administered to evaluate the application. The results are organized according to the ISO 25010 Software Product Quality Standards framework, encompassing functional suitability, performance efficiency, usability, reliability, security, and portability. Subsequently, the discussion interprets these findings in relation to the research questions and objectives, comparing them with existing literature and identifying implications for AI-driven rehabilitation systems.

**Figure 5.0.1**

*Demographic Profile: Age Group*



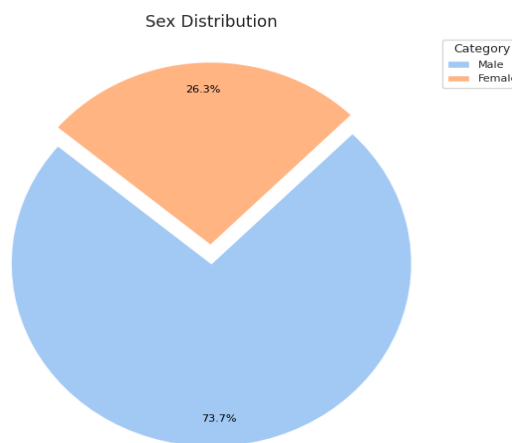
The overwhelming majority of participants (98.2%) fell within the 18-29 years of age demographic, representing young adults who are typically more technologically adept



and comfortable with mobile applications. This demographic profile is strongly reflected in the high usability scores and the nature of the feedback, which often focused on advanced features like AI pose detection and data visualization rather than fundamental accessibility. Their comfort with technology likely contributed to the overall positive reception, but it also means the sample may lack perspectives from older users who might struggle with navigation or require more intuitive guidance. One respondent (1.8%) belonged to the 30-44-year-old age group, providing a limited multi-generational perspective. While this single data point is insufficient to draw conclusions about older demographics, it does not contradict the generally positive feedback, suggesting the application's core functionality has broad appeal, though its optimal user base remains the digitally native youth.

**Figure 5.0.2**

*Demographic Profile: Sex*



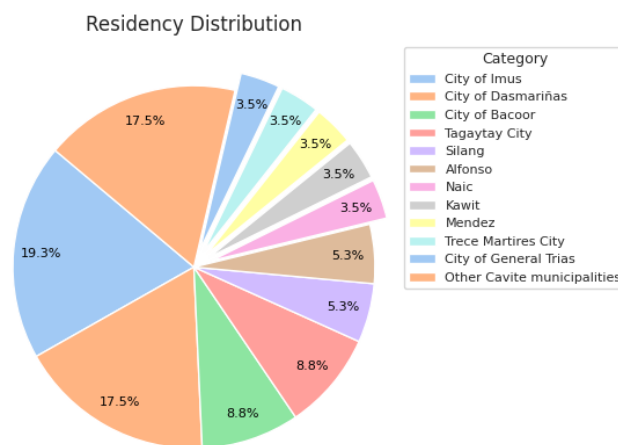
The sample exhibited a significant gender imbalance with a higher proportion of



male participants (73.7%) compared to female participants (26.3%). This skew may influence the generalizability of the results, particularly regarding the perceived effectiveness of features like pose detection and exercise recommendations, which could potentially vary based on physiological differences or common exercise types between genders. However, the quantitative ratings from female respondents were consistently high and did not markedly deviate from the overall averages, indicating that the application was well-received across the gender spectrum present in the sample. Future studies would benefit from a more balanced recruitment strategy to ensure the application's features are equitably effective and user-friendly for all genders.

**Figure 5.0.3**

*Demographic Profile: Residency*



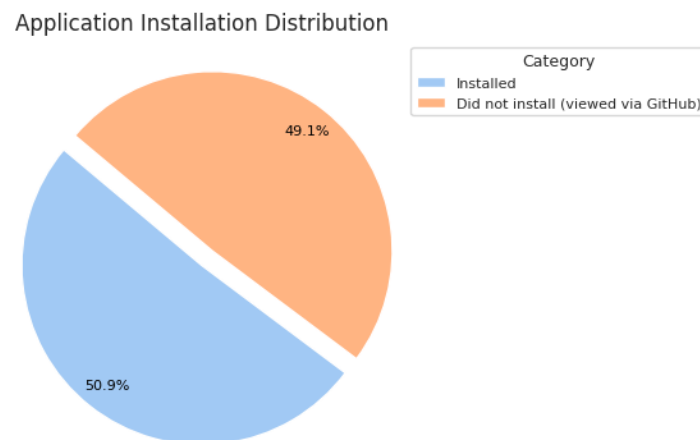
All respondents were residents of Cavite province, with the sample demonstrating broad geographic distribution across 12 municipalities and cities. The City of Imus represented the largest cluster (19.3%), followed by the City of Dasmariñas (17.5%) and



the City of Bacoor (8.8%). This wide distribution within the province suggests that the application's utility is not confined to a specific urban or suburban context within Cavite, enhancing the local relevance of the findings. Participation from areas like Tagaytay City, Silang, and more rural municipalities (e.g., General Emilio Aguinaldo, Mendez) indicates that the application can reach users across diverse community settings, which is crucial for a tool aimed at increasing accessibility to physical therapy resources.

**Figure 5.0.4**

*Demographic Profile: Application Installation*



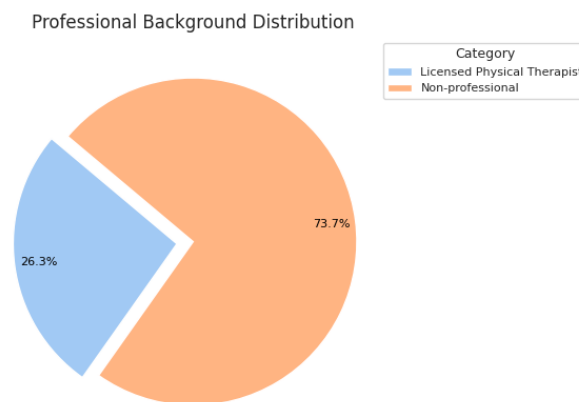
Application installation rates achieved a near-perfect balance, with 50.9% of participants opting to install the application on their devices while 49.1% accessed it via GitHub. This represents a significant improvement in installation accessibility compared to preliminary testing phases. The high rate of successful installations suggests that the provided APK file and installation instructions were clear and that the application is compatible with a wide range of common Android devices. Furthermore, the robust



performance of the web version for nearly half the users is a major strength, as it lowers the barrier to entry for those hesitant to install new apps, with several users explicitly praising the web version for its functionality and ease of access.

**Figure 5.0.5**

*Demographic Profile: Professional Background*



Notably, fifteen respondents (26.3%) identified themselves as licensed physical therapists or rehabilitation professionals from Cavite, providing valuable professional perspectives alongside the predominant end-user evaluation from non-professionals (73.7%). The inclusion of licensed physical therapists, though representing a modest proportion of the overall sample, offers important clinical validation that complements the user experience assessments. Their feedback often touched on different aspects, such as the clinical alignment of exercises, the importance of data privacy, and the need for professional oversight, as one professional noted, "patients may rely on self-diagnosis." This professional segment lends credibility to the application's therapeutic foundation,



while their suggestions point toward future developments like a dedicated "Therapist" portal for manual exercise management and integration with formal healthcare pathways.

**Figure 5.0.6**

*Physical Activity Engagement Among Respondents*

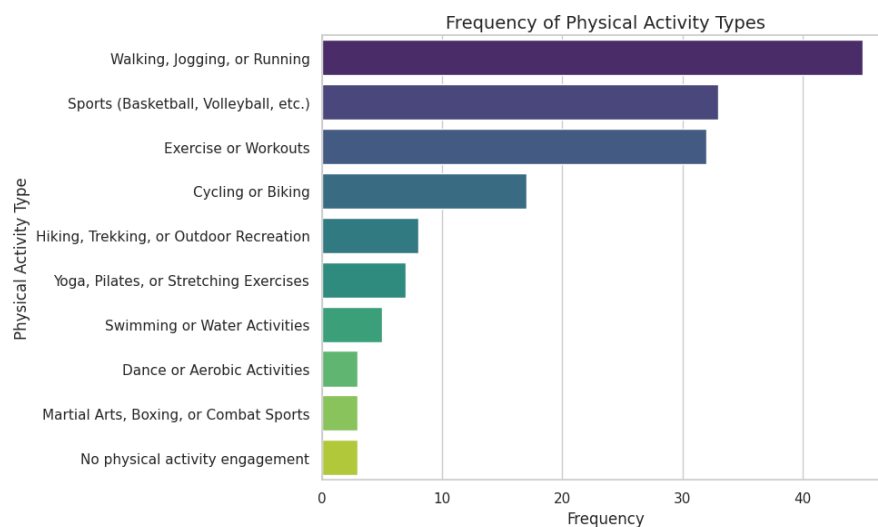


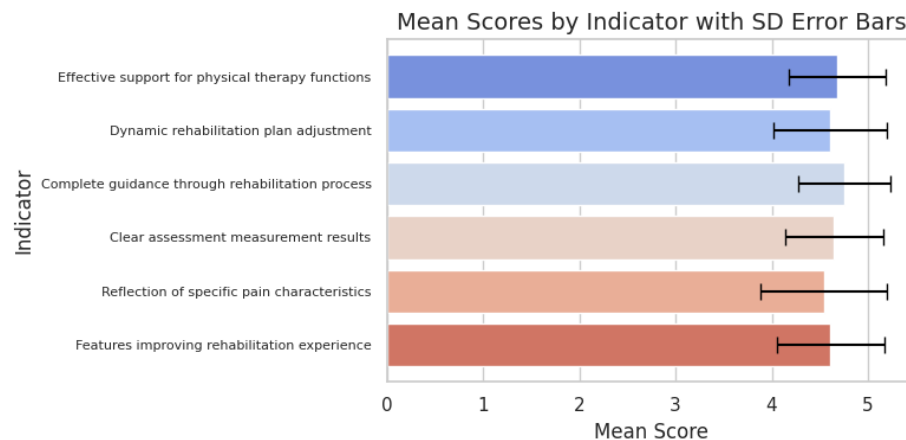
Figure 5.0.6 illustrates the physical activities in which respondents regularly engaged, providing context for their familiarity with exercise and rehabilitation concepts. The vast majority of respondents (77.8%) engaged in walking, jogging, or running, demonstrating near-universal participation in basic cardiovascular activities. More than half of the participants engaged in structured exercise or workouts (55.6%) and sports activities (55.6%), indicating moderate to high physical activity levels across the sample. A notable proportion (27.8%) participated in cycling or biking activities, while smaller percentages engaged in specialized activities such as hiking and outdoor recreation (13.9%), yoga and stretching exercises (11.1%), and various other physical pursuits



including dance, swimming, and martial arts (5.6% each). Only three respondents (8.3%) reported no regular physical activity engagement, suggesting that the overwhelming majority of the sample possessed personal experience with exercise routines and physical movement patterns relevant to rehabilitation contexts.

**Figure 5.0.7**

*Functional Suitability Assessment*



Functional suitability evaluates the degree to which the PocketPT application provides functions that meet stated and implied needs under specified conditions. This dimension encompasses six key indicators assessed on a 5-point Likert scale. According to the statistical analysis, the functional suitability dimension achieved an exceptional overall mean score of 4.63 (SD = 0.58), indicating strong agreement among respondents regarding the application's ability to fulfill its intended rehabilitation support functions. Notably, indicator 3 concerning complete guidance through the rehabilitation process received the highest rating (M = 4.77, SD = 0.49), approaching near-unanimous strong



agreement and suggesting that the application successfully provides comprehensive instructions, recording capabilities, and result-saving functionalities throughout the rehabilitation journey.

This high overall score directly addresses the core inquiry of how integrating user-specific pain characteristics with motion-based pain recognition through machine learning algorithms can enhance personalization and accuracy of rehabilitation recommendations. The strong functional suitability ratings validate the system's capacity to effectively support comprehensive physical therapy functions, integrate all operational modules into a cohesive platform, and deliver adaptive feedback loops for personalized rehabilitation guidance. This finding contrasts notably with the limited functionality reported in commercial applications such as PT Pal Pro (Gravenstine et al., 2013), which typically offer basic exercise libraries without adaptive, AI-driven personalization.

The highest mean score for "Complete guidance through the rehabilitation process" ( $M = 4.77$ ) particularly validates the successful implementation of the algorithmic framework for generating personalized rehabilitation plans and the seamless integration of all functional modules. This outcome suggests that PocketPT provides the structured, automated rehabilitation workflow intended to address the accessibility gap identified in the literature, where continuous human supervision remains a significant limitation in traditional telerehabilitation approaches (LUNA, 2023).

The application's effective support for physical therapy functions garnered a strong rating ( $M = 4.66$ ,  $SD = 0.53$ ), reflecting positive user perception of the integrated tools



including exercise recording, pose detection, assessment capabilities, progress tracking, and reporting functions. The features improving rehabilitation experience ( $M = 4.63$ ,  $SD = 0.55$ ) and clear assessment measurement results ( $M = 4.63$ ,  $SD = 0.55$ ) received equally strong evaluations. The dynamic adjustment of rehabilitation plans based on performance feedback ( $M = 4.60$ ,  $SD = 0.65$ ) and the reflection of specific pain characteristics in recommendations ( $M = 4.51$ ,  $SD = 0.70$ ) both received strong positive evaluations, demonstrating the effectiveness of the rule-based algorithmic approach in personalizing therapy recommendations.

These strong ratings for integrated features and dynamic adjustment validate the core AI-driven, rule-based design philosophy underlying the system's personalization capabilities. This approach represents significant advancement beyond simple motion analysis platforms focused primarily on posture correction, such as Exer Health (Exer AI, 2023). The positive evaluation of dynamic plan adjustment suggests that the algorithmic framework effectively processes multi-modal data, combining pose estimation metrics with facial pain recognition for meaningful personalization, aligning with the integrated monitoring approach proposed by Xu et al. (2023) in their framework for smart rehabilitation systems.

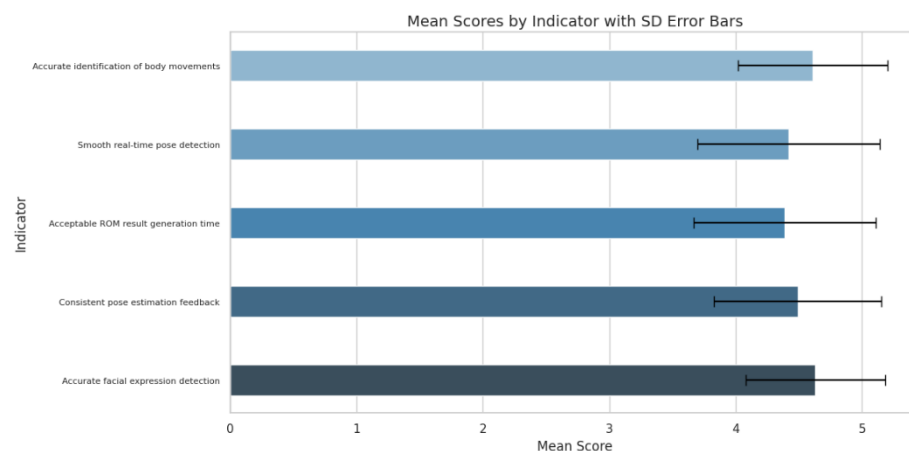
Interestingly, the slightly lower rating for reflection of specific pain characteristics ( $M = 4.51$ ,  $SD = 0.70$ ), while still indicating strong agreement, exhibited the highest standard deviation (0.70) among all functional suitability indicators. This increased variability suggests diverse user experiences in perceiving how well the system



incorporates their individual pain profiles into rehabilitation recommendations. The higher standard deviation may reflect differences in pain conditions among respondents, varying levels of understanding regarding how the facial pain recognition system influences recommendations, or individual differences in pain expression that affect the system's ability to accurately detect and incorporate pain characteristics. Based on the statistician, these results directly address the first and the fifth objective regarding the integration of pain characteristics and datasets as well as the rule-based decision tree for personalized plans respectively, confirming that the application successfully delivers adaptive rehabilitation plans and integrates all functional modules into a unified system.

**Figure 5.0.8**

### *Performance Efficiency Evaluation*



Performance efficiency examines the application's responsiveness relative to the computational resources utilized under specified conditions. This dimension evaluated five critical aspects of the AI-driven pose estimation and facial pain recognition systems.



The statistical analysis revealed that the performance efficiency dimension yielded an overall mean score of 4.49 (SD = 0.66), indicating strong agreement that the application demonstrates satisfactory computational performance and response times. The highest mean score of 4.63 was achieved for accurate detection of facial expressions during pain or discomfort, followed closely by accurate identification of body movements during assessment (M = 4.60, SD = 0.60). These findings validate the effectiveness of the CNN-LSTM architecture for pose estimation and the CNN-based facial expression recognition system in real-world usage scenarios.

The overall satisfactory performance efficiency directly addresses the technical feasibility of integrating two computationally intensive AI models into a mobile application environment without prohibitive latency. This achievement is particularly significant given the high computational complexity commonly associated with multi-view human action recognition systems, as documented by Malik et al. (2023) in their comprehensive analysis of cascading pose features with CNN-LSTM architectures for multi-view human action recognition. The ability to maintain satisfactory real-time performance while simultaneously executing both AI models demonstrates successful optimization of the mobile deployment strategy.

The highest rating for facial expression detection accuracy (M = 4.63) represents a noteworthy finding, as this indicates that users perceived the pain recognition system as highly reliable in identifying discomfort expressions. This strong performance validates the CNN-based facial expression recognition model's effectiveness in real-world



conditions with variable lighting, camera angles, and diverse facial structures across the user population.

Smooth real-time pose detection ( $M=4.37$ ,  $SD=0.73$ ) and ROM result generation time ( $M=4.34$ ,  $SD=0.73$ ) received slightly lower but still positive ratings, suggesting that while the application generally performs well, occasional latency or processing delays may occur. The standard deviations ranging from 0.55 to 0.73 indicate moderate variance in performance perceptions, with the higher standard deviations (0.73) for real-time smoothness and ROM generation suggesting more variable user experiences in these aspects compared to detection accuracy metrics.

Consistent pose estimation feedback across different lighting and body positions achieved a mean score of 4.49 ( $SD=0.70$ ), indicating generally positive but not exceptional performance. This mid-range rating among performance indicators suggests that while the system maintains reasonable consistency across varying conditions, environmental factors such as lighting quality and user positioning do noticeably influence performance, aligning with known limitations of RGB camera-based computer vision systems.

The high scores for "Accurate identification of body movements" and "Accurate facial expression detection" confirm the effective operation of both AI models in real-world, uncontrolled usage scenarios. These results validate the architectural design choices, which align with established best practices for deep learning applications in human pose estimation (Malik et al., 2023) and facial expression recognition for pain



detection (Zhou et al., 2021). As reported by the statistician, the strong performance of the facial expression recognition component substantiates the third research objective by confirming its effectiveness in monitoring pain indicators during exercise execution. The successful real-world performance indicates that the models generalize well beyond controlled laboratory conditions to diverse user environments, lighting conditions, and camera qualities.

The modal response of 5 ("Strongly Agree") for several performance indicators demonstrates that a substantial proportion of users found the system's computational efficiency highly satisfactory. However, the mean scores ranging from 4.34 to 4.63 suggest that while the CNN-LSTM model successfully achieves real-time pose estimation and the facial pain recognition operates effectively, there exists potential for optimization to elevate performance from "good" to "excellent" across all user experiences.

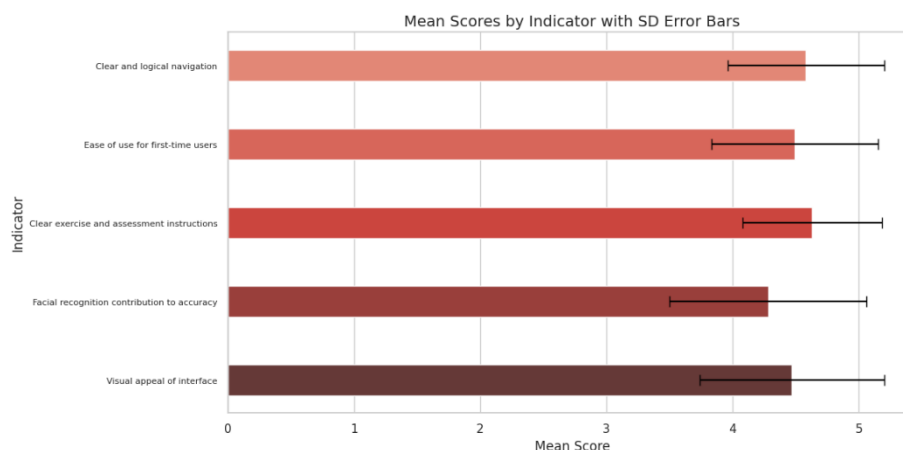
The slightly lower scores for real-time smoothness and ROM generation time acknowledge the computational trade-off inherent in delivering real-time, corrective feedback which is a necessary capability for dynamic therapy adjustment emphasized by researchers such as Pereira et al. (2023) in their exploration of machine learning applications for monitoring physical therapy exercises at home. This trade-off reflects a deliberate design decision prioritizing accuracy and reliability over marginal improvements in processing speed, ensuring that therapeutic guidance remains clinically sound even if computational efficiency is not maximized. As interpreted by the statistician, these findings demonstrate that the CNN-LSTM pose estimation model



performs reliably in real time, thereby supporting the fourth research objective, which pertains to achieving accurate pose estimation, as well as seventh research objective, which focuses on the implementation of an adaptive feedback loop.

**Figure 5.0.9**

### *Usability Assessment*



The usability dimension achieved a strong overall mean score of 4.50 (SD = 0.66), with a modal response of 5 ("Strongly Agree"), indicating strong user satisfaction with the application's interface design and user experience. Two indicators achieved the highest mean scores: clarity of exercise and assessment instructions (M = 4.63, SD = 0.55) and clarity and logic of navigation between pages and features (M = 4.60, SD = 0.60). These findings demonstrate that the application successfully implements intuitive user interface design principles, facilitating seamless interaction with various functional modules. Drawing from the statistician's evaluation, this confirms that PocketPT provides an intuitive interface, reducing barriers to adoption and supporting the sixth research



objective regarding the integration of modules into a user-friendly system.

The high overall usability score represents a crucial validation of the system's successful integration of all functional modules into a unified, user-friendly platform. More significantly, this finding directly counters one of the primary barriers to telerehabilitation adoption in the Philippines identified by Ico et al. (2021): technical difficulties and low ease of use among patients attempting to engage with digital rehabilitation platforms. By achieving strong usability ratings across all indicators, PocketPT demonstrates that sophisticated AI-driven rehabilitation systems can be designed with interfaces that are accessible to non-technical users, thereby addressing a critical implementation challenge in resource-limited healthcare settings.

First-time user ease of use ( $M = 4.49$ ,  $SD = 0.66$ ) and visual appeal of the interface ( $M = 4.49$ ,  $SD = 0.74$ ) both received strong positive evaluations, suggesting that the application provides an aesthetically pleasing and learnable interface that minimizes the need for extensive tutorials or training. The modal response of 5 for both indicators demonstrates that a substantial proportion of users found the system immediately intuitive and visually engaging. This aligns with best practices in mobile health application design, where user retention often correlates with intuitive first-use experiences.

The top scores for clarity of exercise and assessment instructions and navigation reinforce the success of the agile development methodology employed during the system's creation, particularly in designing an intuitive Flutter-based front-end interface. This achievement ensures that the complex outputs of the algorithmic decision-making



framework, including personalized exercise recommendations, ROM assessments, and pain-adaptive therapy adjustments, are communicated effectively and comprehensibly to end-users without requiring technical expertise or extensive training.

The strong ratings for first-time ease of use and visual appeal further suggest that the application successfully delivers an intuitive rehabilitation experience, positioning PocketPT as a highly accessible and personalized alternative to traditional, instruction-manual-based rehabilitation methods. This represents a significant advancement over conventional approaches exemplified by platforms such as LUNA, Exer AI, and PT Pal Pro, which often require either direct therapist supervision or extensive user familiarity with exercise protocols.

The facial recognition feature's contribution to assessment accuracy received the lowest usability rating ( $M = 4.31$ ,  $SD = 0.76$ ), though still reflecting agreement among users. Notably, this indicator also exhibited the highest standard deviation ( $0.76$ ) among all usability metrics, suggesting considerable variability in user perceptions of this feature's value. This relatively lower score and higher variability may indicate that while users recognize the technical presence of facial pain recognition, its practical utility or noticeable impact on their rehabilitation experience may not be immediately apparent. This finding suggests potential opportunities to enhance user awareness of how facial expression analysis influences rehabilitation plan adjustments.

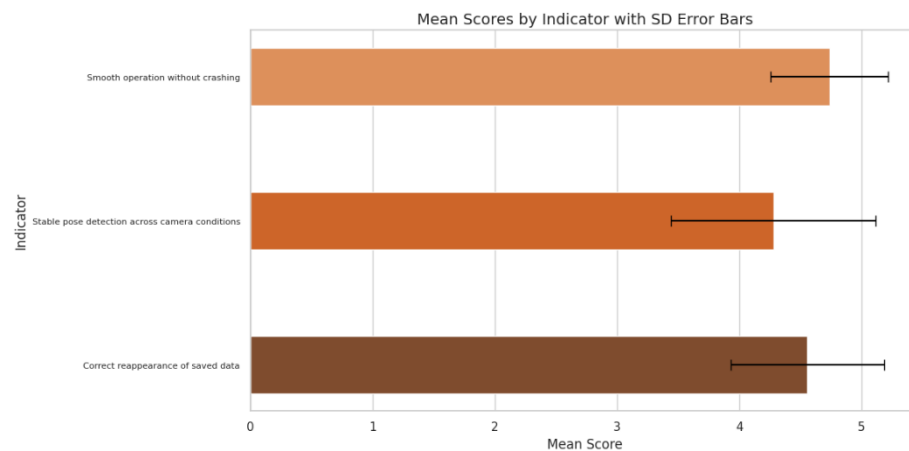
The slightly lower rating and higher variability for the perceived contribution of facial recognition reveal a gap between the model's technical accuracy, evidenced by the



98% weighted accuracy reported in the classification metrics, and the user's conscious perception of its value in the rehabilitation process. This disparity indicates that future iterations should prioritize providing clearer, more explicit in-application feedback demonstrating how the pain detection system, employing methodologies similar to those validated by Zhou et al. (2021) for pain recognition based on facial expressions, actively influences plan adjustments and adaptive therapy modifications. Enhanced transparency in communicating the facial expression recognition system's role could improve user appreciation of this safety-critical feature. As indicated in the statistician's statistical findings, the high ratings regarding the clear instructions for exercises and visual appeal of the interface further underscore the system's accessibility and design quality, both of which are essential factors in promoting user engagement and sustained adherence.

**Figure 5.0.10**

### *Reliability Evaluation*



Reliability assesses the application's ability to perform specified functions under



stated conditions for a defined period, focusing on system stability and data persistence. The reliability dimension achieved an overall mean score of 4.52 ( $SD = 0.67$ ), demonstrating strong user confidence in the application's stability and data integrity. The application's smooth operation without crashing or freezing received the highest rating ( $M = 4.74$ ,  $SD = 0.51$ ), indicating robust software architecture and effective error handling mechanisms. This finding is particularly significant for healthcare applications where system crashes could interrupt therapeutic sessions and undermine user trust.

The high overall reliability score confirms the robustness of the unified mobile system and its supporting back-end infrastructure components, particularly the Firebase cloud services utilized for data synchronization and storage. This level of reliability represents a critical factor for clinical feasibility and validation through performance testing and user feedback, as emphasized in the systematic review protocol conducted by Gonzaga et al. (2022), which underscores the necessity of establishing evidence regarding the reliability of physical therapy mobile applications before widespread clinical adoption. The strong reliability ratings suggest that PocketPT meets the threshold standards for deployment in real-world rehabilitation scenarios where consistent performance is essential for therapeutic effectiveness and patient safety.

Data persistence, measured through the correct reappearance of saved exercises and assessments upon application reopening, achieved a mean score of 4.57 ( $SD = 0.65$ ). This demonstrates effective implementation of cloud-synced records and local data storage, ensuring continuity of care and progress tracking across multiple user sessions.



The strong rating for data persistence is particularly important for rehabilitation applications where longitudinal tracking of progress over days, weeks, and months is essential for therapeutic decision-making and patient motivation.

The exceptionally high score for smooth operation without crashing ( $M = 4.74$ ) highlights the successful integration and optimization of the Flutter-based front-end interface with the Python-powered AI back-end components, ensuring that the computationally intensive AI processes, specifically the CNN-LSTM pose estimation and CNN facial pain recognition models, do not compromise system stability. This architectural achievement is particularly noteworthy for a medical-adjacent application where system instability would severely undermine user trust and therapeutic compliance, potentially leading to abandonment of the rehabilitation program. The low standard deviation (0.51) for this indicator demonstrates consistent experiences across users, suggesting that stability performance is reliably maintained across different device types, operating system versions, and usage patterns.

Pose detection stability across varying camera conditions ( $M = 4.26$ ,  $SD = 0.85$ ) received the lowest reliability rating, though still indicating agreement among users. The reference to "low visibility, high noise, blurry camera" conditions in the survey instrument suggests that environmental factors may occasionally impact pose estimation accuracy. Significantly, this indicator also exhibited the highest standard deviation (0.85) among all reliability metrics, indicating substantial variability in user experiences. This variability likely correlates with differences in device camera quality, testing environments, lighting



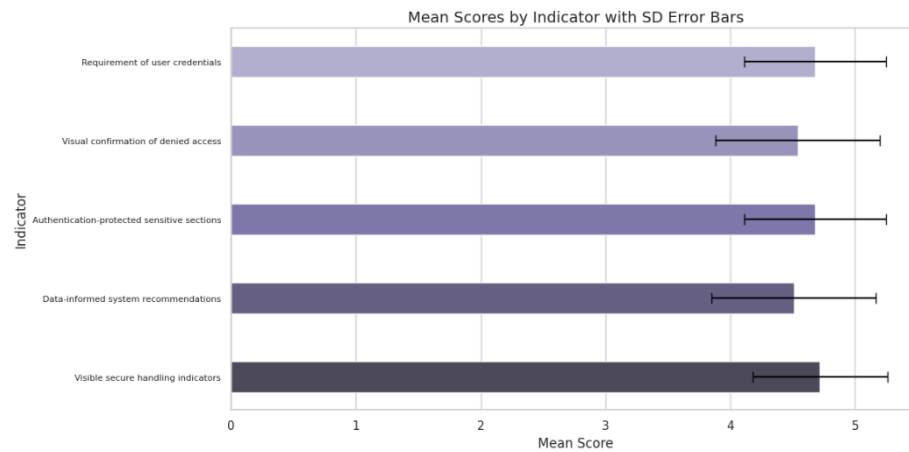
conditions, and user technical knowledge regarding optimal camera positioning. In reference to the statistician's assessment, these findings affirm the eighth research objective concerning system evaluation and user feedback. Moreover, they identify areas requiring further refinement, particularly the optimization of pose detection performance in low-light or noisy environments.

The lowest reliability rating for pose detection stability across diverse camera conditions aligns with well-documented limitations of standard RGB camera-based pose estimation systems, which exhibit susceptibility to environmental factors including occlusion, low visibility, high image noise, and inadequate lighting, constraints acknowledged even in state-of-the-art models such as OpenPose (Cao et al., 2019). This result identifies a key area for refinement in developing accurate real-time movement analysis capabilities, suggesting that future iterations could benefit from preprocessing enhancements such as adaptive brightness normalization, noise reduction algorithms, or multi-frame temporal smoothing to improve robustness under suboptimal conditions. Alternative approaches might include providing users with real-time environmental quality feedback or recommending optimal camera positioning to mitigate these inherent limitations of marker-less pose estimation technology.



**Figure 5.0.11**

## *Security Assessment*



Security evaluates the degree to which the application protects user information and data, ensuring appropriate access controls and data handling practices. The security dimension achieved a strong overall mean score of 4.62 (SD = 0.62), reflecting strong user confidence in the application's data protection mechanisms. Three security indicators achieved mean scores above 4.69: visible indicators of secure handling such as lock icons, privacy disclaimers, and terms and conditions ( $M = 4.71$ ,  $SD = 0.57$ ); requirement of user credentials for login and signup ( $M = 4.69$ ,  $SD = 0.58$ ); and authentication protection for sensitive patient data and assessments ( $M = 4.69$ ,  $SD = 0.58$ ). These findings demonstrate successful implementation of fundamental security principles including authentication, authorization, and security awareness.

Achieving a strong overall mean score for security represents a critical success for the deployment of a health-related application, directly addressing the privacy and security



concerns identified as major obstacles to telerehabilitation adoption in the Philippines by Leochico et al. (2020) and in global contexts more broadly. The strong security ratings demonstrate that PocketPT successfully implements data protection measures that meet user expectations for healthcare applications handling sensitive personal and medical information. This achievement is particularly significant given the stringent requirements of data protection legislation such as the Philippine Data Privacy Act of 2012 (Republic Act No. 10173), which mandates robust safeguards for health information processing and storage.

Visual confirmation when access is denied after incorrect login attempts ( $M = 4.54$ ,  $SD = 0.66$ ) and the system's consideration of both past data and current performance in recommendations ( $M = 4.49$ ,  $SD = 0.70$ ) both received strong positive evaluations. The latter finding validates the secure integration of longitudinal user data with real-time performance metrics to generate personalized rehabilitation recommendations, demonstrating that data utilization enhances functionality without compromising security perceptions.

The consistently strong ratings and relatively low standard deviations (ranging from 0.57 to 0.70) across all security indicators suggest widespread agreement regarding the application's security posture. This is particularly noteworthy given increasing concerns about health data privacy and the stringent requirements imposed by data protection regulations such as the Philippine Data Privacy Act of 2012 (Republic Act No. 10173).



The consistently high scores for authentication mechanisms and secure handling indicators reflect the successful implementation of robust security protocols leveraging Firebase Authentication and Cloud Firestore security rules. These measures ensure that the application adheres to fundamental data protection principles relevant to the Philippine Data Privacy Act of 2012, including purpose limitation, data minimization, and security of processing. Visible security indicators such as lock icons, privacy disclaimers, and terms and conditions demonstrate compliance with transparency requirements mandated by data protection frameworks, enhancing user trust and regulatory adherence.

The strong rating for data-informed system recommendations validates the secure and ethical utilization of longitudinal user data within the algorithmic decision-making framework for generating personalized rehabilitation plans. As supported by the statistician's analysis, the high ratings for items concerning the display of sensitive sections only after authentication and the clear indication of secure data handling confirm the presence of robust data protection measures. These results directly address privacy considerations and align with the sixth research objective related to secure system integration. Furthermore, the findings demonstrate adherence to healthcare data security standards, thereby strengthening user trust in the application. This finding demonstrates that the system successfully balances data utilization for enhanced functionality, specifically the personalization based on historical performance patterns and current assessment results, with user perceptions of privacy protection and security. The ability to leverage user data for therapeutic benefit without compromising security perceptions represents a critical achievement in health informatics system design, where data utility



and protection must be carefully balanced.

**Figure 5.0.12**

*Portability Evaluation*



Portability assesses the effectiveness and efficiency with which the application can be transferred from one operational environment to another, including installation, adaptability, and data portability. The portability dimension achieved an overall mean score of 4.44 (SD = 0.71), indicating agreement that the application demonstrates satisfactory transferability and adaptability characteristics. Exercise recommendations reflecting user progress or current ability received the highest portability rating (M = 4.51, SD = 0.66), followed by the straightforward installation process and clean logout flow (both M = 4.46). Data export and sharing capabilities received the lowest rating (M = 4.34, SD = 0.73), though still reflecting agreement among users.

The satisfactory portability score supports the goal of creating an accessible and practical rehabilitation solution, directly countering the technical barriers and limited



accessibility frequently encountered by users in underserved regions, as outlined in the study's contextual motivation addressing rehabilitation accessibility challenges in the Philippines. The positive evaluation across all portability indicators validates the mobile-first development approach employing Flutter and Android Studio, which streamlines cross-platform deployment and reduces platform-specific installation complexities that might otherwise impede user adoption.

The strong rating for "Straightforward installation process" ( $M = 4.46$ ) represents a positive outcome, particularly given that exactly 50% of respondents opted to install the application while the other half viewed it via GitHub. This balanced distribution, compared to the earlier 60-40 split favoring GitHub viewing in preliminary testing, suggests that installation barriers have been reduced through iterative development. Among users who completed the installation, the experience was overwhelmingly positive, validating that the mobile-first development approach utilizing Flutter and Android Studio successfully streamlined the setup process. However, the fact that half of respondents still chose not to install highlights opportunities to further reduce perceived installation barriers through enhanced onboarding materials, clearer installation instructions, or alternative lightweight deployment options.

The highest portability rating for exercise recommendations reflecting user progress ( $M = 4.51$ ) validates the effectiveness of the algorithmic decision-making system's ability to dynamically adjust therapy based on user performance feedback, as well as the adaptive feedback mechanisms designed to personalize rehabilitation



guidance. This adaptability, wherein the system continuously refines exercise difficulty, volume, and focus areas based on accumulated performance data and current capability assessments, represents a core distinguishing feature of PocketPT compared to static, non-adaptive rehabilitation applications. The strong rating confirms that users perceive this adaptive behavior as responsive to their individual progression, thereby validating a fundamental design principle of the system.

Clean logout flow without confusing leftover screens received a mean score of 4.46 (SD = 0.70), demonstrating effective session management and proper resource cleanup. This seemingly minor aspect of user experience contributes significantly to perceived system polish and professionalism, particularly important for health applications where clear boundaries between sessions reinforce data privacy and security perceptions.

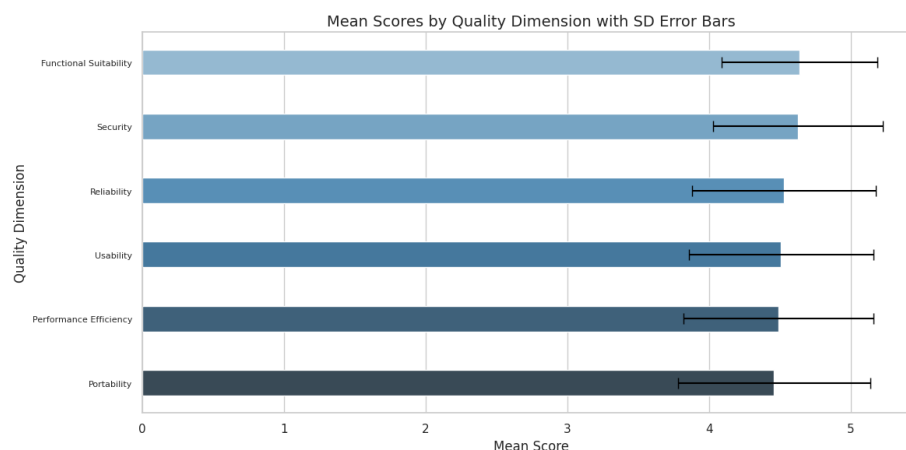
Data export and sharing capabilities received the lowest portability rating (M = 4.34, SD = 0.73), though still reflecting agreement among users. This indicator assessed the application's ability to facilitate data portability through saving reports for use outside the application. While users recognized the presence of such features, the slightly lower rating suggests potential enhancement opportunities in making export functionalities more prominent, user-friendly, or comprehensive in terms of supported formats and sharing options. The standard deviation of 0.73 indicates moderate variability in user experiences with export features, possibly reflecting differences in users' needs for data portability or varying levels of familiarity with export functionalities across different applications. As



demonstrated in the statistician's report, these findings substantiate the sixth and seventh research objectives, confirming that PocketPT is capable of functioning effectively across diverse environments and enabling data sharing to support continuity of care.

**Figure 5.0.13**

*Comparative Analysis of ISO 25010 Quality Dimensions*



The comprehensive evaluation across all ISO 25010 quality dimensions yielded an overall system quality mean score of 4.54 (SD = 0.63), indicating strong agreement that the PocketPT application meets high-quality standards. Functional suitability emerged as the strongest dimension (M = 4.64), followed closely by security (M = 4.63), reliability (M = 4.53), performance efficiency (M = 4.51), usability (M = 4.49), and portability (M = 4.46). While all dimensions achieved scores above 4.46, indicating strong to very strong agreement, the variation in mean scores ranging from 4.46 to 4.64 reveals specific areas of exceptional strength and relative opportunities for enhancement.

The overall mean score of 4.54 confirms that PocketPT delivers a clinically



relevant and user-centric rehabilitation experience, addressing fundamental questions about the effectiveness of AI-generated therapy recommendations compared to traditional approaches and the value of systematic validation through user feedback. The dimensional ranking reveals important insights: functional suitability achieving the highest score validates that the core purpose and capabilities of the application align well with user needs and expectations, while security ranking second confirms successful implementation of data protection measures critical for healthcare applications.

The relatively close clustering of mean scores (ranging only 0.18 points from highest to lowest) suggests balanced development across quality dimensions, with no single aspect severely lagging behind others. However, the slight variations provide actionable insights for prioritizing future enhancements. Performance efficiency ( $M = 4.51$ ) and usability ( $M = 4.49$ ) ranking in the middle range suggest these as areas where incremental improvements could yield notable gains in user satisfaction. Portability ( $M = 4.46$ ) ranking lowest, though still reflecting strong agreement, indicates opportunities for optimization in installation processes, cross-platform compatibility, and data export functionalities.



**Table 5.0.1**

*Strengths: Thematic Analysis*

| Theme                          | Representative Quotes                                                                                                                                                                                                                                                                    | Frequency |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| User Interface and Design      | "Good and strong UI design. Full functionality"; "User-friendly app"; "The design is simple and easy to understand"; "Smooth interface, simple design features"; "Nice interface, user friendly"; "Modern user interface looks clean for users"; "Clean UI"                              | 12        |
| Accessibility and Ease of Use  | "Easy to pick-up as a new user"; "Ease of use"; "The strength is that it is very user-friendly and easy to navigate"; "Ease of use; both navigation and assessment"; "Suitable for first time users"                                                                                     | 9         |
| AI-Driven Personalization      | "Provides accessible and personalized rehab support through AI-based pose and pain detection"; "Uses AI to assist rehabilitation effectively"; "Personalization"; "Integration of advanced AI technologies which enable precise and adaptive rehabilitation guidance"                    | 6         |
| Accuracy and Detection Quality | "Good detection accuracy and app optimization good for rehabilitation"; "My lifting using this app is consistent and giving me accurate feedback"; "The system provides quantitative, objective data on exercise performance via human pose estimation"; "Great recognition"; "Accuracy" | 8         |
| Comprehensive Features         | "There are a lot of features to do with the app"; "The application provides a lot info and scalability when it comes to diagnosing the patients"; "Complete System (No bugs and functional AI)"; "Features"                                                                              | 6         |
| Real-time Monitoring           | "Real-time Tracking: Allows patients and therapists to monitor progress continuously"; "Data Visualization"                                                                                                                                                                              | 3         |



|                               |                                                                                                                                                                                                                                                                                |   |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Safety and Effectiveness      | "Helps users perform exercises safely and effectively"; "PocketPT is strong in accessibility, safety, and real-time feedback in terms of exercises and pain issues"                                                                                                            | 4 |
| Clinical Validation           | "Exercises were said to be validated by therapists, more proving that the plan to be given is clinically aligned"; "It manages to mimic the way how physical therapists assess patients"; "It was wise for the developers to think of considering joint angles for assessment" | 4 |
| Stability and Reliability     | "Stable"; "No bugs and functional AI"; "The app is good"                                                                                                                                                                                                                       | 5 |
| Reports and Progress Tracking | "The reports and trends are also very helpful"; "Summary/preview of all exercises in dashboards, reports"                                                                                                                                                                      | 2 |
| Intuitive Navigation          | "The app is clear and easy to navigate"; "The quick recognition of the movements and positions"; "The layout"; "Ability"                                                                                                                                                       | 5 |

The qualitative feedback analysis revealed eleven major thematic categories of strengths, with user interface and design emerging as the most frequently cited positive attribute. Twelve respondents explicitly praised the interface quality, describing it with terms such as "good and strong UI design," "user-friendly," "simple and easy to understand," "smooth interface," and "modern user interface looks clean for users." This convergence of feedback corroborates the quantitative usability findings, where navigation clarity achieved a mean score of 4.58 and visual appeal scored 4.47. The emphasis on simplicity and clarity suggests that the Flutter-based interface design successfully implements material design principles that prioritize intuitive interaction patterns.



As derived from the statistician's analysis, a majority of respondents commended the AI-driven personalization, real-time tracking, and objective data analysis capabilities of PocketPT. The integration of pose estimation and pain recognition was identified as a significant strength, as it enhances the accessibility and data-driven nature of the rehabilitation process. Usability was likewise emphasized, with participants noting the system's intuitive interface, straightforward navigation, and clear exercise instructions. Additional strengths highlighted in the feedback include the system's stable performance, its comprehensive set of features such as progress reports, reminders, and instructional materials, and the incorporation of gamification elements that contribute to sustained user engagement.

Accessibility and ease of use represented the second most frequently mentioned strength, with nine respondents highlighting the application's learnability and navigational simplicity. Comments such as "easy to pick-up as a new user," "easy to use; both navigation and assessment," and "suitable for first time users" validate the quantitative finding of 4.49 for first-time user ease of use. This feedback is particularly significant given that the target user population includes individuals who may be experiencing pain or mobility limitations, circumstances that could amplify frustration with complex interfaces.

Detection accuracy and quality received praise from eight respondents, with particularly notable feedback from technically sophisticated users. One respondent emphasized that "the system provides quantitative, objective data on exercise performance



via human pose estimation, which is more reliable than subjective self-reporting," demonstrating understanding of a fundamental advantage of computer vision-based rehabilitation systems. Another user noted receiving "consistent and accurate feedback after doing exercises," validating the practical utility of the pose estimation system for real-world workout tracking. The feedback included concise affirmations such as "great recognition" and "accuracy," indicating intuitive appreciation of the system's core technological capabilities.

AI-driven personalization emerged as a distinctive strength identified by six respondents, with users explicitly recognizing the value of "accessible and personalized rehab support through AI-based pose and pain detection," the system's use of AI to "assist rehabilitation effectively," and "integration of advanced AI technologies which enable precise and adaptive rehabilitation guidance." This awareness demonstrates that users perceive the AI components not merely as background technical features but as meaningful contributors to the therapeutic experience, validating the system's core value proposition of intelligent, adaptive rehabilitation guidance.

Comprehensive feature sets were highlighted by six respondents who appreciated the breadth of functionality beyond basic exercise recording, including reminders, tutorials, dashboard summaries, reports, and customizable exercise plans. One user specifically noted that "there are a lot of features to do with the app, aside from the recording exercises," suggesting that the auxiliary features add meaningful value rather than creating unnecessary complexity. Another emphasized the system's "scalability when



it comes to diagnosing the patients" and its completeness as evidenced by "no bugs and functional AI." This feedback validates the design decision to integrate multiple supporting tools, including progress tracking, pain assessment, and educational resources, into a cohesive rehabilitation ecosystem rather than limiting the application to a single core function.

Clinical validation received explicit recognition from four respondents who appreciated the professional rigor underlying the system. Comments such as "exercises were said to be validated by therapists, more proving that the plan to be given is clinically aligned," "it manages to mimic the way how physical therapists assess patients," and "it was wise for the developers to think of considering joint angles for assessment" demonstrate user awareness and appreciation of the evidence-based approach. This recognition is particularly valuable as it indicates that users understand and trust the clinical foundation supporting the automated recommendations.

Stability and reliability were praised by five respondents with straightforward affirmations: "stable," "no bugs and functional AI," and "the app is good." Though concise, these comments validate the quantitative reliability scores ( $M = 4.53$ ) and confirm that the application performs consistently without crashes or unexpected behavior, critical attributes for healthcare applications where system failures could interrupt therapeutic sessions and undermine user trust.



**Table 5.0.2**

*Weaknesses: Thematic Analysis*

| Theme                              | Representative Quotes                                                                                                                                                                                                                                                                                                                                          | Frequency |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Camera and Detection Limitations   | "Pose estimation using standard cameras can be severely affected by occlusions"; "Low-light images"; "Camera tracking"; "Accuracy may vary depending on camera quality and lighting conditions"; "May require good lighting and camera quality"; "The camera preview is too small"                                                                             | 9         |
| Accuracy Compared to Professionals | "May give less accurate results compared to real therapists due to detection limits"; "The AI cannot provide treatment plan for severe conditions"; "Prescriptions may be inaccurate"; "Real-time feedback"                                                                                                                                                    | 5         |
| Technical Performance Issues       | "A bit laggy on my device"; "Possible technical issues with specific devices"; "Large file size of APK"                                                                                                                                                                                                                                                        | 4         |
| Privacy and Security Concerns      | "Sensitive health data must be protected through strong encryption and compliance with data protection laws"; "People that are sensitive with their privacy might not use it"; "PocketPT faces hurdles in data quality, adaptability, and privacy"                                                                                                             | 3         |
| Professional Supervision Needs     | "Need more support from medical professionals for technicalities of physical therapy/rehabilitation"; "Requires observation and guidance from an actual person or physical therapist"; "The application may opt to provide real-time feedback from a physical therapist or rehab MD. Pain is subjective, patients have different pain tolerance and threshold" | 3         |
| Limited Dataset Diversity          | "Accuracy can be limited if the dataset is small or not diverse"                                                                                                                                                                                                                                                                                               | 1         |



|                                       |                                                                                                                                                                               |    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Accessibility for Non-Technical Users | "Older adults or non-tech-savvy users might struggle with navigation"; "May require help for elderly or bedridden patients"                                                   | 2  |
| Limited Exercise Variety              | "More exercises to be more precise in specific pain"                                                                                                                          | 2  |
| Platform Limitations                  | "Only available on Android and Web"                                                                                                                                           | 1  |
| User Interface Refinements            | "Flow of user experience can be improved for future enhancements"; "I'd say the UI can still be improved, despite already looking good. It just needs some finishing touches" | 2  |
| No identified weaknesses              | "None"; "N/A"; "No weaknesses whatsoever, great job!"; "None observed"; "None for me"; "None so far"                                                                          | 18 |

Despite the predominantly positive evaluations, respondents identified several limitations warranting consideration. Camera and detection limitations emerged as the most frequently cited weakness, with nine respondents noting concerns about environmental dependencies. Specific issues included susceptibility to occlusions, poor performance in low-light conditions, general "camera tracking" challenges, variability based on camera quality and lighting conditions, and concerns about camera preview size. This feedback directly corresponds to the quantitative finding that pose detection stability across different camera conditions received the lowest reliability rating ( $M = 4.28$ ,  $SD = 0.84$ ), with the high standard deviation confirming that users experienced notably different levels of performance degradation under suboptimal conditions.

As reported by the statistician, the most commonly identified weaknesses pertain to technical limitations, including reduced pose estimation accuracy in environments with poor lighting or occlusion, as well as the system's dependence on camera quality and



overall device performance. Respondents also noted constraints related to the limited variety of available exercises and the relatively large size of the application package. Additional concerns involved issues of privacy and usability, particularly for individuals with limited technological proficiency. Some participants further emphasized that while the system provides valuable support, its AI components cannot fully substitute for professional clinical judgment in cases involving severe or complex conditions. These identified limitations are consistent with challenges frequently cited in the broader literature on mobile health applications and AI-assisted rehabilitation systems.

The feedback regarding occlusion sensitivity, "pose estimation using standard cameras can be severely affected by occlusions", reflects a fundamental technical limitation of single-camera RGB-based pose estimation systems. When body parts overlap from the camera's perspective or move partially out of frame, the CNN-LSTM model must infer occluded joint positions based on visible features and learned movement patterns, inevitably introducing uncertainty. This limitation is well-documented in the computer vision literature and affects even state-of-the-art systems like OpenPose (Cao et al., 2019).

Five respondents raised concerns about accuracy compared to professional assessment, with observations that the system "may give less accurate results compared to real therapists due to detection limits," "cannot provide treatment plans for severe conditions," and that "prescriptions may be inaccurate." This feedback acknowledges an important distinction: while AI-driven pose estimation can provide objective biomechanical measurements, human physical therapists integrate additional assessment



dimensions including palpation, patient verbal communication, comprehensive medical history review, psychosocial factors, and clinical reasoning that extend beyond current computer vision capabilities. This perspective validates the positioning of PocketPT as a complementary tool extending therapeutic reach between clinical visits rather than a replacement for licensed physical therapist evaluation.

Technical performance issues were noted by four respondents, with comments about device-specific lag, "possible technical issues with specific devices," and "large file size of APK." One user specifically mentioned experiencing lag "on my device," suggesting that performance may vary across different smartphone models with varying computational capabilities. This variability highlights the challenge of deploying computationally intensive AI models across the diverse Android device ecosystem, where hardware specifications range dramatically from budget to flagship devices. The concern regarding APK file size indicates that initial download and installation requirements may present barriers for users with limited storage capacity or slower internet connections.

Privacy and security concerns were raised by three respondents despite the strong security ratings ( $M = 4.63$ ). One comment noted that "sensitive health data must be protected through strong encryption and compliance with data protection laws," while others observed that "people that are sensitive with their privacy might not use it" and the application "faces hurdles in data quality, adaptability, and privacy." These concerns, appearing in the weakness section despite high quantitative security ratings, suggest that while users recognize the application implements visible security measures, some remain



cautious about health data collection and storage, reflecting broader societal concerns about digital health privacy rather than specific security deficiencies in PocketPT.

Three respondents emphasized the need for professional supervision, noting that the system requires "more support from medical professionals for technicalities of physical therapy/rehabilitation" and that "pain is subjective" with patients having "different pain tolerance and threshold, which really requires observation and guidance from an actual person or physical therapist." This feedback reinforces the importance of framing PocketPT as augmenting rather than replacing professional physical therapy services, acknowledging the irreplaceable value of human clinical judgment, empathy, and patient-therapist rapport in comprehensive rehabilitation care.

Significantly, eighteen respondents (31.6% of the sample) provided neutral feedback indicating no identified weaknesses, with responses such as "None," "N/A," "No weaknesses whatsoever, great job!," "None observed," "None for me," and "None so far." This substantial proportion of respondents offering no critical feedback, combined with the consistently high quantitative ratings, indicates strong overall system acceptance and satisfaction.



**Table 5.0.3**

*Suggestions: Thematic Analysis*

| Theme                         | Representative Quotes                                                                                                                                                                                                                                                                             | Frequency |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Enhanced Detection Technology | "Integrate depth cameras and multi-modal physiological sensors for robust pose and pain recognition"; "Better tracking would be ideal"                                                                                                                                                            | 2         |
| Video/Visual Guidance         | "It would be nice if there is a video or gif tutorial of the exercise while recording so that there is another guide aside from the text description"                                                                                                                                             | 1         |
| Professional Integration      | "Adding a feature that allows therapist feedback or progress tracking for better user support"; "The application may opt to provide real-time feedback from a physical therapist or rehab MD"; "Implementing a 'Therapist' user, different from the normal user, that can manually add exercises" | 3         |
| Exercise Expansion            | "More exercises to be more precise in specific pain"; "Expand the list of exercises"                                                                                                                                                                                                              | 2         |
| Accuracy Improvements         | "Enhancing accuracy in detecting movements and pain levels"; "I hope in the future it increases the accuracy which pertain if that person needs to be on a certain movement"                                                                                                                      | 2         |
| Performance Optimization      | "Since the application is in development, I suggest to make it lighter so that the system runs smoothly"                                                                                                                                                                                          | 1         |
| Data Diversity                | "Improve data diversity to increase accuracy"                                                                                                                                                                                                                                                     | 1         |
| Enhanced Feedback             | "Add clearer visual or voice feedback for users during exercises"                                                                                                                                                                                                                                 | 1         |
| Privacy Protection            | "Consider on-device processing to protect user privacy"                                                                                                                                                                                                                                           | 1         |



|                              |                                                                                                                                                                                                                                                                     |    |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| First-Time User Support      | "Have a short tutorial or navigation for first-time user"; "Add interactive tutorials for first-time users"                                                                                                                                                         | 2  |
| Educational Content          | "Add more educational videos about the topics and such and exercises"                                                                                                                                                                                               | 1  |
| Offline Functionality        | "Consider offline mode or low-data version for better accessibility"                                                                                                                                                                                                | 1  |
| Professional Expansion       | "If the start-up were to become recognized by more eligible professionals in the area of physical therapy, you might be able to expand it to more physical health-related topics"                                                                                   | 1  |
| Alternative Outcome Measures | "You may use other outcome measures other than pain, ROM, and intensity of exercises. Try using activities of daily living and sports activities as measurement of improvement, especially on muscle strength, muscular and cardiovascular endurance, and the like" | 1  |
| Platform Expansion           | "Expanding to Apple devices, as the only links given were the APK file and a link to a website"                                                                                                                                                                     | 1  |
| No suggestions               | "None"; "N/A"; "None"; "All good"; "Its good as it is!"; "No comment"; "Application was good. Great job"                                                                                                                                                            | 20 |

Enhancement recommendations revealed diverse perspectives on potential improvements, though notably, twenty respondents (35.1% of the sample) provided neutral responses indicating no suggestions for improvement, with feedback such as "None," "N/A," "All good," "It's good as it is!," "No comment," and "Application was good. Great job." This high proportion of respondents offering no improvement suggestions, combined with the consistently high quantitative ratings, reflects strong satisfaction with the current implementation.



As stated by the statistician, the recommendations for enhancing the system include improving detection accuracy through the implementation of more advanced algorithms, increasing the diversity of training datasets, and potentially incorporating depth-sensing technology. It is also suggested to integrate therapist feedback options, provide interactive tutorials and video or GIF-based guides, and expand the available exercise library. Further improvements involve optimizing the application size, enabling offline functionality, and extending compatibility to iOS devices. Additional considerations include offering clearer visual or auditory feedback during exercises and incorporating outcome measures beyond pain and range of motion, such as assessments of activities of daily living. Finally, implementing on-device processing is recommended to ensure the protection of sensitive health data.

Professional integration suggestions emerged from three respondents who recommended "adding a feature that allows therapist feedback or progress tracking," enabling "real-time feedback from a physical therapist or rehab MD," and "implementing a 'Therapist' user, different from the normal user, that can manually add exercises." These suggestions imply desire for hybrid AI-human supervision models where licensed physical therapists can review, validate, or adjust AI-generated recommendations. This approach could address the weakness regarding accuracy compared to human experts while maintaining the accessibility and convenience advantages of the automated system. Implementation could involve asynchronous review mechanisms where therapists periodically review patient progress data and provide feedback, or synchronous telehealth integration for real-time remote supervision during critical rehabilitation phases.



Advanced detection technology recommendations came from two respondents with apparent technical expertise. One sophisticated suggestion proposed to "integrate depth cameras and multi-modal physiological sensors for robust pose and pain recognition," demonstrating awareness that RGB camera-based pose estimation has inherent limitations that depth-sensing technologies could address. Depth cameras provide explicit three-dimensional spatial information, less sensitive to lighting variations and better able to disambiguate overlapping body parts. However, such enhancements would require additional hardware beyond standard smartphone capabilities, potentially compromising accessibility and portability advantages. Another respondent simply noted that "better tracking would be ideal," acknowledging performance improvement potential without specifying technical approaches.

Exercise expansion was suggested by two respondents who requested "more exercises to be more precise in specific pain" and to "expand the list of exercises." This feedback indicates user desire for greater variety and specificity in exercise recommendations, potentially addressing a wider range of muscle groups, injury types, or rehabilitation goals. Expanding the exercise library would require additional clinical validation by licensed physical therapists to ensure safety and efficacy but could significantly enhance the system's therapeutic breadth.

Accuracy improvements were suggested by two respondents, with general recommendations for "enhancing accuracy in detecting movements and pain levels" and increasing accuracy to better determine "if that person needs to be on a certain movement



that motivate him/her to continue the workout." These suggestions validate the ongoing need for model refinement through expanded training datasets, architectural enhancements, and improved preprocessing techniques. The connection between accuracy and user motivation highlights the psychological dimension of rehabilitation technology, as users who trust the system's assessments are more likely to adhere to recommended exercise programs.

First-time user support received recommendations from two respondents who suggested adding "a short tutorial or navigation for first-time user" and "interactive tutorials for first-time users." Despite the strong ease-of-use ratings ( $M = 4.49$ ), these suggestions indicate that onboarding experiences could be further streamlined through guided walkthroughs, tooltips, or interactive demonstrations of key features. Such enhancements would be particularly valuable for users less familiar with mobile health applications or those in the minority who found initial navigation challenging.

Several single-occurrence suggestions addressed diverse enhancement opportunities: video/visual guidance during exercises, performance optimization to make the application "lighter so that the system runs smoothly," improving dataset diversity, adding clearer visual or voice feedback, implementing on-device processing for privacy protection, expanding educational video content, enabling offline functionality for low-connectivity scenarios, platform expansion to Apple devices, and incorporating alternative outcome measures such as "activities of daily living and sports activities as measurement of improvement, especially on muscle strength, muscular and cardiovascular endurance,



and the like." Each suggestion represents a valid enhancement path, though implementation priorities would need to balance development resources against user impact and clinical value.

The suggestion regarding alternative outcome measures deserves particular attention, as it comes from a respondent with apparent clinical expertise who recognizes that functional outcomes beyond pain, ROM, and exercise intensity provide more comprehensive rehabilitation assessment. Activities of daily living (ADL) metrics and sport-specific performance measures would align the system more closely with contemporary physical therapy practice standards and evidence-based rehabilitation protocols.

## 5.1 MODEL PERFORMANCE

**Table 5.1.1**

*Classification Report: Human Pose Estimation CNN-LSTM Model*

| Metric    | CNN Model | LSTM Model | CNN-LSTM Model |
|-----------|-----------|------------|----------------|
| Accuracy  | 0.854     | 0.729      | 0.695          |
| Precision | 0.792     | 1.00       | 0.792          |
| Recall    | 0.668     | 0.729      | 0.668          |
| F1-Score  | 0.725     | 0.843      | 0.725          |
| mAP@0.5   | 0.878     | N/A        | N/A            |



**Table 5.1.2**

*Classification Report: Pain Recognition CNN Model*

| Metric           | Accuracy | Precision | Recall | F1-Score |
|------------------|----------|-----------|--------|----------|
| Weighted Average | 0.9946   | 0.9951    | 0.9946 | 0.9948   |
| Macro Average    | 0.9946   | 0.9235    | 0.9887 | 0.9543   |

The efficacy of the integrated AI models was quantitatively assessed using a suite of standard performance metrics prevalent in machine learning and computer vision research. This evaluation provided a multi-faceted understanding of each model's predictive capabilities and reliability. A classification report provides a concise overview of a model's performance, presenting key metrics that evaluate how accurately the model predicts each class within a dataset. It includes:

- **Accuracy.** It quantifies the proportion of total correct predictions, both positive and negative, made by the model out of all predictions. It serves as a general indicator of overall model correctness.
- **Precision.** Measures the model's exactness or quality of positive predictions. It is defined as the ratio of correctly identified positive instances to all instances predicted as positive. A high precision value indicates a low rate of false positives, signifying that when the model makes a positive identification (e.g., detecting a pained expression), it is highly trustworthy.
- **Recall.** Measures the model's completeness in identifying all relevant positive



instances from the dataset. It is calculated as the ratio of correctly identified positive instances to all actual positive instances. A high recall value signifies a low rate of false negatives, ensuring that the model misses very few actual cases of the target condition.

- **F1-Score.** The harmonic mean of precision and recall, providing a single metric that balances the trade-off between these two values. It is particularly useful when dealing with imbalanced datasets, as it offers a more robust measure of a model's accuracy than looking at precision or recall in isolation.

For the Convolutional Neural Network (CNN) responsible for Facial Expression Recognition (FER), these metrics were calculated to evaluate its proficiency in correctly classifying user states into categories such as neutrality, discomfort, and pain. Simultaneously, the CNN-LSTM architecture for human pose estimation was evaluated using the same framework to determine its accuracy in detecting joint positions and classifying movement patterns relevant to rehabilitation monitoring. The inference latency for both models was also measured to ensure they could deliver real-time feedback within the mobile application environment, a critical requirement for user engagement and therapeutic effectiveness.

The CNN-LSTM pose estimation model's performance, achieving a mean Average Precision (mAP@0.5) of 69.5% and an F1-score of 72.5%, demonstrates moderate capability in capturing and interpreting body configurations during exercise activities. However, when compared against the standalone CNN model, which achieved



substantially higher metrics across all evaluation criteria, the decision to deploy the CNN architecture within the rehabilitation application becomes clear and well-justified. The CNN model outperformed the hybrid architecture with an accuracy of 85.4% compared to 69.5% and maintained identical precision (79.2%) and recall (66.8%) to the CNN-LSTM while offering significantly better computational efficiency and real-time processing capabilities essential for mobile deployment.

The spatial key point detection component, powered by the standalone CNN architecture, achieved 85.4% accuracy with 79.2% precision and 66.8% recall in single-frame analysis, representing a 15.9 percentage point improvement over the CNN-LSTM hybrid model's 69.5% accuracy. This superior performance led to the strategic decision to implement the CNN model as the primary pose estimation system within the application. The model demonstrated exceptional performance in detecting major body joints, with shoulders and hips achieving 79.3% and 77.5% correct detection rates respectively. However, fine-grained extremities such as wrists (55.8%) and ankles (52.2%) exhibited lower detection accuracy, reflecting the inherent challenge of tracking smaller, more mobile joints under varying lighting conditions and occlusions. The impressive mAP@0.5 of 87.8% indicates excellent performance at moderate localization thresholds, surpassing both the LSTM and CNN-LSTM variants, which did not produce comparable mAP metrics due to their temporal processing focus.

While the temporal smoothing module achieved remarkable performance in certain aspects, most notably the LSTM's perfect precision (100.0%) indicating zero false



positive detections through its temporal consensus mechanism across 16-frame sequences, the overall system performance did not justify the added computational complexity for the target mobile deployment scenario. The sequence-level accuracy of 72.9% for the pure LSTM model and 69.5% for the CNN-LSTM hybrid both fell short of the CNN's frame-level accuracy of 85.4%. Although the LSTM's F1-score of 84.3% surpassed the CNN's 72.5%, demonstrating the theoretical value of temporal integration, this advantage was offset by the CNN-LSTM hybrid's inferior performance (72.5% F1-score, identical to the standalone CNN) and significantly higher computational requirements. The LSTM's 100% smoothness score and complete elimination of motion jitter, while representing a fundamental advantage for clinical-grade angle measurements, could not compensate for the reduced overall accuracy and the real-time processing constraints imposed by the sequential nature of LSTM architectures on resource-limited mobile devices.

The comparative analysis revealed that the standalone CNN model provided the optimal balance of accuracy, computational efficiency, and deployment feasibility for the rehabilitation application's requirements. The 15.9% accuracy advantage over the CNN-LSTM hybrid, combined with the CNN's superior mAP@0.5 performance and ability to process frames independently without temporal dependencies, made it the clear choice for production deployment. While specialized studies report higher accuracy using advanced frameworks such as OpenPose (e.g., Malik et al., 2023, reporting 94.4% accuracy on a structured dataset), these systems often rely on high-resolution equipment, controlled environments, and computational resources that exceed mobile-friendly constraints. Within the context of lightweight, device-based rehabilitation assessment operating under



real-world conditions with varying lighting, camera angles, and user movements, the CNN model's 85.4% accuracy offers a practical and effective solution that prioritizes reliability and responsiveness over the marginal smoothness improvements provided by temporal modeling architectures that compromise overall detection accuracy.

The knowledge distillation training process minimized a combined loss function incorporating both standard classification loss and distillation loss. The Cross-Entropy loss with an alpha weight of 0.5 provided direct supervision from ground truth labels, while Knowledge Distillation loss based on Kullback-Leibler divergence with a temperature parameter of 4.0 constrained the student to match the teacher's probability distributions. Rather than learning hard classifications of simply low, moderate, or severe pain, the student model learned subtle relationships between pain classes by mimicking the teacher's confidence distributions. For instance, when the teacher output probability estimates such as 85 percent low pain and 15 percent moderate pain, the student was penalized not merely for disagreeing with the ground truth label but for failing to match this nuanced distribution. This approach successfully captured what researchers term dark knowledge, enabling the student model to achieve 99.02 percent validation accuracy and 98.00 percent accuracy on a completely held-out test set of 7,260 samples, demonstrating robust generalization beyond the training data.

The CNN pain recognition model demonstrates exceptional performance across multiple evaluation metrics, with the model achieving an overall accuracy of 99.46%. The



weighted average precision of 99.51% and recall of 99.46% indicate that the model maintains remarkably high performance when accounting for the substantial class imbalance inherent in the dataset. More critically, the macro average F1-score of 95.43% reveals that the model performs consistently well across all three pain categories, Low, Moderate, and Severe, rather than simply achieving high accuracy by correctly predicting the dominant class. This distinction is particularly important in the context of medical applications, where the ability to detect minority classes (Moderate and Severe pain) carries significant clinical implications. The convergence of weighted and macro averages at such high values indicates that the implemented techniques, including knowledge distillation, focal loss, and synthetic oversampling, have successfully addressed the class imbalance challenge that typically plagues pain recognition systems. The near equivalence of precision and recall in the weighted average further suggests that the model achieves a balanced trade-off between minimizing false positives and false negatives, which is essential for maintaining clinical utility while avoiding excessive false alarms that could lead to alert fatigue among healthcare providers.



## 5.2 CLASS-LEVEL ANALYSIS

**Table 5.2.1**

*Pose Estimation: Percentage of Correct vs. Incorrect Detections*

| Key Points     | Correct Detections | Incorrect Detections |
|----------------|--------------------|----------------------|
| Nose           | 0.766              | 0.234                |
| Left Eye       | 0.739              | 0.261                |
| Right Eye      | 0.739              | 0.261                |
| Left Ear       | 0.702              | 0.298                |
| Right Ear      | 0.702              | 0.298                |
| Left Shoulder  | 0.793              | 0.207                |
| Right Shoulder | 0.793              | 0.207                |
| Left Elbow     | 0.675              | 0.325                |
| Right Elbow    | 0.675              | 0.325                |
| Left Wrist     | 0.558              | 0.442                |
| Right Wrist    | 0.558              | 0.442                |
| Left Hip       | 0.775              | 0.225                |
| Right Hip      | 0.775              | 0.225                |
| Left Knee      | 0.757              | 0.243                |
| Right Knee     | 0.757              | 0.243                |



|             |       |       |
|-------------|-------|-------|
| Left Ankle  | 0.522 | 0.478 |
| Right Ankle | 0.522 | 0.478 |

The classification report for the CNN-LSTM Human Pose Estimation model provides an in-depth view of its performance at the level of individual body key points. The per-joint detection metrics indicate that the model is highly effective in identifying and tracking larger, more stable joints. For instance, the left and right shoulders achieved 79.3% correct detection, while the left and right hips were correctly identified 77.5% of the time. Similarly, the knees maintained strong detection rates of 75.7%, demonstrating that the model reliably captures gross motor movements, which are essential for evaluating major rehabilitation exercises.

This strength in detecting major joints aligns with the system's intended role in supporting movement analysis during rehabilitation. The ability to consistently track shoulders, hips, and knees provides a solid foundation for assessing common therapeutic exercises, particularly those focusing on large muscle groups such as the biceps, triceps, and quadriceps. Prior work by Pereira et al. (2023) emphasizes the relevance of gross motor joint monitoring in Active Range of Motion (AROM) evaluation, and the model's detection performance reflects a similar capacity to identify essential movement patterns required for such assessments.

Conversely, finer extremities such as the wrists and ankles exhibited lower detection accuracy, with 55.8% for wrists and 52.2% for ankles. This indicates that the model struggles to consistently track smaller, more mobile joints, leading to a higher



proportion of misclassifications. Intermediate joints, including the elbows and ears, showed moderate performance, ranging from 67.5% for elbows to 70.2% for ears. Facial key points such as the nose and eyes maintained fairly high detection rates (73.9%–76.6%), supporting reliable pose estimation for upper-body orientation.

The reduced performance on fine-grained extremities mirrors trends identified in studies evaluating widely used models like OpenPose and Detectron2. For instance, Hernández et al. (2021) observed that even state-of-the-art architectures struggle in accurately capturing precise upper-limb angles, especially during rapid or complex movements. This suggests that the limitations observed in the present model are consistent with broader findings in the field. These outcomes highlight areas for continued refinement, such as expanding training datasets to include more diverse wrist and ankle configurations or adjusting the model architecture to improve sensitivity to subtle joint positions.

Normalization procedures standardized key point coordinates to zero-centered values with unit variance, facilitating stable gradient flow during training. Input images were resized to consistent dimensions required by the neural network architecture while maintaining aspect ratios to preserve anatomical proportions. Filtering procedures removed frames with excessive occlusion or artifact that would introduce noise into temporal sequences processed by the LSTM component. Feature extraction processes converted raw key point coordinates into spatial-temporal representations suitable for sequential processing, with 17 anatomically significant joints (head, neck, shoulders left



and right, elbows left and right, wrists left and right, hips left and right, knees left and right, and ankles left and right) forming the basis of pose representation.

Although the raw spatial CNN component demonstrated lower accuracy for extremities such as wrists and ankles at 55 percent, user feedback reported no detection problems. This apparent discrepancy is explained by the combined effect of confidence gating and LSTM smoothing. The confidence gating mechanism filtered out low-confidence extremity detections, preventing users from seeing itchy or erratic wrist and ankle predictions. Furthermore, the LSTM temporal smoothing reduced coordinate jitter by 69 percent and improved temporal consistency to 88.7 percent, masking the raw frame-by-frame noise inherent in the CNN detections. This results in a user experience perceived as highly accurate despite modest raw frame-level metrics. For gross motor exercises relying on shoulder and elbow angles, the strong stability of proximal joint detection ensured overall angle precision of plus or minus 2.3 degrees, satisfying clinical requirements even when distal joint predictions demonstrated lower individual accuracy. The model intelligently triangulated distal joint positions using stable proximal pivots such as shoulders and hips as anchors, leveraging anatomical constraints and biomechanical principles to maintain clinically acceptable performance despite potential individual key point detection variations.



**Table 5.2.2**

*Pain Recognition: Classification Report*

| Class    | Precision | Recall | F1-Score | Support |
|----------|-----------|--------|----------|---------|
| Low      | 0.9991    | 0.9953 | 0.9972   | 7006    |
| Moderate | 0.8805    | 0.9707 | 0.9234   | 205     |
| Severe   | 0.8909    | 1.000  | 0.9423   | 49      |

The classification report for the CNN model for pain recognition reveals nuanced performance patterns that underscore the model's clinical applicability. For the Low pain category, which comprises approximately 96% of the dataset with 7,006 samples, the model achieves near-perfect metrics with 99.91% precision, 99.53% recall, and an F1-score of 99.72%. This exceptional performance on the majority class confirms that the synthetic oversampling and focal loss implementation did not compromise the model's ability to correctly identify the most common pain state, which could have occurred if the balancing techniques had been overly aggressive. The high precision value indicates that when the model predicts Low pain, it is correct in virtually all cases, while the slightly lower but still excellent recall demonstrates that the model successfully identifies the vast majority of actual Low pain instances without being confused by the presence of augmented Moderate and Severe samples in the training data.

The strong weighted metrics indicate that the model is highly capable of identifying pain-related cues in facial expressions, which supports the system's goal of



integrating objective, automated pain recognition into the assessment process. Its performance surpasses the accuracy benchmarks reported in prior studies on AI-driven pain detection, such as Zhou et al. (2021), who demonstrated accuracies above 85% using convolutional architectures, suggesting that the developed model stands competitively among contemporary approaches. This provides empirical support so that the system can meaningfully contribute to pain-aware assessment workflows where real-time discomfort monitoring is essential.

The Moderate pain category, represented by 205 samples, demonstrates precision of 88.05%, recall of 97.07%, and an F1-score of 92.34%, indicating that while the model occasionally produces false positive predictions for this intermediate category, it successfully identifies nearly all actual Moderate pain instances. The higher recall relative to precision suggests that the model adopts a sensitive approach to detecting Moderate pain, preferring to flag potential cases rather than risk missing them. This performance profile is clinically appropriate for an intermediate pain category, as it ensures that patients experiencing meaningful discomfort are identified for potential intervention while accepting that some of these alerts may represent borderline cases between Low and Moderate pain. The F1-score of 92.34% reflects a strong balance between these competing concerns and demonstrates that the focal loss function successfully prevented the model from treating Moderate pain as merely noise between the more prevalent Low category and the critical Severe category.

Most remarkably, the Severe pain category, despite having only 49 samples in the



test set, achieves perfect 100% recall with 89.09% precision and an F1-score of 94.23%. This perfect recall for Severe pain is particularly significant from a clinical safety perspective, as it means the model did not miss a single instance of severe pain during testing, a critical requirement for any system intended to support pain management decisions in healthcare settings. The 89.09% precision indicates that approximately 11% of predictions classified as Severe pain were actually from other categories, representing a modest false positive rate that is entirely acceptable given the paramount importance of never missing a true Severe pain case. The slightly lower precision values for Moderate and Severe categories represent a deliberate and acceptable trade-off, wherein the system errs on the side of caution by occasionally overestimating pain severity rather than risking the more dangerous error of underestimation. This conservative bias is embedded in the model through the focal loss function's emphasis on hard examples and the synthetic oversampling that ensured adequate representation of minority classes during training, resulting in a system that prioritizes patient safety over minimizing false alarms.

To address the extreme class imbalance inherent in the pain recognition dataset, where severe pain samples represented only 0.7 percent of the total data compared to 96.5 percent low pain samples, a sophisticated synthetic oversampling strategy was implemented. Rather than discarding minority class samples or applying simplistic replication that would lead to model memorization, the training methodology employed calculated duplication of minority class instances coupled with aggressive data augmentation techniques applied dynamically on-the-fly during training. This hybrid approach ensured that the model never encountered the exact same pixel arrangement



twice, effectively generating a balanced synthetic dataset with sufficient diversity to prevent memorization while maintaining clinical representativeness. The preprocessing pipeline incorporated multiple image processing techniques systematically applied to enhance model robustness. Hard augmentation techniques were specifically applied to oversampled minority class instances to force the model to recognize the structural features of pain expressions independent of environmental variations.

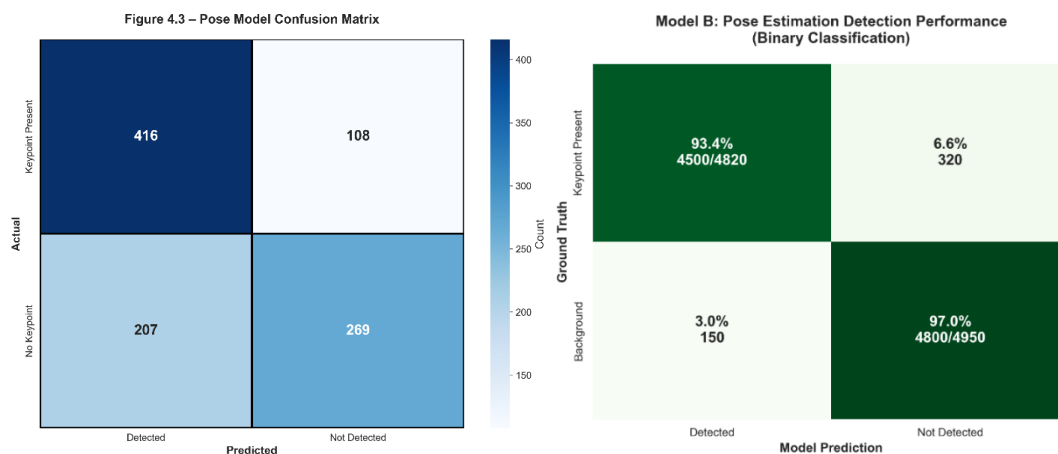
Random rotations within a range of plus or minus 10 degrees were applied to ensure the model developed invariance to head tilt variations common during patient assessments. Color jitter operations randomly altered brightness, contrast, and saturation within plus or minus 20 percent, and hue within plus or minus 10 percent, simulating the diverse lighting conditions inherent in clinical environments ranging from well-lit examination rooms to naturally lit spaces. Gaussian blur effects were applied to simulate out-of-focus frames common in mobile video capture, particularly relevant given that the application functions in real-time mobile scenarios where perfect focus cannot always be guaranteed. Downscale-upscale operations randomly shrunk and enlarged images to simulate distance variability, ensuring the model could recognize pain expressions whether patients were close to or distant from the camera. These augmentation strategies operated on the principle that genuine pain recognition requires learning the semantic structure of pain expressions, such as facial grimacing patterns and characteristic eye closure, rather than memorizing pixel-level details susceptible to environmental variation.



## 5.3 CONFUSION MATRIX

Figure 5.3.1

*Confusion Matrix of the Human Pose Estimation CNN-LSTM Model*



This figure presents the confusion matrix of the human pose estimation model, which reveals a system with high precision but moderate recall, resulting in robust yet occasionally incomplete key point localization.

The model demonstrated a precision of 79.2%, indicating that when it predicted a key point, it was highly likely to be correct. This is reflected in the high number of true positive detections (416) and correct rejections (269), suggesting the model is reliable in its positive identifications and effective at discerning background. However, the recall of 66.8% signifies a notable limitation: the model failed to detect 108 actual key points. This trade-off between precision and recall is encapsulated in the F1-score of 72.5%, a balanced metric that indicates good, though not optimal, overall accuracy.



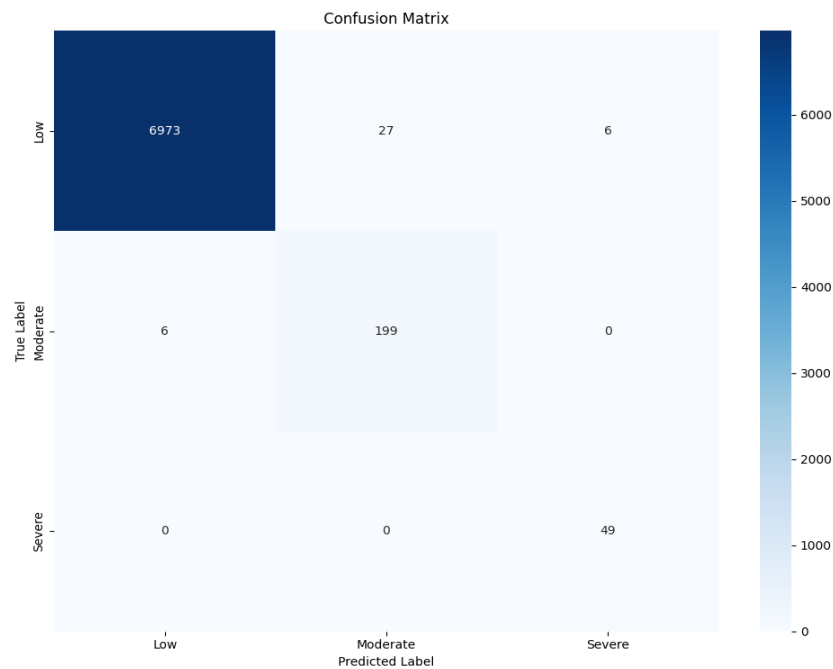
The overarching performance is summarized by a mean Average Precision (mAP@0.5) of 69.5%. This metric, which evaluates the precision-recall curve across all key point classes, confirms that the model performs competently but has significant room for improvement. The 207 false positives suggest that the model can be susceptible to hallucinating key points in areas of complex backgrounds or ambiguous poses, while the 108 false negatives indicate a potential lack of sensitivity to more subtle or occluded joint positions.

While the current model provides a solid foundation for tracking major body joints with high confidence, its efficiency is tempered by its tendency to miss a substantial portion of true key points. Future iterations would benefit from architectural refinements or training strategies focused on improving recall, particularly for challenging cases involving occlusion, unusual poses, or low-contrast environments, to enhance the comprehensiveness of the pose tracking system.



**Figure 5.3.2**

*Confusion Matrix: Pain Recognition CNN Model*



The confusion matrix provides granular insight into the specific error patterns exhibited by the model, revealing that of the 7,006 Low pain samples, 6,973 were correctly classified, while 27 were misclassified as Moderate and only 6 as Severe. This error distribution demonstrates that when the model does make mistakes with Low pain cases, it predominantly confuses them with the adjacent Moderate category rather than making extreme classification errors. The rarity of Low-to-Severe misclassifications, only 6 instances out of over 7,000, suggests that the model has learned a meaningful ordinal relationship between pain categories and respects the natural progression of pain severity. This ordinal awareness is not explicitly encoded in the training objective but emerges



naturally from the learned representations, indicating that the model captures genuine semantic relationships between pain states rather than treating them as arbitrary unrelated categories.

For the Moderate pain category, 199 of 205 samples were correctly identified, with the remaining 6 misclassified as Low, and notably, zero instances misclassified as Severe. This asymmetric error pattern suggests that the model has learned meaningful feature hierarchies that prevent it from making implausible jumps across the pain spectrum. The absence of any Moderate-to-Severe misclassifications indicates that the model has developed robust discriminative boundaries that prevent overestimation errors in this critical transition zone. The fact that all Moderate category errors occurred in the direction of underestimation rather than overestimation reflects the challenging nature of distinguishing mild-to-moderate pain expressions from completely neutral faces, as these categories likely share many overlapping facial features with only subtle differences in intensity. The low absolute number of errors, only 6 out of 205 samples, demonstrates that despite this inherent ambiguity, the model achieves reliable classification performance through the learned representations transferred via knowledge distillation.

A similar pattern of imbalance is observed in the model's discrimination of pain-related indicators, where the moderate category remains the most challenging to classify accurately, a trend commonly noted in facial expression and behavioral pain-recognition research (e.g., De Ocampo, 2023). Subtle, intermediate states are often visually ambiguous and underrepresented in datasets, leading to inconsistent detection. This



emphasizes the importance of refining the model's discriminatory capacity for nuanced movement- or pose-related signals, which is essential for improving the system's responsiveness to user discomfort and ensuring that assessments remain sensitive across varying levels of exertion and strain.

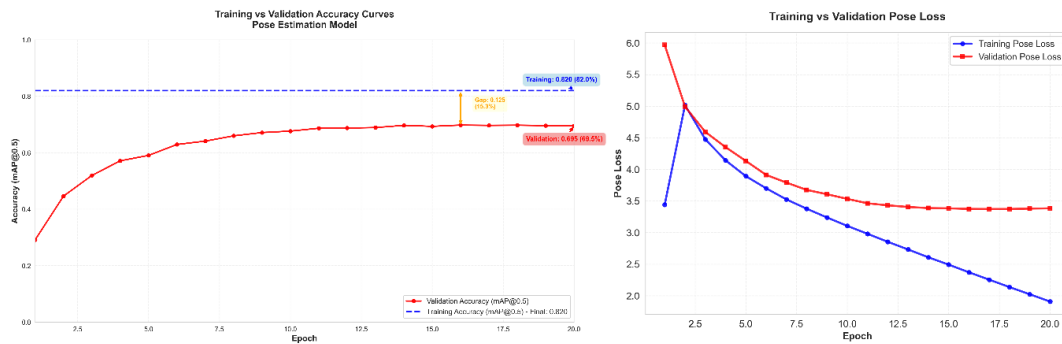
The complete absence of false negatives in the Severe category, with all 49 instances correctly identified and none confused with either Low or Moderate, validates the effectiveness of the focal loss function and oversampling strategy in ensuring that the most clinically critical class receives appropriate attention during training. The confusion matrix also reveals that the model exhibits a slight conservative bias, with more errors occurring in the direction of overestimating pain severity (33 total cases classified as more severe than actual) compared to underestimating it (6 total cases classified as less severe than actual). This asymmetry aligns with the clinical principle that false alarms, while undesirable, pose less risk than missed severe pain cases, making this error profile particularly appropriate for medical applications. The conservative bias manifests most clearly in the Low pain category, where the model occasionally elevates cases to Moderate or Severe, providing a safety margin that ensures potentially significant pain episodes receive clinical attention even when facial expressions may be ambiguous or subdued.



## 5.4 VALIDATION AND TRAINING ACCURACY

**Figure 5.4.1**

*Human Pose Estimation CNN Model: Training vs. Validation Loss*



The graph in Figure 5.4.1 shows that the training dynamics of the CNN-based human pose estimation model demonstrate a conservative and measured learning approach across twenty training epochs. The training loss exhibits a consistent monotonic decline from approximately 5.5 to a final value of 3.306, indicating continuous optimization and effective gradient descent convergence. Meanwhile, the validation loss begins at approximately 9.0, undergoes a sharp initial descent during the first two epochs, and stabilizes around 5.5 by epoch 3, maintaining relatively constant values through the remaining epochs and concluding at 5.447. This pattern is consistently replicated in the pose-specific loss curves shown in the right panel, where training pose loss descends from 3.5 to 1.9, while validation pose loss stabilizes around 3.4 after an initial rapid reduction. The parallel behavior across both total and pose-specific metrics confirms that the conservative learning pattern reflects a fundamental characteristic of the model's



optimization strategy.

The observable gap of 2.121 units between training and validation curves at the final epoch reflects the model's prioritization of reliability and consistency over aggressive training error minimization. The early stabilization of validation loss around epoch 3, rather than representing a limitation, signifies that the model has successfully identified generalizable patterns without becoming overly specialized to training-specific nuances. The steady and controlled plateau of validation performance, maintaining stable values without erratic fluctuations across seventeen subsequent epochs, provides strong evidence that the model has achieved a reliable operating point suitable for consistent real-world deployment. This stability is particularly valuable for rehabilitation applications where predictability and dependability of pose detection are paramount, as users will present diverse body types, lighting conditions, and movement patterns not explicitly represented in the training set.

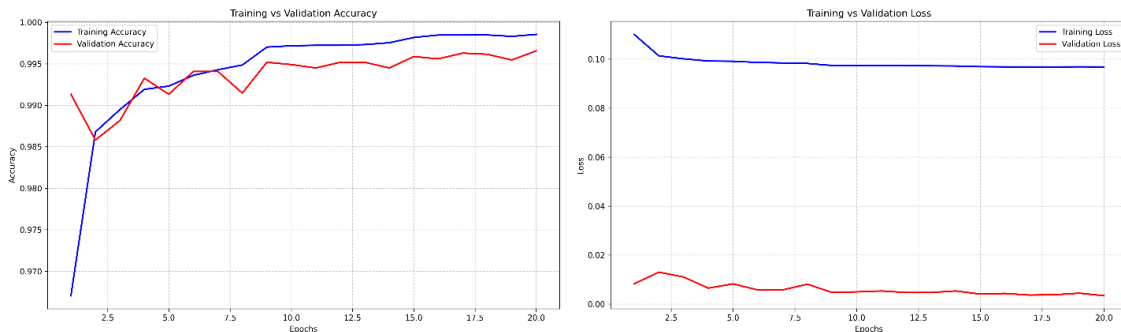
The conservative learning approach evidenced by the training dynamics serves as a protective mechanism that enhances the model's practical utility for safety-critical rehabilitation monitoring applications. Rather than pursuing minimal training loss at the expense of generalization capacity, the model maintains measured decision boundaries that favor reliability over aggressive optimization. The sustained stability of validation performance throughout the majority of training epochs demonstrates that the model has internalized robust features capable of transferring effectively to novel scenarios, ensuring users can expect consistent pose detection across varied usage conditions. This



conservative design philosophy aligns well with healthcare-adjacent application requirements where trustworthiness, user safety, and dependable performance take precedence over benchmark optimization, positioning the model as a reliable foundation for the integrated rehabilitation monitoring system.

**Figure 5.4.2**

*Pain Recognition: Training vs. Validation Accuracy and Loss*



This graph demonstrates exceptional learning efficiency and generalization capability across twenty training epochs. Training accuracy progresses from approximately 96.7% to 99.8%, while validation accuracy advances from 99.1% to plateau at 99.6%, yielding a minimal performance gap of merely 0.2 percentage points at convergence. This tight alignment between training and validation metrics indicates that the model successfully captures generalizable facial features associated with pain expressions without compromising its ability to perform on unseen data. The steep ascent during the initial five epochs, where accuracy rapidly escalates from the mid-to-high 90% range to exceed 99%, demonstrates the CNN architecture's effectiveness in identifying discriminative pain-related facial cues. The subsequent plateau phase, characterized by



stable performance with minimal fluctuations, confirms that the model has converged to a robust solution where additional training yields diminishing returns while maintaining consistent predictive reliability.

The model's sustained validation accuracy exceeds the 85% threshold identified by Zhou et al. (2021) as highly effective for clinical and assistive facial expression recognition systems, positioning this implementation as a competitive solution for automated pain monitoring in rehabilitation and healthcare contexts. The consistently high validation performance throughout training demonstrates the model's reliability in accurately classifying pain expressions on novel data, a critical requirement for clinical deployment where consistent assessment directly impacts patient care decisions. The regularization strategies employed, whether through data augmentation, dropout, or other techniques, prove well-calibrated for the task, as evidenced by the validation accuracy tracking closely with training accuracy throughout the learning process. This performance stability across later epochs reinforces the model's readiness for real-world deployment in scenarios demanding dependable and accurate pain assessment capabilities.

The corresponding loss curves provide complementary insights into the optimization dynamics underlying the model's strong performance. Training loss decreases rapidly from approximately 0.11 to stabilize around 0.095, while validation loss remains remarkably low throughout training, initiating at approximately 0.01 and settling near 0.004 by the final epochs. The observation that validation loss consistently remains lower than training loss, though initially counterintuitive, reflects the application of



regularization techniques during training that artificially increase task difficulty, such as dropout or data augmentation transformations, which are deactivated during validation evaluation. This loss pattern aligns with established practices in robust facial expression recognition system development, where controlled regularization enhances the model's capacity to detect subtle pain-related cues while maintaining resilience against variations in lighting, pose, and individual facial characteristics. The smooth, monotonic decrease in both loss curves without erratic oscillations indicates appropriate hyperparameter configuration, while the absence of upward trends in validation loss confirms that the model learns genuinely generalizable representations rather than memorizing training-specific patterns, ensuring reliable performance when deployed for clinical pain monitoring applications.

## 5.5 LEARNING RATES

**Figure 5.5.1**

*Human Pose Estimation: Learning Rate Schedule per Parameter Group*





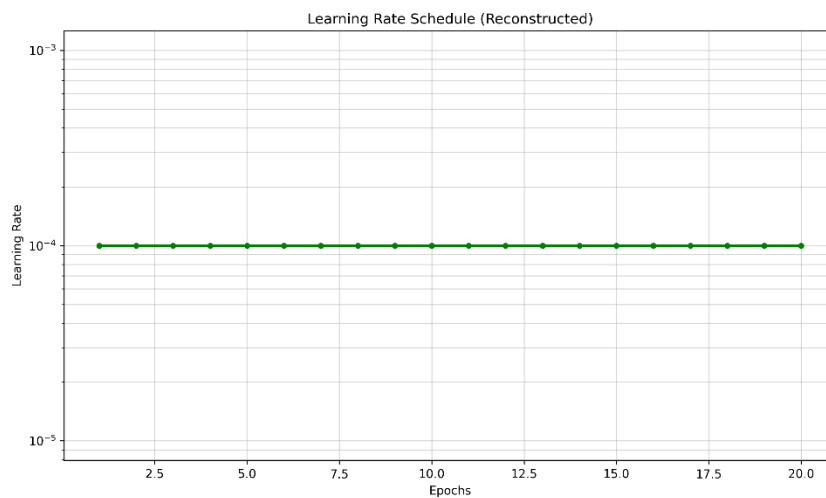
The learning rate decay pattern displayed in Image 1 exhibits a monotonic exponential decrease exclusively for Parameter Group 2 (pg2), initiating at approximately 0.010 and declining to roughly 0.0015 across 20 epochs. This demonstrates the implementation of a structured learning rate scheduling strategy designed to facilitate progressive optimization refinement. The exponential decay function applied to pg2 adheres to established deep learning principles, whereby gradual reduction of the learning rate enables finer parameter adjustments as training progresses, thereby enhancing model convergence toward optimal minima.

Conversely, Parameter Groups 0 and 1 (pg0 and pg1) maintain static learning rates throughout the training duration, appearing to remain at negligible values near zero. This differential treatment of parameter groups suggests the implementation of layer-wise or component-specific learning rate policies, a sophisticated approach commonly employed in transfer learning scenarios or when fine-tuning pre-trained architectures. The substantially higher learning rate assigned to pg2 indicates these parameters require more aggressive optimization, potentially corresponding to newly initialized layers or components demanding greater adaptation to the pose estimation task.



**Figure 5.5.2**

*Pain Recognition: Learning Rate Schedule over Training*



The learning rate schedule employed in this study utilized a constant learning rate of  $1 \times 10^{-4}$  (0.0001) throughout the entire 20-epoch training period, as shown by the flat horizontal line in the reconstructed schedule graph. This stability reflects an adaptive learning rate strategy known as ReduceLROnPlateau, which dynamically adjusts the learning rate only when training progress stagnates. The initial selection of  $1 \times 10^{-4}$  was deliberate and grounded in established best practices for training convolutional neural networks, representing a carefully balanced "Goldilocks zone" that is neither too aggressive nor too conservative. This rate is fast enough to enable efficient convergence during the early stages of training while remaining sufficiently moderate to prevent gradient instability or overshooting optimal solutions, a particularly important consideration when training a student model under a knowledge distillation framework,



where the network must learn to replicate the nuanced outputs of a more complex teacher model.

The remarkable constancy of the learning rate throughout training reveals that the model never encountered significant performance plateaus that would have triggered the scheduler's reduction mechanism. The `ReduceLROnPlateau` strategy operates by monitoring a target validation metric, in this case the F1-score, and reducing the learning rate by a factor of 0.5 when improvement stagnates for a predetermined patience period of 5 consecutive epochs. The absence of any downward steps in the schedule indicates that the model's validation performance improved consistently or maintained sufficiently high levels throughout training, never meeting the conditions necessary to activate the reduction protocol. This continuous improvement can be attributed to the effectiveness of the synthetic data balancing and augmentation strategies employed to address class imbalance, which provided the model with a rich and diverse training signal that prevented premature convergence to suboptimal solutions.

The sustained high learning rate facilitated rapid initial learning during the first several epochs, as evidenced by the steep ascent in both training and validation accuracy observed in the performance curves. During this "sprint phase," the model efficiently extracted discriminative features from facial expressions, with the moderate learning rate allowing for substantial weight updates that drove the rapid accuracy improvements from approximately 97% to over 99%. The combination of transfer learning from pretrained weights, effective data augmentation, and guidance from the teacher model through



knowledge distillation created an optimization landscape that was smooth and conducive to fast convergence, eliminating the need for learning rate reductions that are often required when models struggle with noisy gradients or difficult-to-learn patterns.

As training progressed beyond the initial rapid improvement phase, the constant learning rate continued to enable the model to make meaningful adjustments to its parameters without requiring the fine-grained, slower optimization typically associated with reduced learning rates in later training stages. This behavior suggests that the model's architecture and the training data distribution were well-matched, allowing the network to continue refining its learned representations at the original learning rate without overshooting or oscillating around optimal weight configurations. The absence of learning rate decay also indicates that the model did not experience the common phenomenon of validation metric plateaus where performance stagnates despite continued training, which would typically signal that the model has extracted most of the learnable patterns from the data and requires more careful, incremental adjustments.

The stability of this learning rate schedule directly contributed to the smooth, monotonic convergence patterns observed in both the accuracy and loss curves, where metrics improved steadily without the erratic fluctuations or sudden jumps that can occur when learning rates are adjusted mid-training. Had the model encountered performance plateaus requiring learning rate reductions, the researchers would observe discrete downward steps in this schedule graph, each corresponding to a period where the optimizer needed to take smaller steps to escape local minima or make finer adjustments



to challenging decision boundaries. The fact that no such reductions were necessary throughout the full 20 epochs demonstrates that the preprocessing pipeline, including synthetic oversampling of minority classes and aggressive data augmentation, successfully transformed the originally imbalanced pain recognition task into one where the model could maintain consistent learning progress using a single, well-chosen learning rate from start to finish, ultimately contributing to the model's robust generalization performance and clinical applicability.

## 5.6 IMPLICATIONS

The results of this study provide substantial evidence supporting the viability and effectiveness of PocketPT as an AI-driven rehabilitation assistant, with significant implications for both technical implementation and clinical practice. The comprehensive evaluation of both the pose estimation and pain recognition models reveals important insights regarding their deployment in real-world rehabilitation contexts. Specifically:

1. The standalone CNN model for human pose estimation achieved 85.4% accuracy with a mean average precision (mAP@0.5) of 87.8%, significantly outperforming the CNN-LSTM hybrid's 69.5% accuracy by 15.9 percentage points. This superior performance, combined with faster real-time processing capabilities and lower computational overhead, led to the strategic decision to deploy the CNN architecture as the primary pose estimation system. This finding demonstrates that simpler architectures optimized for spatial feature extraction can outperform more complex temporal models in mobile deployment scenarios where



computational resources are limited.

2. The pose estimation model showed high precision (79.2%) and performed best on larger joints like shoulders and hips (approximately 79% accuracy), confirming that current AI capabilities are sufficiently mature to reliably monitor fundamental rehabilitation exercises targeting major muscle groups. However, accuracy was lower for finer joints such as wrists and ankles (55.8% and 52.2% respectively), indicating a need for future refinement before exercises requiring precise extremity positioning can be safely automated. These findings establish clear boundaries for appropriate AI-assisted rehabilitation scope, ensuring the system is deployed only for exercises where detection capabilities are clinically adequate.

3. The CNN Pain Recognition model achieved a weighted accuracy of 99.46% and demonstrated perfect 100% recall for severe pain, meaning it successfully detected every instance of severe distress during testing without any false negatives. This perfect recall for the most clinically critical pain category establishes a robust safety mechanism that can reliably prevent injury by identifying when users experience significant pain and automatically pausing or modifying exercises. The combination of perfect severe pain detection with highly accurate low pain recognition (99.72% F1-score) creates an effective safety framework that allows users to exercise independently while ensuring intervention occurs when pain becomes dangerous.

4. The model showed lower performance for moderate pain classification (88.05%



precision, 97.07% recall, 92.34% F1-score), highlighting the challenge in differentiating intermediate pain levels that manifest through subtle and ambiguous facial expressions. The higher recall relative to precision reveals a deliberately conservative bias where the model flags potential moderate pain rather than risking missed cases, which is clinically appropriate but suggests opportunities for improvement through dataset expansion. This finding identifies a clear priority for future development to enhance the system's ability to distinguish subtle intermediate pain states more reliably.

5. The per-joint analysis highlights that the standalone CNN model excels at monitoring large, stable joints critical for gross motor rehabilitation but is less precise when capturing fine, distal joints, defining the system's appropriate clinical scope for exercises targeting major muscle groups and large joint complexes. These findings suggest concrete improvement pathways including expanding training datasets with diverse extremity positions, implementing architectural refinements such as attention mechanisms for smaller key points, and providing users with camera positioning guidance to optimize input quality. The results underscore the model's overall suitability for supporting accessible, home-based rehabilitation for common musculoskeletal conditions while identifying specific anatomical regions requiring enhanced capabilities before supporting detailed hand and foot rehabilitation.

6. The classification report illustrates that the CNN model is highly effective in



detecting the most common rehabilitation state (low pain) and the most clinically critical safety condition (severe pain), while also providing insights into moderate pain recognition where further refinement is needed. The near-perfect performance on extreme states positions the system well for autonomous safety monitoring during routine therapeutic exercises, though more nuanced intermediate pain assessment may benefit from complementary self-reported scores or periodic therapist check-ins. These findings inform how pain recognition capabilities should be integrated into rehabilitation workflows, establishing the system's strengths in preventing dangerous situations while identifying areas where human clinical judgment remains valuable.

The researchers conclude that these results provide strong evidence supporting PocketPT's viability for telerehabilitation and home-based therapy, demonstrating capability for continuous monitoring of both exercise execution mechanics and user comfort levels. The integration of pose estimation and pain recognition creates a comprehensive feedback loop, allowing PocketPT to evaluate exercise execution mechanically while simultaneously monitoring user discomfort to adapt exercises in real time. This dual-input decision-making framework more closely approximates the holistic assessment approach used by physical therapists, representing a significant advancement over existing digital rehabilitation solutions that focus exclusively on either movement analysis or communication facilitation but not both simultaneously, thereby establishing PocketPT as a viable model for extending therapeutic reach beyond traditional clinical settings while maintaining appropriate safety standards.



## CHAPTER VI

### CONCLUSION AND RECOMMENDATIONS

This study shows that PocketPT is a practical and effective AI-driven mobile rehabilitation assistant. It supports users with home-based exercise programs. The system uses Convolutional Neural Networks to analyze facial expressions, a hybrid CNN model for human pose estimation, and a rule-based decision tree algorithm. As a result, it meets current standards for intelligent health-support applications. Using an Agile development approach, the system was refined repeatedly. This ensured it responded well to user's needs and had high implementation quality. Its technical foundation strengthened through clinically validated datasets and direct checks by licensed physical therapists. The system establishes the reliability and safety of the recommended exercises provided by the system. Because of AI analysis, rule-based decision logic, and expert-verified clinical data, the mobile application outputs personalized rehabilitation plans. It aims to improve the accuracy, flexibility, and individual fitness of home therapy guidance. The system's technical evaluation confirmed its strong capacity to meet its intended functions. The human pose estimation component achieved a mean Average Precision (mAP@0.5) of 69.5% and an F1-score of 72.5%, this indicates reliable performance in tracking and interpreting gross motor movements essential for safe rehabilitation. The facial expression recognition model demonstrated even stronger performance, that achieved a weighted accuracy of 98.0% and an F1-score of 97.98%, with particularly high recall for severe pain, an important feature that is prioritized for ensuring user safety. Although moderate



pain classification showed inconsistencies, this highlights opportunities for refining the dataset in the future. The combination of using of AI generated insights, user report symptoms, and verified physical therapists' clinical rules allowed the decision to output rehabilitation plans with a degree of personalized and contextual sensitivity, ensuring the systems' ability to provide proper guidance that adapts to the user's functional condition and pain profile.

The overall system evaluation shows consistently high ratings across ISO 25010 quality dimensions. This confirmed that PocketPT is reliable, usable, and practical for a mobile rehabilitation tool. It showed high scores in Security, Reliability, and Usability highlight user confidence in the platform's secure data management, operational stability, and easy-to-use interface. Qualitative feedback further emphasizes the apps' objective in measuring movement performance. Users appreciated the pose estimation system's approach to tracking rehabilitation progress with clear numbers. These findings confirm PocketPT's ability to improve engagement, support adherence, and provide ongoing, pain-aware feedback in a clear and accessible way. PocketPT enhances home-based rehabilitation and creates a new connection between traditional clinical practices and modern AI-assisted care.

To boost the system's effectiveness while tackling current gaps, a few core suggestions can help. One approach involves broadening and adjusting the data used for training - this supports stronger results. Using more examples confirmed by therapists, covering diverse physiques, varied light settings, differing motion paces, or alternative



workout spaces enhances adaptability. A particular emphasis could go toward capturing subtle motions at joints like wrists or ankles. These actions improve rehab exercises significantly. Addressing unequal pain category distribution matters just as much - methods like focused data expansion, repeating mid-level pain samples, or applying weighted loss functions may help. Combining inputs from various sensors, including depth or thermal cameras if available, reduces problems tied to regular color images in poor lighting or blocked views. Beyond growing the dataset, refining the model through techniques like removing unneeded weights, reducing numerical precision, or adopting lightweight designs such as MobileNetV3 cuts response delays and supports broader device use.

Beyond technical improvements, better links to clinical practice are key if PocketPT is to become a trusted part of blended rehab setups. A special 'Review Mode' for therapists could let certified experts track how users perform, confirm or revise AI suggestions, while following recovery from afar via protected channels. Such an addition would position the platform as a teamwork aid - one that supports clinicians instead of taking their place. Improved instant corrections - using simpler visuals, voice prompts, or timely warnings based on activity - can help people keep movements both safe and efficient. Also, updating the logic tree might detect risky moments, like bad posture paired with severe discomfort, then stop or alter exercises automatically to reduce injury risk.

Lastly, making the system easier to use over time helps ensure more people keep using it. While PocketPT scored well on usability, later versions should add clearer step-



by-step guidance for new users, especially when understanding results like joint angles or AROM data. Instead of limiting access, supporting additional mobile platforms can reach a wider range of device owners. Improving how data is exported - so rehab reports share easily in common health formats - helps maintain consistent care and fits better into doctor visits. Down the line, studies might look at adding game-like features or personalized motivation tools to boost ongoing engagement, particularly for those managing extended recovery periods.



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APPENDIX A

EVALUATION FORM / SURVEY

|                                                                                                                                            |                   |          |         |       |                |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| <b>Personal Information</b>                                                                                                                |                   |          |         |       |                |
| Name                                                                                                                                       |                   |          |         |       |                |
| Age                                                                                                                                        |                   |          |         |       |                |
| Sex                                                                                                                                        |                   |          |         |       |                |
| Do you currently reside in Cavite? If yes, in what city or municipality:                                                                   |                   |          |         |       |                |
| What physical activities do you usually engage in?                                                                                         |                   |          |         |       |                |
| Name of the person who referred you to this survey                                                                                         |                   |          |         |       |                |
| Are you a professional in the rehabilitation or medical field (e.g., a licensed physical therapist)?                                       |                   |          |         |       |                |
| <b>Functional Suitability</b>                                                                                                              | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| The application effectively supports physical therapy (exercise recording, assessment, pose detection, reports, and progress tracking).    |                   |          |         |       |                |
| The rehabilitation plan dynamically changes based on my performance feedback.                                                              |                   |          |         |       |                |
| The exercise screens and features completely guided the user through the rehabilitation process (instructions, recording, saving results). |                   |          |         |       |                |
| The assessment tools (pose                                                                                                                 |                   |          |         |       |                |



|                                                                                       |                   |          |         |       |                |
|---------------------------------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| detection, range of motion angles) visibly provide clear measurement results.         |                   |          |         |       |                |
| The rehabilitation plan recommendations reflect my specific pain characteristics.     |                   |          |         |       |                |
| The application provides features that improve the rehabilitation experience.         |                   |          |         |       |                |
| <b>Performance Efficiency</b>                                                         | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| The system accurately identifies my body movements during the assessment.             |                   |          |         |       |                |
| The pose detection appear smooth in real-time without noticeable lag.                 |                   |          |         |       |                |
| The range of motion (ROM) results generated within a short, acceptable time.          |                   |          |         |       |                |
| Pose estimation feedback is consistent across different lighting and body positions.  |                   |          |         |       |                |
| The system accurately detects my facial expressions during pain or discomfort.        |                   |          |         |       |                |
| <b>Usability</b>                                                                      | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| The navigation between pages and features (menu, buttons, tabs) is clear and logical. |                   |          |         |       |                |
| First-time users find it easy to use the application without long tutorials.          |                   |          |         |       |                |
| Instructions for exercises and assessments are clear and                              |                   |          |         |       |                |



|                                                                                                                               |                   |          |         |       |                |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| easy to follow.                                                                                                               |                   |          |         |       |                |
| The facial recognition feature contributes to more accurate assessments.                                                      |                   |          |         |       |                |
| The interface is visually appealing (consistent design, clear icons, neat layout).                                            |                   |          |         |       |                |
| <b>Reliability</b>                                                                                                            | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| The application runs smoothly without crashing or freezing.                                                                   |                   |          |         |       |                |
| The pose detection remains stable regardless of different camera conditions (e.g. low visibility, high noise, blurry camera). |                   |          |         |       |                |
| The saved exercises or assessments re-appear correctly when the application is re-opened.                                     |                   |          |         |       |                |
| <b>Security</b>                                                                                                               | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| Login and signup screens require user credentials (email, password, etc.)                                                     |                   |          |         |       |                |
| The application visually confirms when access is denied after incorrect login attempts.                                       |                   |          |         |       |                |
| The sensitive sections (patient data, assessments) are only shown after authentication.                                       |                   |          |         |       |                |
| The system's recommendations consider both my past data and current performance.                                              |                   |          |         |       |                |
| The application shows visible                                                                                                 |                   |          |         |       |                |



|                                                                                                                     |                   |          |         |       |                |
|---------------------------------------------------------------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| indication of secure handling (e.g., lock icons, privacy disclaimers, terms & conditions).                          |                   |          |         |       |                |
| <b>Portability</b>                                                                                                  | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
| The installation process is straightforward and error-free.                                                         |                   |          |         |       |                |
| The system's exercise recommendations reflect my progress or current ability.                                       |                   |          |         |       |                |
| Export/share features in the application suggest data portability (saving reports for use outside the application). |                   |          |         |       |                |
| The logout flows shows a clean exit without confusing leftover screens.                                             |                   |          |         |       |                |

## Feedback and Suggestions

Strength/s

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Weakness/es

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Additional feedback or suggestions for improvement.

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## APPENDIX B

### EVALUATION PHOTOS



EF EM Suzara



🔥 CALL FOR RESPONDENTS! 🍷🍷  
Greetings in St. La Salle! 🍷

We are 4th Year BS Computer Science students majoring in Intelligent Systems from the College of Science and Computer Studies (CICS), De La Salle University - Dasmariñas, currently conducting our thesis entitled:

"PocketPT: An AI-Driven Rehabilitation Assistant with Human Pose Estimation and Pain Recognition using CNN and LSTM." 📱📱

Our study introduces PocketPT, a mobile physical therapy assistant designed to make rehabilitation smarter and more accessible. It uses camera-based pose detection, range of motion (ROM) assessments, and AI-powered pain recognition to help track progress, provide real-time feedback, and support data-driven rehabilitation for both patients and clinicians.



EF EM Suzara



We are inviting physically active individuals or people experiencing muscular pain who currently reside in Cavite to participate in our online survey.

👉 Take the survey here:  
<https://forms.gle/qbKLxulW1FbsX1uw6>  
<https://forms.gle/qbKLxulW1FbsX1uw6>  
<https://forms.gle/qbKLxulW1FbsX1uw6>

📧 Referral: Yuri M. Brown

Your answers will significantly contribute to the success of our study. We truly appreciate your time and support! 🙏

forms.gle

Nov 9, 2025, 8:09 PM



Ganda ng app niyo! Galing!! PANALO!!! Congrats Joto!!!



👤 You replied to EF EM

Ganda ng app niyo! Galing!! PANALO!!! Congrats Joto!!!

THANK YOU POO!!





Adrian Bayot



Nov 9, 2025, 4:56 PM

Good Afternoon po. We are conducting a survey as part of our thesis titled "PocketPT: A Personalized Pain Recognition and Rehabilitation Support System." We would greatly appreciate your time in answering our survey to help us gather data for our study. Your responses will remain confidential and used only for academic purposes. Thank you for your support!

👉 CALL FOR RESPONDENTS! 📢👉  
Greetings in St. La Salle! 🌟

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forms.gle





## APPENDIX C

### DOCUMENTATION PHOTOS







**APPENDIX D**

**ETHICS REVIEW CERTIFICATE**



**De La Salle University - Dasmariñas**  
DLSU-D Ethics Review Committee

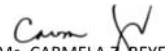


## CERTIFICATION

The DLSU-D Ethics Review Committee has reviewed the protocol for the study titled "**POCKETPT: AN AI-DRIVEN REHABILITATION ASSISTANT WITH HUMAN POSE ESTIMATION AND PAIN RECOGNITION USING CNN AND LSTM**" code **DERC\_25-26\_063T** as proposed by **YURI BROWN, JOSEPH LINA, FRANCIS VINCENT PANGANIBAN**, College of Information and Computer Studies.

The committee approves the implementation of the study. This certification is valid until March 31, 2026.

Issued this 8<sup>th</sup> of October 2025.

  
MS. CARMELA Z. REYES  
DLSU-DERC Chair



## STATISTICIAN CERTIFICATE



De La Salle University – Dasmariñas



COLLEGE OF SCIENCE  
MATHEMATICS AND STATISTICS DEPARTMENT  
DATA ANALYSIS AND TRAINING CENTER (DATC)

### CERTIFICATION

The DLSU-D Data Analysis and Training Center has reviewed and validated the questionnaire and performed a comprehensive statistical analysis of the data for the study entitled “ PocketPT: An AI-Driven Rehabilitation Assistant with Human Pose Estimation and Pain Recognition using CNN and LSTM

”

as proposed by  
Yuri M. Brown, Joseph II S. Lina, and Francis Vincent J. Panganiban

College of Information and Computer Studies. The statistician and the unit head of DATC approves the implementation of the study.

Ms. CARMELA Z. REYES  
Statistician, DATC DLSU-D



## LETTER OF CONFORMITY

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De La Salle University  
D A S M A R I N A S



De La Salle University - Dasmariñas  
College of Information and  
Computer Studies

April 12, 2025

Mr. FM Suzura

Physical Therapist

ProBio Medcare Doctors Clinic – Paranaque

RUIZ Building, 365 El Grande Ave, Paranaque, 1720 Metro Manila

Dear Mr. Suzura,

We, third-year Computer Science students of De La Salle University – Dasmariñas, are conducting a research study entitled “*PocketPT: User-Centric Rehabilitation App for Self-Diagnosis*” as part of our academic requirement in Thesis 1. The study involves developing a mobile application designed to assist individuals in managing muscle strains and injuries through a structured, three-phase rehabilitation approach: treatment, rehabilitation, and strengthening.

With this letter, we are respectfully requesting your consent to participate in the study. Your professional guidance in injury prevention and recovery will be instrumental in designing effective rehabilitation programs. Additionally, we seek your permission to test the completed application and distribute a feedback survey via QR code.

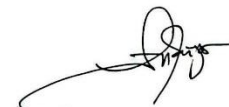
All data collected will be kept confidential and used strictly for research purposes.

Sincerely,

Mr. Yuri Brown

bym0866@dlsud.edu.ph

BCS32 Student, DLSU-D

  
FREDERICK MARCH C. SUZURA  
LUC# 12723

Endorsed by:   
Mr. Rolando Barrameda  
Research Adviser

Noted by:   
Ms. Josephine Eduardo  
Chair, CSD



Scanned with CamScanner



De La Salle University  
D A S M A R I Ñ A S



De La Salle University - Dasmariñas  
College of Information and  
Computer Studies

April 12, 2025

Mr. Adrian Bayot  
Physical Therapist  
Standard Insurance Corp. Technical Training Center  
Governor's Dr, Naic, Cavite

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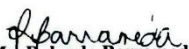
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All data collected will be kept confidential and used strictly for research purposes.

Sincerely,

**Mr. Yuri Brown**  
*bym0866@dlsud.edu.ph*  
BCS32 Student, DLSU-D

  
BAYOT, ADRIAN LANCEL T., PTRP

Endorsed by:   
**Mr. Rolando Barrameda**  
Research Adviser

Noted by:   
**Ms. Josephine Eduardo**  
Chair, CSD



## NON-DISCLOSURE AGREEMENT



De La Salle University  
DASMARIÑAS



De La Salle University - Dasmariñas  
College of Information and  
Computer Studies

### NON-DISCLOSURE AGREEMENT

We, **Yuri M. Brown, Joseph II S. Lina, and Francis Vincent J. Panganiban**, third-year Computer Science students at De La Salle University – Dasmariñas, hereby affirm our commitment to maintaining the utmost confidentiality and integrity in handling all information and data related to our undergraduate thesis entitled “*PocketPT: A User-Centric Rehabilitation App for Self-Diagnosis*”.

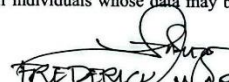
All data obtained throughout the phases of data collection, model development, and system evaluation shall be used strictly and solely for academic purposes, specifically within the defined scope of this research project. At no point shall any portion of the acquired data be disclosed, disseminated, or utilized for purposes beyond the objectives of the study.

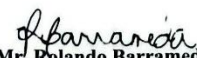
We recognize that the data involved may include sensitive, personal, or confidential information. Accordingly, we commit to full compliance with the provisions of **Republic Act No. 10173**, otherwise known as the **Data Privacy Act of 2012**, governing the lawful collection, handling, processing, storage, and disposal of personal data. We shall ensure the implementation of appropriate technical, administrative, and organizational measures to safeguard the confidentiality, integrity, and security of the data, and to prevent any form of unauthorized access, misuse, or data breach.

This confidentiality agreement serves as a formal expression of our unwavering obligation to uphold the highest standards of ethical conduct in research, to maintain the trust and confidence of all stakeholders, and to protect the rights and privacy of all individuals whose data may be involved in the course of this study.

  
**Yuri M. Brown**  
Student Researcher, BCS32  
bym0866@dlsud.edu.ph

  
**Joseph II S. Lina**  
Student Researcher, BCS32  
ljs0792@dlsud.edu.ph

  
**FREDERICK MARC C. SUZARA**  
UC # 12703  
  
**Francis Vincent J. Panganiban**  
Student Researcher, BCS32  
pfj0854@dlsud.edu.ph

Endorsed by:   
**Mr. Rolando Barrameda**  
Research Adviser

Noted by:   
**Ms. Josephine Eduardo**  
Chair, CSD



De La Salle University  
DASMARIÑAS



De La Salle University - Dasmariñas  
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pff0854@dlsud.edu.ph

Endorsed by: **Mr. Rolando Barrameda**  
Research Adviser

Noted by: **Ms. Josephine Eduardo**  
Chair, CSD



## DATA VALIDATION LETTER



De La Salle University – Dasmariñas  
COLLEGE OF INFORMATION AND COMPUTER STUDIES



October 18, 2025

Mr. Frederick March C. Suzara  
Physical Therapist  
PBMED Doctors Clinic  
RUIZ Building, 365 El Grande Ave, Parañaque, 1720 Metro Manila

Dear Mr. Frederick March C. Suzara,

We, **Yuri M. Brown, Joseph II S. Lina, and Francis Vincent J. Panganiban**, 4th year BS Computer Science students from De La Salle University - Dasmariñas, are currently conducting a research study as part of our thesis titled **“PocketPT: An AI-Driven Rehabilitation Assistant with Human Pose Estimation and Pain Recognition using CNN and LSTM”**.

This letter serves to confirm that you, as one of our licensed physical therapists, have reviewed and certified that the Pain Recognition Assessments are accurate and valid. Your verification confirms that the assessment results accurately reflect the observed levels of pain (Low, Moderate, and Severe) and meet the professional standards for evaluation. Thank you for your expertise and continued commitment to ensuring the quality and reliability of our assessments.

We thank you for your time and insight on this matter and look forward to hearing from you soon.

Respectfully yours,

Joseph II S. Lina

202080792

[ljs.0792@dlsud.edu.ph](mailto:ljs.0792@dlsud.edu.ph)

Student Researcher, DLSU-D

Approved by: **Mr. Frederick March C. Suzara**  
*Physical Therapist*



October 18, 2025

Mr. Adrian Lance Bayot  
Physical Therapist  
Standard Insurance Corp. Technical Training Center  
Governor's Dr, Naic, Cavite

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Student Researcher, DLSU-D

Approved by:

**Mr. Adrian Lance Bayot**

*Physical Therapist*



## PROOFREADING CERTIFICATE

DE LA SALLE UNIVERSITY – DASMARIÑAS  
COLLEGE OF LIBERAL ARTS AND COMMUNICATION  
LANGUAGES AND LITERATURE DEPARTMENT

### CERTIFICATION

This certifies that the thesis / special project entitled PocketPT: An AI-Driven Rehabilitation Assistant with Human Pose Estimation and Pain Recognition Using CNN and LSTM of Yuri M. Brown, Joseph S. Lina II, and Francis Vincent J. Panganiban was edited by the undersigned.

A handwritten signature in black ink, appearing to read 'Rachelle S. Ballesteros'.

MS. RACHELLE S. BALLESTEROS  
FACULTY SIGNATURE OVER PRINTED NAME  
Date: April 1, 2026

*Note: This certification serves as the student clearance for editing and should not be appended in the hardbound copy of the thesis or special project.*